HADLEY CIRCULATION

I N James, University of Reading, Reading, UK

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

The so-called 'Hadley circulation' is perhaps the earliest attempt to account for the global-scale distribution of winds in the Earth's atmosphere in terms of basic physical processes. Halley in 1685 and Hadley in 1735 both proposed that the 'Trade Winds' that blow toward the Equator at low latitudes could be understood as the lower branch of an axially symmetric convection cell driven by the temperature difference between the Equator and poles of the Earth. Their ideas were ahead of their time, especially as there was then no prospect of determining winds at upper levels of the atmosphere and thus verifying their hypothesis. When routine upper-air observations became available in the mid-twentieth century, the ideas of Halley and Hadley were essentially confirmed. Today, the term 'Hadley circulation' refers to the thermally driven meridional overturning motions in the low-latitude troposphere.

Figure 1 is the traditional, and somewhat oversimplified, schematic view of the global atmospheric circulation that will be found in many elementary text books. It divides the Earth into a set of climate zones, with the Trade Wind regime confined to the tropics. The Trade Winds are simply the low-level part of the overturning 'Hadley circulations', with ascent near the Equator, descent in the subtropics and a poleward return flow at upper levels. The more disturbed midlatitudes are characterized by generally westerly winds, with irregular growing and decaying eddies, the cyclonic and anticyclonic weather systems generated by baroclinic instabilities. When averaged around entire latitude circles, this turbulent midlatitude flow averages to a weak 'Ferrel circulation', in which warmer air at lower latitudes sinks and colder air at high latitude rises. There is some evidence of a very weak 'polar cell' at high latitudes.

The energy that drives the Hadley circulation comes from the conversion of heat energy to mechanical energy in the tropical atmosphere: the Hadley circulation is a prototypical example of a thermodynamic 'heat engine'. Such heat engines are ultimately responsible for maintaining all motions in the atmosphere against the dissipative effects of friction. The operation of the atmospheric heat engine is shown in **Figure 2**, which is a classic thermodynamic diagram in which temperature is plotted against specific entropy. The thermodynamic state of an air parcel – that is, its temperature, pressure, density and so on – are represented by a point on the thermodynamic diagram, and any change of its thermodynamic state by a curve on the diagram. The area under the process curve is

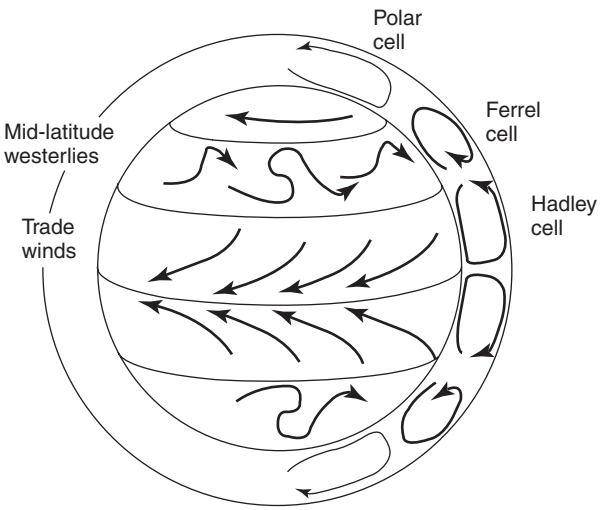

Figure 1 A schematic view of the mean circulation of the troposphere. The arrows on the globe show the winds near the Earth's surface. The circulations at the side show the zonal mean circulation cells at various latitudes.

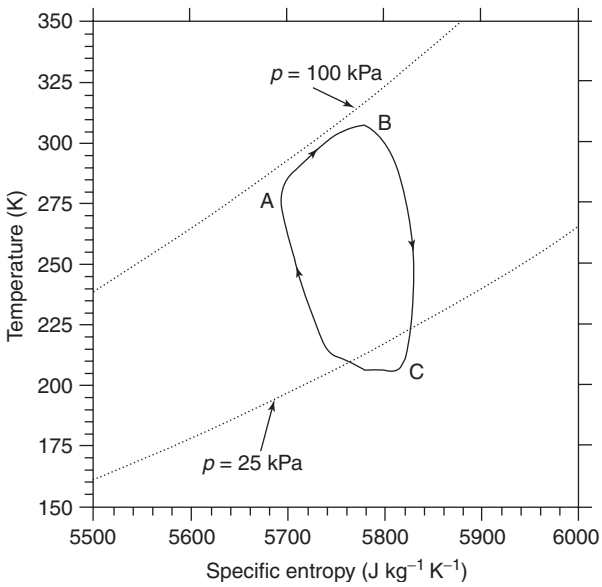

Figure 2 A schematic thermodynamic diagram for the Hadley circulation.

proportional to the heat energy entering an air parcel. The diagram also shows two different lines of constant atmospheric pressure, one near the Earth's surface and one in the tropical upper troposphere. Near the surface, air flows toward the Equator, along the segment marked AB, gaining heat from the surface (this may be in the form of direct or sensible heat, or in the form of latent heat as water evaporates into the air). Near the Equator, it rises almost adiabatically (that is, with little heat entering or leaving the air) along the segment BC. It then moves poleward along segment CA, cooling (that is, losing entropy) by emitting infrared radiation to space and descending. Eventually it returns to its original location A. During this cyclic process, more heat is added to the air along AB than is removed along CA. The excess heat is converted to mechanical energy associated with the circulation of the tropical air.

The condition for such an energy conversion to take place is that heat should on average be added at higher pressure than it is removed. Equivalently, one can say that air must rise on average when it is warmer, and descend when it is cooler. A circulation with these properties is called a 'thermally direct circulation'. A thermally indirect circulation, in contrast, must be driven by a source of mechanical energy; a refrigerator cycle is an example of such a thermally indirect circulation. In the schematic diagram of **Figure 2**, the Hadley circulation is thermally direct, and therefore generates mechanical energy. In contrast, the Ferrel circulation of midlatitudes is thermally indirect and consumes mechanical energy.

The observed annual mean meridional circulation is shown in **Figure 3**. The contours are parallel to the northward and upward winds averaged around latitude circles and in time. The contour values have been scaled to have units of $\mathrm{kg\,s^{-1}}$. They may be thought of as denoting the mass flux across a line from the edge of the plot to that point. The most striking feature is the strong rising motion near the Equator, and sinking motion at latitudes of about 25° N and S, defining two overturning cells, the 'Hadley cells', one in each hemisphere. However, the actual winds associated with these circulations are not particularly strong: they barely exceed $5\,\mathrm{m\,s^{-1}}$. The Hadley cells are thermally direct. Weaker, thermally indirect Ferrel cells are seen at higher latitudes. Considerably weaker thermally direct polar cells are seen at high latitudes.

The diagram also reveals that there is a close relationship between the westerly component of the wind, shown by the shading, and the meridional flow. The westerly component is much stronger, with values up to $40\,\mathrm{m\,s^{-1}}$. These maximum winds, the so-called 'subtropical jet', are found in the upper troposphere, just where the circulations associated with the Hadley cells meet those associated with the Ferrel cells. There is also a close relationship between the zonal winds and the temperature fields: they are linked, to a very good approximation by the thermal wind relationship, which can be written as eqn [1].

$$\frac{\partial[\bar{u}]}{\partial p} = \frac{R}{pf}\frac{\partial[\bar{T}]}{\partial y} \qquad [1]$$

That is, a strong vertical wind shear is associated with a strong poleward temperature gradient. In the deep tropics where the Coriolis parameter f is small, this relationship indicates that the temperature gradients must be small, whatever the wind field. But in the subtropics and midlatitudes, the increasing westerly wind with height is associated with the fall of temperature toward the poles.

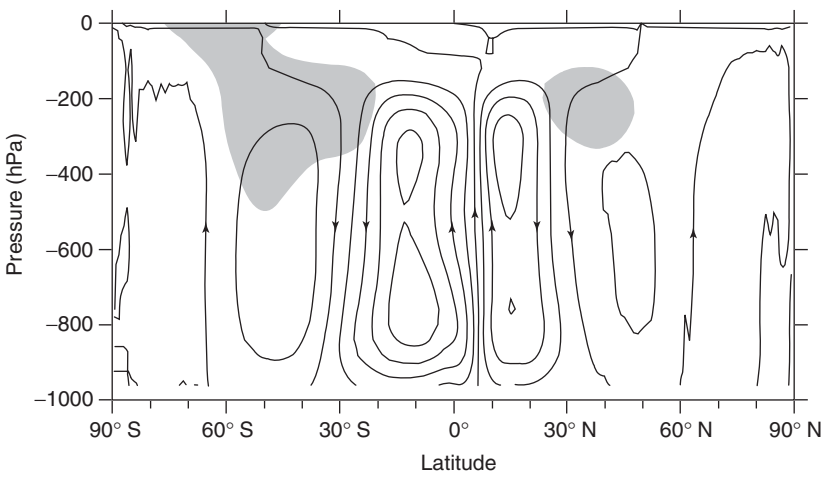

Figure 3 The annual mean meridional streamfunction. Contour interval $2 \times 10^9\,\mathrm{kg\,s^{-1}}$. Shading shows zonal winds greater than $20\,\mathrm{m\,s^{-1}}$. Based on an analysis of 20 years of ECMWF analyses.

The Held–Hou Model

An elegant model due to Held and Hou gives considerable insight into the Hadley circulation and the factors that determine its extent. **Figure 4** illustrates the model. The atmosphere is divided into two layers. The lower layer is affected by friction at the ground, and flow within it is supposed to be generally small. Friction is effectively zero in the upper layer and so at this level rings of air conserve their angular momentum as they move poleward. Assuming that such rings start at the Equator with zero motion relative to the solid Earth, the wind at higher latitudes in the upper layer is given by eqn [2], where Ω is the rotation rate of the Earth, a is the radius of the Earth, and y is the distance from the equator, proportional to the latitude.

$$U_{\mathrm{M}} = \frac{\Omega}{a} y^2 \qquad [2]$$

Using the principle of thermal wind balance in the form of eqn [3], the formula for U_{M} can be used to predict the variation of temperature with latitude, $\theta_{\mathrm{M}}(y)$.

$$\frac{\partial u}{\partial z} = \frac{ga}{2\Omega y T} \frac{\partial T}{\partial y} \qquad [3]$$

This is to be compared with the hypothetical 'radiative equilibrium' temperature distribution $\theta_{\mathrm{E}}(y)$ of an atmosphere that is not permitted to circulate. Where the actual temperature is less than radiative equilibrium there is net heating, and vice versa. In a steady state, this heating and cooling should exactly balance in the Hadley circulation and this requirement fixes the meridional extent and strength of the Hadley circulation.

Figure 5 illustrates a graphical solution of the Held–Hou model. The actual temperature varies very little with latitude in the tropics but drops rapidly in the subtropics and mid-latitudes. The radiative equilibrium temperature has a maximum at the Equator. The temperature on the equator is set by requiring that there be no net heating of air parcels as they circulate, that is, that the two shaded areas must be equal. The poleward limit of the Hadley circulation is at the latitude where these curves cross for the second time. A formula for the distance of the poleward edge of the Hadley cell from the equator results from this solution (eqn [4]).

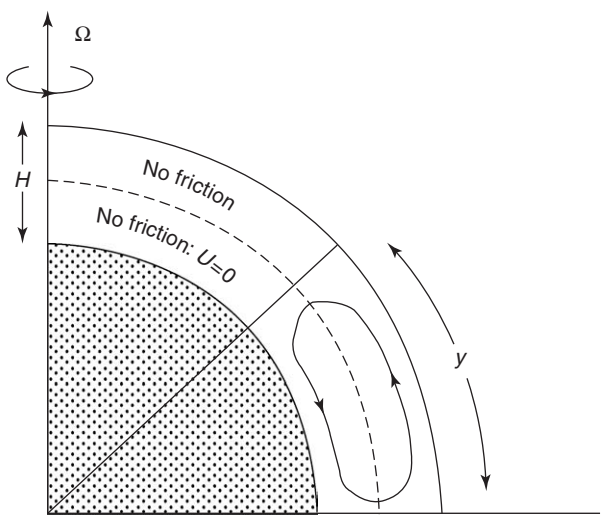

Figure 4 The configuration of the Held–Hou model.

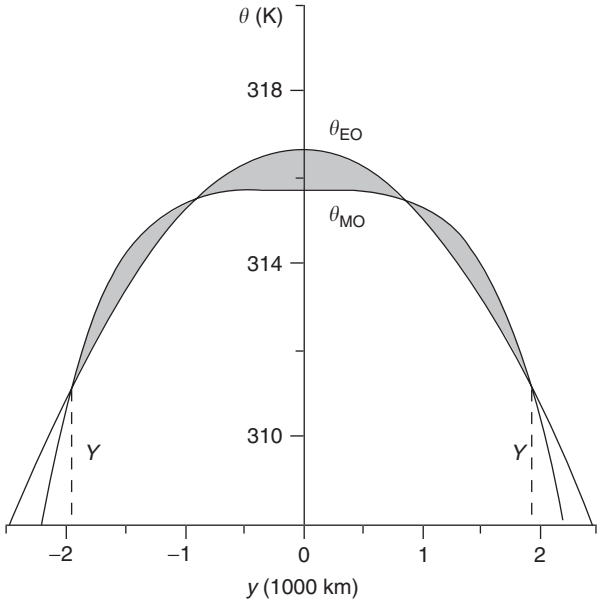

Figure 5 Solution of the Held–Hou model.

$$Y = \sqrt{\frac{5gH\,\Delta\theta}{3\Omega^2\theta_0}} \qquad [4]$$

This formula suggests a value for Y of about 2500 km, in remarkably good agreement with observations considering the simplicity of the model.

The model can be elaborated. For example, the vertical motions, proportional to the heating in the regions of ascent and descent, can be estimated. The model predicts a vertical circulation that is rather weaker than that observed. The effect of latent heat release in cumulonimbus clouds, which leads to intensified but narrower regions of ascent, and broad regions of descent, can be represented. But the basic physics, which predicts that the Hadley circulations are confined to 2500 km or so of the equator, remains relevant.

Seasonal Effects

The annual mean circulation shown in **Figure 3** is in fact the average of two quite different circulation

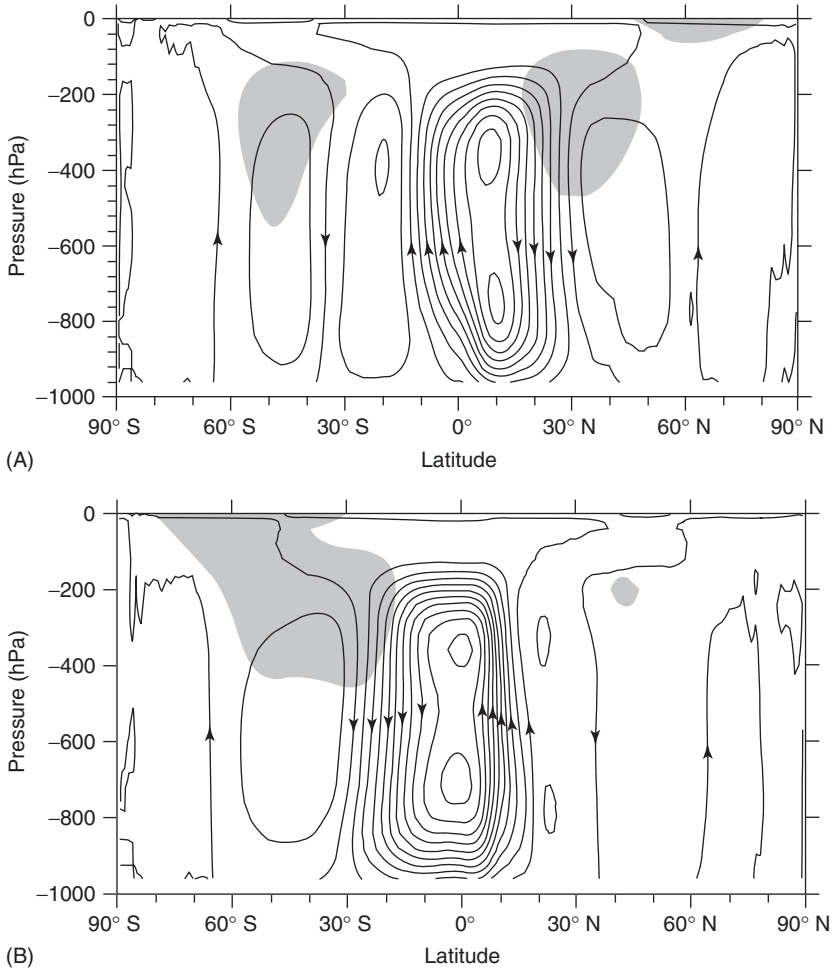

Figure 6 The mean meridional circulation for (A) the December–January–February season and (B) the June–July–August season. Other details are as for **Figure 3**.

regimes that persist around the solstices. **Figure 6** shows the circulation for the mean Northern Hemisphere winter and summer seasons. In both cases, there is a single strong thermally direct Hadley cell with rising motion in the summer hemisphere and descent in the winter hemisphere. Weaker, thermally indirect Ferrel cells are seen at middle latitudes in both hemispheres, but there is little or no sign of a Hadley cell in the summer hemisphere. Looking at the mean circulation for shorter periods reveals that the transition between a circulation like that of **Figure 6A** and one like that of **Figure 6B** is quite abrupt. At most times, there is just a single tropical Hadley cell whose circulation links the two hemispheres: at some point in the spring and autumn its direction of circulation switches abruptly as the temperature maximum crosses the Equator.

The Held–Hou model can be adapted to the situation where the heating is not symmetric about the Equator. Assume that the maximum radiative equilibrium temperature is no longer at the Equator, but at some latitude y_0. As well as the latitude of the northern and southern edges of the Hadley cells, the latitude y_c of the streamline that divides circulation into the summer and winter hemispheres, and which is not the same as y_0, must be determined. The algebra is more complicated, but the steps in the argument are just the same as for symmetric Hadley cell described in the previous section.

Figure 7 shows the results. For even small y_0, the summer cell shrinks drastically and the winter cell intensifies. Almost all the circulation is associated with ascent in the summer hemisphere and with descent in the winter. The strength of the circulation is indicated by the area between the temperature curves and the radiation equilibrium curve. For y_0 of only 500 km, the winter cell has intensified by a factor of about 10 compared to the symmetric case, while the summer cell has weakened by a similar factor. The winter cell is therefore some 100 times as intense as the summer cell. Such a highly nonlinear response to the latitude of the heating maximum means that the annual mean

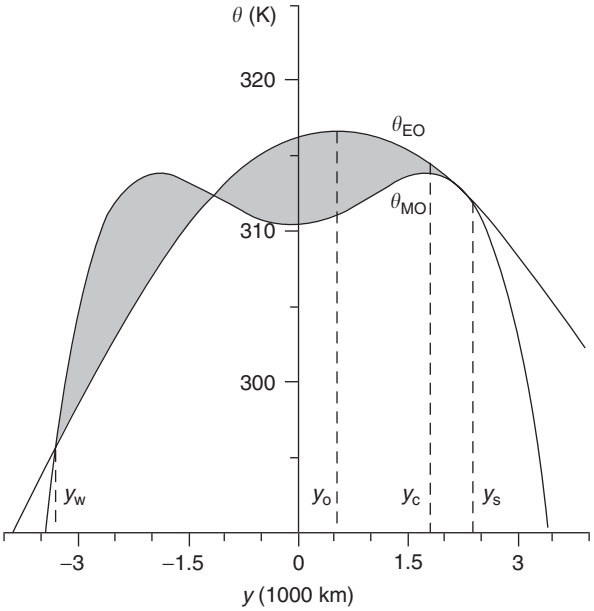

Figure 7 As **Figure 5**, but for a situation in which the heating maximum is located away from the Equator. y_c is the latitude of maximum radiative equilibrium temperature, y_c is the latitude dividing the winter and summer Hadley circulation cells, y_w and y_s designate the limits of the winter and summer Hadley cells, respectively.

meridional circulation is much more intense than the circulation derived from the annual mean heating. This is a particularly pointed example of the problem of 'nonlinear averaging', which is ubiquitous in the study of climate. This result also reconciles the weak circulations of the Held–Hou model with the stronger observed circulation: we should interpret the annual mean circulation as the average of the two solsticial circulations, not as the response to the annual mean thermal forcing.

A Lagrangian View

The diagrams of the meridional circulation shown so far have all been based on so-called 'Eulerian averages'. That is, the winds have been averaged at fixed points in space to produce the time-mean, zonal-mean circulation. At all points in space, the winds and temperatures fluctuate to some degree as weather systems pass across the observing site. An alternative is to follow individual elements of fluid as they move around in the atmosphere, and average their properties to define a mean circulation. Such a mean is called the 'Lagrangian mean', and in many ways is a much preferable way to describe the circulation. For example, the laws of physics applied to the atmosphere all refer to the properties of discrete, identifiable lumps of fluid. However, the Lagrangian mean is very difficult to calculate in practice, not least because individual elements of fluid rapidly become distorted and eventually thoroughly mixed with neighboring elements.

An approximation to the Lagrangian meridional mean circulation can easily be calculated, and is shown in **Figure 8**. In constructing this diagram, the wind data were averaged not on surfaces of constant pressure (as in **Figures 2–7**) but on surfaces of constant 'potential temperature'. The potential temperature of an air parcel generally remains more or less constant for periods of less than a few days. It follows that surfaces of constant potential temperature move up and down in response to the movement of the air. Averaging on potential temperature surfaces is equivalent, to the degree that potential temperature is indeed conserved, to taking the Lagrangian average.

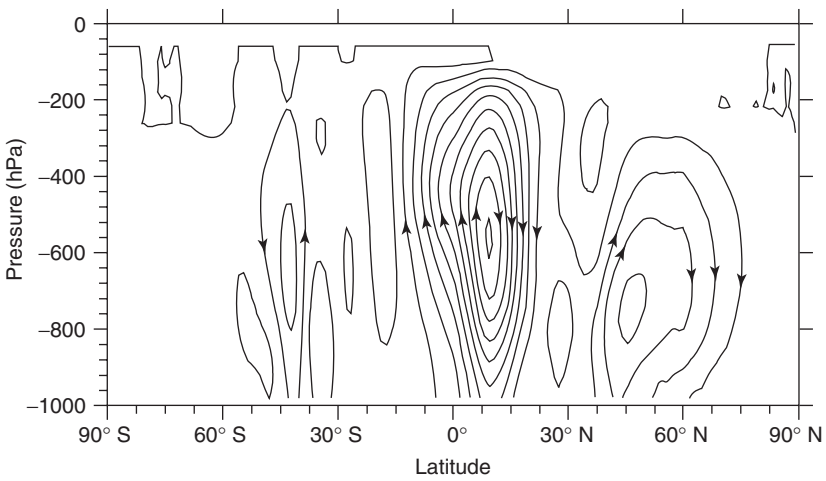

Figure 8 The mean meridional circulation for the December–January–February season, but with the data zonally averaged on surfaces of constant potential temperature rather than on surfaces of constant pressure. Other details are as for **Figure 3**.

Figure 8 differs dramatically from the corresponding Eulerian mean circulation shown in **Figure 6B**. The tropical Hadley cell is still present, but the mid-latitude, thermally indirect Ferrel is more or less eradicated. Instead, a thermally direct circulation extends all the way from the tropics to the pole in the winter hemisphere. The original picture of the global circulation suggested by Halley and Hadley is largely vindicated if one views the circulation in Lagrangian terms.

The thermally indirect Ferrel cell actually transports heat against the temperature gradient, from high latitudes to low latitudes. At the same time, fluctuations in the flow (often termed 'eddies') more than compensate by transporting heat down the temperature gradient, from low latitudes to high. In fact, the partitioning of the flow into mean and eddy parts is arbitrary. The Lagrangian circulation, dominated by thermally direct circulations at nearly all latitudes, is a more natural and less arbitrary description. In the Lagrangian view, thermally direct Hadley circulations dominate the large-scale tropospheric circulation. The Lagrangian view is remarkably like that originally suggested by Halley and Hadley.

See also

Boundary Layers: Convective Boundary Layer. **Coriolis Force**. **General Circulation:** Energy Cycle. **Lagrangian Dynamics**.

Further Reading

Hadley G (1735) Concerning the cause of the general trade winds. *Philosophical Transactions of the Royal Society of London* 39: 58–62.

Halley E (1686) An historical account of the trade winds, and monsoons, observable in the seas between and near the tropicks, with an attempt to assign the physical cause of the said winds. *Transactions of the Royal Society of London* 16: 153–168.

Held IM and Hou AY (1980) Nonlinear axially symmetric circulations in a nearly inviscid atmosphere. *Journal of the Atmospheric Science* 37: 515–533.

James IN (1995) *Introduction to Circulating Atmospheres*. Cambridge: Cambridge University Press.

Lindzen RS and Hou AY (1988) Hadley circulations for zonally averaged heating centred off the equator. *Journal of Atmospheric Sciences* 45: 2417–2427.

Peixóto JP and Oort AH (1992) *Physics of Climate*. New York: American Physical Society.

HAIL AND HAILSTORMS

C Knight and N Knight, National Center for Atmospheric Research, Boulder, CO, USA

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

Hailstones are balls of ice that typically fall from cumulonimbus clouds. By convention, they must be greater than 5 mm in diameter but their composition, size, and shape are variable. The largest hailstones can have longest dimensions of 15 cm or more.

Hailstone amounts are also highly variable, but generally the largest hailstones and heaviest hailfalls are from the most severe storms; that is, storms with the strongest updrafts, tallest cloud tops, and largest size. Thus hail is correlated with tornadoes, and also with lightning, though many storms produce lightning with no hail at the ground. Hail is not as well correlated with flooding, which often results from long-lasting and slow-moving precipitation systems that do not produce hail and may not have exceptionally strong updrafts.

Hailstones include various amounts of air bubbles, often in layers that indicate growth stages, but when larger than about 2 cm in diameter their densities are usually within 5% of that of solid ice, 0.91 g cm^{-3}. However, hail may be slushy, containing significant amounts of liquid water, and, especially at small sizes, the air content may be great enough that the hail is soft. Soft hail is distinguished from graupel (accumulations of rime on snow particles or small frozen water drops) only by size, and since nearly all hail falls through a thick layer of air above the freezing point before reaching the ground, soft hail is often slushy, because of melting, when it falls. Much rainfall from cumulonimbus clouds in temperate climates is melted graupel and small hail.

Fundamental Concepts of Hail Formation

Hail forms by the accretion of water droplets onto ice particles falling through supercooled cloud. The basic elements needed for understanding the principles of hail formation are as follows.

The Updraft and its Consequences

Humid air rising in the cores of cumulus clouds cools as it rises. The cooling causes the condensation and

growth of water droplets to form the cloud, and the droplets supercool when the temperature falls below the freezing point. These water droplets are very small, and rise in the updraft almost as fast as the air rises, typically several tens of meters per second in hailstorms, because their terminal fall speeds are less than $10\,\mathrm{cm\,s^{-1}}$. An ice particle in such an updraft that is big enough to have a higher fall speed collides with and collects supercooled droplets, which freeze upon impact and stick to it. This is the basic mechanism of hailstone growth, and the initiating particle may be a snow crystal or snowflake or a frozen water drop. The other main role of the updraft in hail formation is to be strong enough and long-lasting enough to hold the hailstones aloft, within supercooled cloud above the freezing level, long enough for them to grow to their final sizes. If they are to reach the ground as hail, they must be big enough not to melt on the way down.

Terminal velocities of hailstones are described by

$$V_T = \sqrt{\frac{4g\rho_i D}{3C_D\rho_a}} \qquad [1]$$

In eqn [1], where C_D is the drag coefficient, a dimensionless quantity that expresses how the drag force (the air resistance) relates to the velocity and the fluid properties. The densities of ice and air are indicated by ρ; g is the acceleration due to gravity; and D is the diameter of the hailstone. Numerical values are plotted in **Figure 1** for different values of C_D and, with $C_D = 0.55$, for both sea level and 500 hPa pressure. A pressure of 500 hPa corresponds very roughly to $-10°C$ and 6 km above sea level, with considerable variability depending upon local conditions. The large range of values for C_D comes about because of the highly variable shapes of hailstones, which influence fall speed considerably. Hailstone diameter is usually defined as the diameter of a sphere of equivalent mass. A hailstone growing within an updraft may ascend or descend depending upon whether its terminal fall speed is less or greater than the updraft speed.

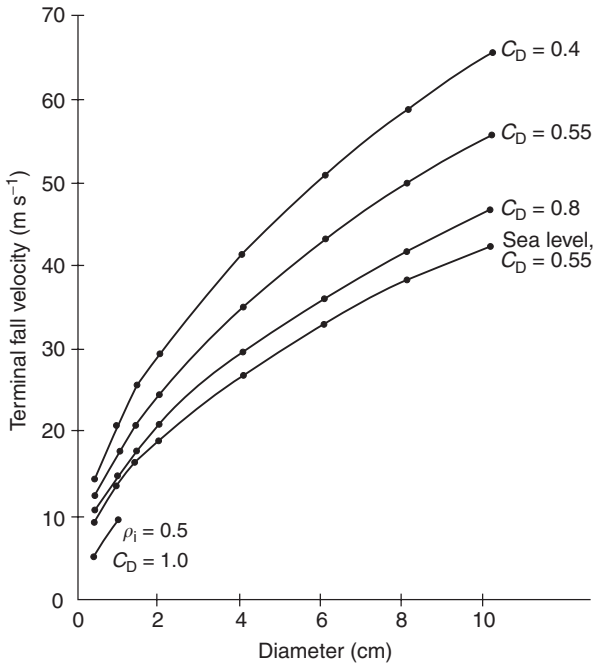

Figure 1 Terminal fall velocities of hailstones plotted against diameter, for the drag coefficients (C_D) indicated, calculated for a pressure of 500 hPa (about 6 km above sea level) and a hailstone density of $0.9\,\mathrm{g\,cm^{-3}}$, except where otherwise indicated. (From Knight CA and Knight NC (2001). In: *Severe Convective Storms*. Boston: American Meteorological Society, by permission of AMS.)

The Thermodynamics of Hailstone Growth

Water remains in the liquid state when cooled below 0°C unless it contains a particle of a solid material that is an ice nucleus, with the property of initiating ice. The content of ice nuclei in the atmosphere is variable, but as a rough average there might be one that acts at $-20°C$ per liter of air and only one per many cubic meters active at $-5°C$. Thus clouds that are not too much colder than 0°C are often composed almost entirely of supercooled water droplets, since the droplet populations in clouds vary between about 100 and 1000 per cubic centimeter. Hailstones grow within such clouds.

The droplets freeze when they collide with ice, releasing latent heat ($\sim 334\,\mathrm{J\,g^{-1}}$). However, since the specific heat of water is $4.2\,\mathrm{J\,g^{-1}\,K^{-1}}$, a drop supercooled to $-10°C$ is warmed to 0°C when only about 1/8 of it is frozen. If the remaining 7/8 is to freeze, the rest of the latent heat must be absorbed by the environment, which in this case is the surrounding air, at $-10°C$. Growing hailstones are thus warmer than the surrounding air, and in extreme cases their temperature may be 0°C and they may grow as a mixture of ice and water, called spongy hail.

Hailstone layering consists of shells with varying air bubble contents, which arise because of air temperature variations (changes in growth altitude) or cloud water content variations, which cause changes in the hailstone temperature.

Hail Falling Behavior, Shape, and Growth

Since hail grows by accreting and freezing water droplets, it grows mostly on its underside where the collisions occur. Small hailstones, falling at only $10–15\,\mathrm{m\,s^{-1}}$, usually maintain a single falling orientation. This often leads to a roughly conical shape, with the growth 'center' on top and consecutive,

broadening layers beneath that constitute most of the bulk. When hailstones grow larger, the higher fall speed creates more turbulence and they tumble. Often their growth shapes are then flattened, most of the collisions occurring around a perimeter because of a rapid and symmetrical, but complicated, tumbling motion. **Figure 2** shows two sections through such a hailstone, illustrating the conical center that grew into a rather thick, somewhat elongated disk. The tumbling has important consequences for shape and also for terminal fall speed and heat exchange, and it may aid in shedding any unfrozen water that otherwise might accumulate.

Hailstones larger than a few centimeters in diameter are often lumpy. When the hail grows wet with liquid water, lumps may form like icicles, from water flowing over the surface. Hailstones of this kind are seen in **Figure 3**. The lumpy shapes often suggest aggregation of smaller hail, but sections through lumpy hail never have shown this to be the case. The lumpiness makes the tumbling more chaotic and the shapes more complex, and also influences the fall speed by increasing the drag coefficient.

The rate of hailstone growth is expressed as eqn [2], where D is the diameter, W is the cloud water content in mass per unit volume, and E is the collection efficiency.

$$\frac{\mathrm{d}D}{\mathrm{d}t} = \frac{V_{\mathrm{T}} W E}{2\rho_{\mathrm{i}}} \qquad [2]$$

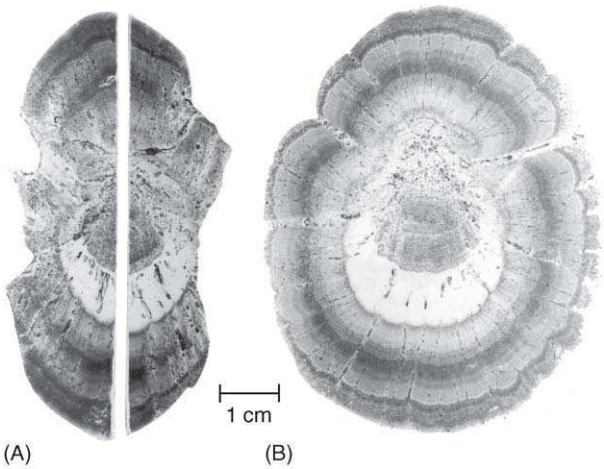

Figure 2 Two slices through a large, oblate hailstone: (A) perpendicular to the short axis of the stone and (B) parallel to the short and the long axes. The photographs were taken with a bright background, so clear ice appears white and ice with small air bubbles is darker. The hailstone grew in a constant falling orientation and developed a conical shape up to 2–3 cm in length. After that the growth was fastest around the perimeter of the flattened hailstone, owing to a rapid, and complicated but symmetrical tumbling motion. Note the growth layers (shown by differing air bubble content) that signify changes in the growth environment.

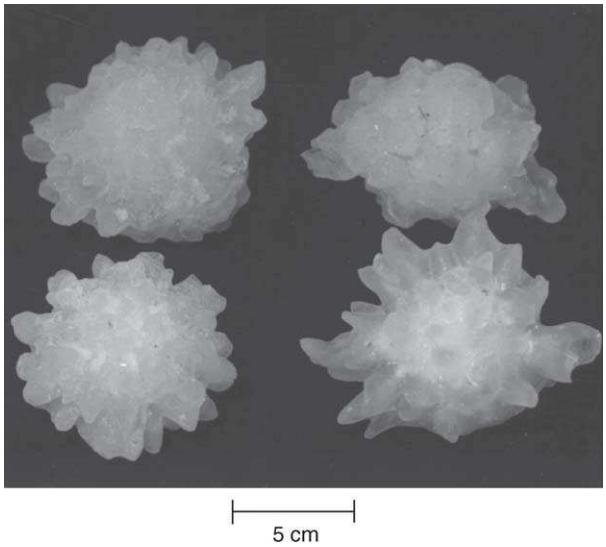

Figure 3 Four hailstones greater than 5 cm in diameter that show prominent projections. These form as the hail grows while wet, with liquid water flowing over the hailstone surface, and grow in a manner similar to the formation of icicles.

$E = 1$ for the case of simple sweep-out – the collection of all water droplets in the volume of air traversed by the hailstone. In reality, E is usually between roughly 0.3 and 0.8 depending upon the sizes of both the droplets and the hailstone, because some droplets in the path of the hailstone are carried around it by the airflow. **Figure 4** shows the typical range of values of

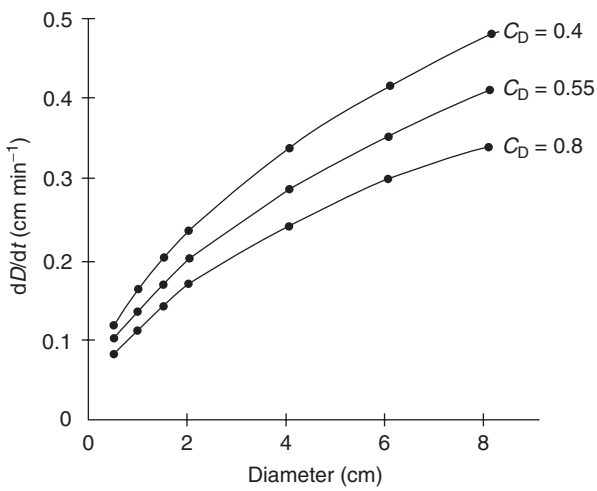

Figure 4 Hailstone growth velocities, in cm min^{-1}, as a function of diameter calculated for the three top curves in **Figure 1**, assuming an effective cloud liquid water content WE of 2.5 g m^{-3} and no shedding of liquid water. Cloud water contents vary from zero to perhaps twice this value, and growth rates vary accordingly (see eqn [2]). (Reproduced with permission from Knight CA and Knight NC (2001) In: *Severe Convective Storms*. Boston: American Meteorological Society.)

growth rates as a function of hail size, for $WE = 2.5\,\mathrm{g\,m^{-3}}$ and $\rho_i = 0.9\,\mathrm{g\,cm^{-3}}$.

Hailstorms

The necessary and sufficient conditions for hail formation in a thunderstorm are easy to state in general terms. The storm needs to provide a strong updraft containing supercooled water droplets as the environment for hailstone growth, and into this environment must come hailstone 'embryos'. These are particles of ice with terminal fall speeds of a few meters per second that can grow in a few tens of minutes into hailstones. Their growth must be fairly rapid because the growth environment may disappear as the updraft weakens (the updrafts in most thunderstorms are intermittent), or the supercooled water content of the updraft may become depleted, either by evaporation as the updraft air mixes with drier surrounding air or by conversion to many small ice particles. If the ascent rate of the potential hail embryo is too great, it may be elevated out of the growth region before it grows big enough to fall as hail; if the ascent rate is too slow, it may fall out before it reaches large enough size. The growth region extends from 0 to -40°C, the temperature at which all the supercooled water droplets freeze spontaneously; a distance of 5 to 6 km within cloud.

The formation of hail, then, depends upon a critical interrelation between the three-dimensional air flow within storms, the fate of the supercooled droplets that occur within the updrafts, and the trajectories of the potential hail embryos. These factors work together to determine how much of the water vapor that condenses reevaporates in the middle and upper atmosphere, how much of it falls as rain, and how much as hail. The percentage of it that falls as hail is difficult to measure, but is usually very small.

Hail in Supercell and Multicell Storms: Steady and Time-Dependent Concepts

There are at least two specific hypotheses for hail formation that refer to two idealized storm types: the supercell storm and the multicell storm. The supercell is a type of storm that can last a long time and travel considerable distances, often producing tornadoes and often producing long swaths of large, damaging hailstones. The diagrams in **Figure 5** represent a typical case in the central United States, a horizontal cross-section view looking down at the storm and a vertical cross-section view, both indicating possible growth trajectories. The humid air entering the storm comes in a vigorous flow at the surface from the south or south-east, rises up through the middle of the storm,

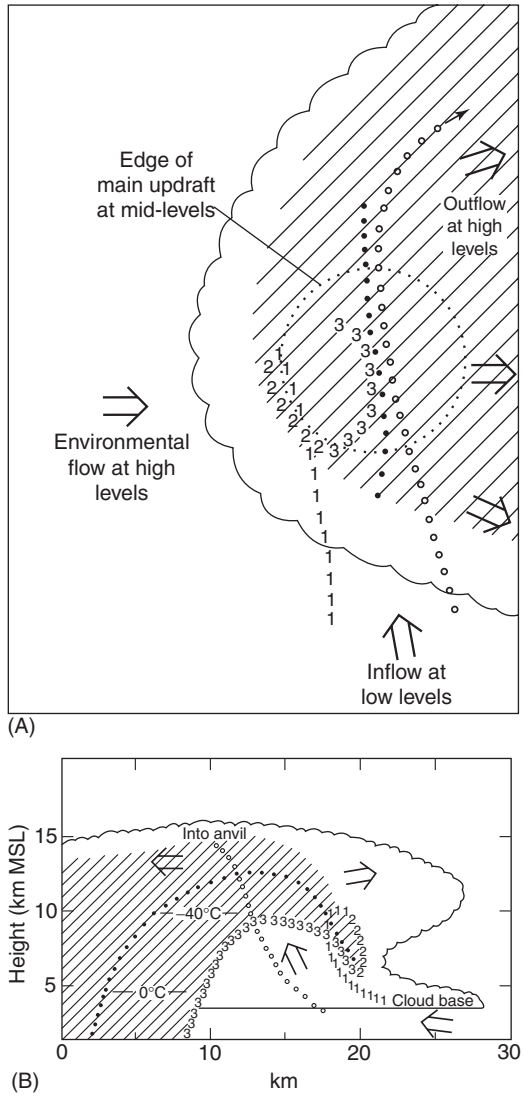

Figure 5 Schematic of airflow and some hydrometeor trajectories deduced in a supercell storm. (A) Plan view, looking down at the storm, showing low-level inflow entering from the south (the bottom of the drawing), the main updraft within the storm shown by the dotted circle, upper-level environmental air flow from the west and storm outflow to the east. The area within the cloud containing precipitation-sized hydrometeors is hatched. (B) Cross-section south to north parallel to the inflow, with the upper level flow away from the reader. The trajectory indicated by open circles in both views represents direct growth in the main updraft: condensation, freezing, and some riming leading to small ice particles that travel more or less with the air and out into the upper-level anvil. The trajectory indicated by the solid circles indicates particles that start in weaker upward flow on the south side of the updraft, grow to larger sizes and higher fall velocities and fall as small hail or rain to the north of the location of the main updraft. The trajectories indicated by the numbers 1, 2, 3 represent large hail formation: (1) slow growth rising at the west side of the updraft, (2) descent back into the inflow while traveling in the environmental flow around the south side of the updraft, and (3) final growth as hail within the strong updraft and fallout at its northern edge. (Adapted with permission from Browning KA (1977) In: Foote GB and Knight CA (eds.) *Hail: A Review of Hail Science and Hail Suppression*. Boston: American Meteorological Society.)

and exits to the east because the flow in the upper levels is from the west. That carries the southern part of the outflow in the upper levels over the top of the inflowing air at the surface. The updraft in the central part of the storm can be 30–50 m s^{-1}, so small ice particles that grow within the updraft do not have time to grow very large and attain much fall speed before being ejected into the outflow at the top of the storm, producing the extensive anvils characteristic of supercells. However, in the outflow there is little vertical air motion, and the idea is that some of these particles fall back into the inflow at low levels. They may become hail embryos, getting a chance to grow further while rising a second time within the updraft. Falling faster with respect to the air, they ascend more slowly and have time to grow into fully fledged hailstones.

This is a particularly simple and organized recycling of precipitation particles to form hail embryos in a storm: simplified because most supercells are not especially at steady state but evolve in various ways.

Multicell storms, on the other hand, are composed of individual convective cells that grow and decay in proximity to one another, sometimes in a systematic way. The individual cells may last 45–60 min, as their updrafts increase up to a maximum and then die out. This can be enough time to produce significant hail and, if the initial updraft is not too strong, embryos may have time to grow and attain a fall speed of several meters per second without being elevated too high in the cloud. Now as the updraft strengthens the embryos may already be there, in the right locations and ready to form hailstones.

Multicell hailstorms typically produce spotty, discontinuous hailfalls, which may be organized within a larger hailswath if the multicell is an organized one or may be distributed irregularly.

Hail Suppression

Hailfall is a widespread phenomenon in temperate zones around the world. In some areas it represents a major loss to agricultural production, and in recent years the insurance losses in large, urban areas have increased greatly, mostly from hail damage to roofs and automobiles.

Suppression of hail by seeding clouds with artificial ice nuclei has been practiced in many parts of the world for several decades. It is still widespread, although it is controversial and there has been no definitive demonstration of positive effects. The main idea is that furnishing ice nuclei might increase the number of potential hail embryos in the clouds, thereby depleting the supercooled water and reducing the hailstone size while increasing the number. If the size is reduced enough, most or all of the hail may then melt before reaching the ground. Ideally, the hail would be suppressed and the rainfall increased. There has been a great deal of discussion about optimizing the seeding materials and the timing and location of seeding, but the knowledge of the natural evolution of the ice contents within hailstorms is still rudimentary, so it may be many years before a consensus is reached on the prospects of hail suppression.

Hail Climatology

Hailfall at the ground is a small-scale phenomenon generally affecting areas of one to a few tens of square kilometers, though much larger hailfalls have been documented. It is highly variable and poorly resolved by routine weather observation networks. Many small hailfalls are missed altogether. Thus data on hail climatology are statistical in nature and much of the information derives from insurance records.

Probably the most general climatological correlation for hailstorms is that they often occur at the lee sides of mountain ranges. In South America, the United States, and Canada there is a strong tendency for maximum hail frequencies to occur in the lee of the Andes and the Rocky Mountains – locations also of exceptionally high thunderstorm frequency. In terms of hail damage to crops, however, areas of moderate hail frequency may be more important if they contain crops that are exceptionally sensitive to hail damage and have high value per unit area, such as wine grapes, tobacco, tea, and some fruits and vegetables.

Hail becomes rarer toward the poles, simply because the frequency of convective storms decreases with increasing latitude. Hail is also relatively rare in the tropics and in maritime regions, but the reasons for this are not as clear-cut. Part of the reason may be a thunderstorm frequency effect, but not all. Rain forms more readily in clouds in the tropics than in temperate climates because higher humidity and warmer cloud bases provide on the average much more liquid water in the clouds below the freezing level. Thus rain forms by the coalescence of water drops without involving the ice phase, and the fallout of this rain decreases the amount of supercooled water higher in the clouds where the temperature is appropriate for hailstone growth. Likewise, rain forms more easily in maritime cumulus because maritime air contains fewer nuclei for condensation, which reduces the droplet concentrations. A reduced droplet concentration increases droplet sizes, leading again to a faster formation and fallout of rain and a decreasing likelihood of hail.

Hail Detection by Radar

Radar methods for remote sensing of hail within clouds have recently come into use for research but not yet for the operational radar networks. These rely on varying the polarization of the transmitted pulses and sensing different polarizations of the backscattered radiation. Horizontal and vertical, linear polarizations are often used, and sometimes circular polarization. Intense radar echoes may be caused either by heavy rain or by hail, and until recently their interpretation has been ambiguous. However, big raindrops are consistently flattened, with larger horizontal than vertical dimensions, and this produces a substantially stronger echo with a horizontally than with a vertically polarized radar. Hail gives a more nearly equal echo strength at the two polarizations because small hail is not flattened and larger hail tumbles so that its elongations are about equally distributed it space. Thus the difference, termed the differential reflectivity, is used as a hail signal when the radar echo itself is intense.

Widespread application of this and other advanced radar techniques will greatly increase the knowledge of hail production as related to storm structure and behavior, especially if techniques are developed to provide information about hail size and amount within storms. Radar sensing of hail, if developed to be sufficiently quantitative and used routinely, would also contribute greatly to hail climatology and hail research.

See also

Cloud Microphysics. **Radar:** Precipitation Radar. **Severe Storms**. **Snow (Surface)**. **Weather Modification:** Inadvertant. **Weather Prediction:** Severe Weather Forecasting.

Further Reading

Doswell CC III (ed.) (2001) *Severe Local Storms*. Meteorological Monographs vol. 28, No. 50. Boston: American Meteorological Society.

Foote GB and Knight CA (eds) (1977) *Hail: A Review of Hail Science and Hail Suppression*. Meteorological Monographs vol. 16, no. 38. Boston: American Meteorological Society.

Rogers RR and Yau MK (1989) *A Short Course in Cloud Physics*, 3rd edn. Oxford: Pergamon Press.

HALOES

See **OPTICS, ATMOSPHERIC: Optical Phenomena**

HAMILTONIAN DYNAMICS

T G Shepherd, University of Toronto, Toronto, Ontario, Canada

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

Hamiltonian dynamics describes the evolution of conservative physical systems. Originally developed as a generalization of Newtonian mechanics, describing gravitationally driven motion from the simple pendulum to celestial mechanics, it also applies to such diverse areas of physics as quantum mechanics, quantum field theory, statistical mechanics, electromagnetism, and optics — in short, to any physical system for which dissipation is negligible. Dynamical meteorology consists of the fundamental laws of physics, including Newton's second law. For many purposes, diabatic and viscous processes can be neglected and the equations are then conservative. (For example, in idealized modeling studies, dissipation is often only present for numerical reasons and is kept as small as possible.) In such cases dynamical meteorology obeys Hamiltonian dynamics. Even when nonconservative processes are not negligible, it often turns out that separate analysis of the conservative dynamics, which fully describes the nonlinear interactions, is essential for an understanding of the complete system, and the Hamiltonian description can play a useful role in this respect. Energy budgets and momentum transfer by waves are but two examples.

Hamiltonian dynamics is often associated with conservation of energy, but it is in fact much more than that. Hamiltonian dynamical systems possess a mathematical structure that ensures some remarkable properties. Perhaps the most important is the connection between symmetries and conservation laws known as Noether's theorem. Well-known examples are the fact that conservation of energy is linked to symmetry in time, and conservation of momentum to symmetry in space. Less well-known is the fact that material conservation of potential vorticity, so crucial to the theory of dynamical meteorology, is also connected to a symmetry by Noether's theorem, but to a symmetry that is invisible in the Eulerian formulation of the governing equations. It turns out that one can exploit the underlying Hamiltonian structure of a system through the relevant conservation laws even if the explicit form of that structure is not known, which is useful for applications. As is shown in detail below, symmetry-based conservation laws provide a general theory of available potential energy, and show why it is that Rossby waves carry negative zonal momentum, thereby explaining both the maintenance of the westerlies and the stratospheric Brewer–Dobson circulation. Such laws also provide a powerful way of deriving stability criteria.

Dynamical meteorologists use a variety of theoretical models, ranging from the fully compressible equations through the hydrostatic primitive, Boussinesq, and quasi-geostrophic equations to the barotropic equations. With such a zoo of models, it is crucial to know the extent to which theories developed for one model carry over to another. Hamiltonian dynamics provides this unifying framework. All the models just mentioned are in fact Hamiltonian, and models can be grouped into families according to their Hamiltonian structure. In this way it becomes immediately apparent, for example, that the Charney–Stern stability theorem for baroclinic quasi-geostrophic flow is the counterpart to Rayleigh's inflection-point theorem for barotropic flow, and that an analogous stability theorem will exist for any balanced model having a similar Hamiltonian structure, no matter what the definition is of the potential vorticity. Thus, it is precisely through its abstract character that Hamiltonian dynamics has many powerful applications in theoretical dynamical meteorology.

Canonical and Noncanonical Dynamics

In classical mechanics, canonical Hamiltonian dynamical systems are those described by Hamilton's equation (eqns [1]).

$$\frac{\mathrm{d}q_i}{\mathrm{d}t} = \frac{\partial \mathcal{H}}{\partial p_i}, \qquad \frac{\mathrm{d}p_i}{\mathrm{d}t} = -\frac{\partial \mathcal{H}}{\partial q_i} \qquad (i = 1, \ldots, N) \quad [1]$$

$\mathcal{H}(\mathbf{q}, \mathbf{p})$ is the Hamiltonian function, $\mathbf{q} \equiv (q_1, \ldots, q_N)$ are the generalized coordinates, and $\mathbf{p} \equiv (p_1, \ldots, p_N)$ the generalized momenta. For so-called natural systems with $\mathcal{H} = (|\mathbf{p}|^2/2m) + U(\mathbf{q})$, where m is the mass and U the potential energy, eqns [1] immediately lead to eqn [2], which is Newton's second law for a conservative system.

$$m\frac{\mathrm{d}^2 q_i}{\mathrm{d}t^2} = -\frac{\partial U}{\partial q_i} \qquad (i = 1, \ldots, N) \quad [2]$$

Conservation of energy follows directly from eqns [1], for any $\mathcal{H}$, by the chain rule (repeated indices are summed):

$$\begin{aligned}\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} &= \frac{\partial \mathcal{H}}{\partial q_i}\frac{\mathrm{d}q_i}{\mathrm{d}t} + \frac{\partial \mathcal{H}}{\partial p_i}\frac{\mathrm{d}p_i}{\mathrm{d}t} \\ &= \frac{\partial \mathcal{H}}{\partial q_i}\frac{\partial \mathcal{H}}{\partial p_i} - \frac{\partial \mathcal{H}}{\partial p_i}\frac{\partial \mathcal{H}}{\partial q_i} = 0 \end{aligned} \quad [3]$$

Symplectic Formulation

The theory of canonical transformations suggests that there is nothing special about the qs and ps, and Hamilton's equations [1] can be written in the so-called symplectic form, eqn [4].

$$\frac{\mathrm{d}u_i}{\mathrm{d}t} = J_{ij}\frac{\partial \mathcal{H}}{\partial u_j} \qquad (i = 1, \ldots, 2N) \quad [4]$$

In eqn [4], $\mathbf{u} = (q_1, \ldots, q_N, p_1, \ldots, p_N)$ and J is given by eqn [5], where I is the $N \times N$ identity matrix.

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \quad [5]$$

J has certain mathematical properties, including skew-symmetry. More generally, one can take those properties to be the definition of Hamiltonian structure, with J not necessarily of the form of eqn [5]. The skew-symmetry of J nevertheless guarantees energy conservation (eqn [6]).

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \frac{\partial \mathcal{H}}{\partial u_i}\frac{\mathrm{d}u_i}{\mathrm{d}t} = \frac{\partial \mathcal{H}}{\partial u_i}J_{ij}\frac{\partial \mathcal{H}}{\partial u_j} = 0 \quad [6]$$

There is an important distinction between systems with a nonsingular (or invertible) J, which can always be transformed into the canonical form of eqn [5], and those with a singular (or noninvertible) J. The latter, known as noncanonical systems, possess a special class

of invariant functions known as Casimir invariants. These are the solutions of eqn [7] (for canonical systems the solutions are just constants).

$$J_{ij}\frac{\partial \mathcal{C}}{\partial u_j} = 0 \qquad (i = 1, \ldots, 2N) \qquad [7]$$

That they are necessarily conserved in time then follows from the skew-symmetry of J (eqn [8]).

$$\frac{\mathrm{d}\mathcal{C}}{\mathrm{d}t} = \frac{\partial \mathcal{C}}{\partial u_i}\frac{\mathrm{d}u_i}{\mathrm{d}t} = \frac{\partial \mathcal{C}}{\partial u_i}J_{ij}\frac{\partial \mathcal{H}}{\partial u_j} = -\frac{\partial \mathcal{H}}{\partial u_i}J_{ij}\frac{\partial \mathcal{C}}{\partial u_j} = 0 \qquad [8]$$

The best-known example of a noncanonical Hamiltonian system is Euler's equations for rigid-body dynamics. Having an odd number of evolution equations (three in this case), the system is necessarily noncanonical because any skew-symmetric matrix of odd dimension must be singular. There is one Casimir invariant for Euler's equations, the total angular momentum.

Noether's Theorem

For a canonical system, if a particular generalized coordinate q_j does not appear in the Hamiltonian, then the Hamiltonian is invariant under changes in that coordinate; in other words, there is a coordinate symmetry. Translational and rotational symmetries are common examples. Hamilton's equations [1] then immediately imply that the corresponding generalized momentum is conserved: $\mathrm{d}p_j/\mathrm{d}t = 0$.

This connection between symmetries and conservation laws has a more general and far more powerful form. Given a function $\mathcal{F}(\mathbf{u})$, define $\delta_{\mathcal{F}}u_i = \varepsilon J_{ij}(\partial\mathcal{F}/\partial u_j)$, where ε is an infinitesimal parameter; $\delta_{\mathcal{F}}\mathbf{u}$ is called the infinitesimal variation in $\mathbf{u}$ generated by $\mathcal{F}$. (In the canonical case, $\delta_{\mathcal{F}}\mathbf{u}$ is an infinitesimal canonical transformation.) It then follows that the infinitesimal variation in $\mathcal{H}$ generated by $\mathcal{F}$ is given by eqn [9].

$$\delta_{\mathcal{F}}\mathcal{H} = \frac{\partial \mathcal{H}}{\partial u_i}\delta_{\mathcal{F}}u_i = \varepsilon\frac{\partial \mathcal{H}}{\partial u_i}J_{ij}\frac{\partial \mathcal{F}}{\partial u_j} \qquad [9]$$

On the other hand, the time evolution of $\mathcal{F}$ is given by eqn [10].

$$\frac{\mathrm{d}\mathcal{F}}{\mathrm{d}t} = \frac{\partial \mathcal{F}}{\partial u_i}\frac{\mathrm{d}u_i}{\mathrm{d}t} = \frac{\partial \mathcal{F}}{\partial u_i}J_{ij}\frac{\partial \mathcal{H}}{\partial u_j} \qquad [10]$$

Using the skew-symmetry of J, eqns [9] and [10] then imply that $\delta_{\mathcal{F}}\mathcal{H} = 0$ if and only if $\mathrm{d}\mathcal{F}/\mathrm{d}t = 0$. This connects symmetries and conservation laws: the Hamiltonian is invariant under the variation generated by $\mathcal{F}$ (i.e., that variation represents a symmetry of the Hamiltonian) if and only if $\mathcal{F}$ is a conserved quantity. This result, known as Noether's theorem, is one of the central results of Hamiltonian dynamics and underpins many of its applications to dynamical meteorology.

Casimir invariants are special because $\delta_{\mathcal{C}}\mathbf{u} = 0$. This suggests that they correspond to invisible symmetries. For example, in rigid-body dynamics the total angular momentum is a conserved quantity in any description of the motion. In the original canonical description it corresponds to the rotational symmetry of the dynamics, but in Euler's equations, where angles have been eliminated, it enters as a Casimir because the underlying physical symmetry is no longer explicit.

Barotropic Dynamics

In what sense are the models of dynamical meteorology Hamiltonian? Consider what is probably the simplest such model, the barotropic vorticity equation (eqn [11]), which describes two-dimensional, nondivergent flow.

$$\frac{\partial \omega}{\partial t} = -\mathbf{v}\cdot\nabla\omega = -\partial(\psi, \omega) \qquad [11]$$

Here $\omega(x, y, t) = \hat{\mathbf{z}}\cdot(\nabla\times\mathbf{v}) = \nabla^2\psi$ is the vorticity, $\hat{\mathbf{z}}$ is the unit vector in the vertical direction, $\mathbf{v}(x, y, t) = \hat{\mathbf{z}}\times\nabla\psi$ is the horizontal velocity, $\psi(x, y, t)$ is the streamfunction, and $\partial(f, g) \equiv f_x g_y - f_y g_x$ is the two-dimensional Jacobian. The candidate Hamiltonian is the conserved energy of this system, which is just the kinetic energy. The obvious dynamical variable is the vorticity. In order to cast eqn [11] in the form of eqn [4], we need to regard every point (x, y) in space as indexing a degree of freedom analogous to the index i; the sum over i then becomes an integral over space, functions become functionals, and partial derivatives become functional or variational derivatives. Thus we write eqn [12].

$$\begin{aligned}\delta\mathcal{H} &= \delta\iint \frac{1}{2}|\nabla\psi|^2\,\mathrm{d}x\,\mathrm{d}y \\ &= \iint \nabla\psi\cdot\delta\nabla\psi\,\mathrm{d}x\,\mathrm{d}y \\ &= \iint \{\nabla\cdot(\psi\delta\nabla\psi) - \psi\delta\omega\}\,\mathrm{d}x\,\mathrm{d}y \qquad [12]\end{aligned}$$

Assuming for now that the boundary terms vanish, we identify the variational derivative as $\delta\mathcal{H}/\delta\omega = -\psi$. The need to integrate by parts reflects the fact that the effect of a vorticity perturbation on the kinetic energy density is nonlocal; thus, partial derivatives at fixed points in space make no sense and variational derivatives are essential. Equation [11] can now be cast in Hamiltonian form as eqn [13].

$$\frac{\partial \omega}{\partial t} = J\frac{\delta\mathcal{H}}{\delta\omega} \quad \text{where} \quad J = -\partial(\omega, \cdot) \qquad [13]$$

Note that J is now a differential operator rather than a matrix. It is evidently skew-symmetric: $\iint fJg\,\mathrm{d}x\,\mathrm{d}y = -\iint gJf\,\mathrm{d}x\,\mathrm{d}y$ (under suitable boundary conditions) for arbitrary functions f, g.

Conservation Laws

The form of J in eqn [13] is clearly singular: any function of ω inserted in the argument gives zero. These then represent Casimir invariants of the system: functionals of the form [14], where $C(\cdot)$ is an arbitrary differentiable function, evidently satisfy $J(\delta\mathcal{C}/\delta\omega) = 0$.

$$\mathcal{C} = \iint C(\omega)\,\mathrm{d}x\,\mathrm{d}y \quad \text{with} \quad \frac{\delta\mathcal{C}}{\delta\omega} = C'(\omega) \qquad [14]$$

The fact that such functionals are conserved in time corresponds to the material conservation of vorticity expressed by eqn [11].

To identify the momentum invariants, we need to apply Noether's theorem to the various spatial symmetries. Suppose that the domain is unbounded, with decay conditions at infinity, so that there is symmetry in all directions. The variation in ω corresponding to a translation by δx in the coordinate x is given by $\delta\omega = -(\partial\omega/\partial x)\delta x$. Setting $\varepsilon = \delta x$, we then need to solve for the momentum invariant $\mathcal{M}$ according to eqn [15].

$$-\varepsilon\frac{\partial\omega}{\partial x} = \delta_{\mathcal{M}}\omega = \varepsilon J\frac{\delta\mathcal{M}}{\delta\omega} = -\varepsilon\partial\left(\omega, \frac{\delta\mathcal{M}}{\delta\omega}\right) \qquad [15]$$

To within the addition of a Casimir, the solution of eqn [15] is given by $\delta\mathcal{M}/\delta\omega = y$. Hence we may choose $\mathcal{M}$ as in eqn [16], where $\mathbf{v} = (u, v)$.

$$\mathcal{M} = \iint y\omega\,\mathrm{d}x\,\mathrm{d}y = \iint y\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\mathrm{d}x\,\mathrm{d}y$$
$$= \iint u\,\mathrm{d}x\,\mathrm{d}y \qquad [16]$$

The first, elementary form of $\mathcal{M}$ given by eqn [16] is known as Kelvin's impulse. It represents the y 'center-of-mass' of the vorticity distribution, and is in many ways the preferable form because it is local in ω. The final form, however, shows that the invariant $\mathcal{M}$ corresponding to symmetry in x is ultimately just the x-momentum, as expected.

The same argument applied to translation in the coordinate y yields eqn [17].

$$\mathcal{M} = -\iint x\omega\,\mathrm{d}x\,\mathrm{d}y = \iint v\,\mathrm{d}x\,\mathrm{d}y \qquad [17]$$

Finally, rotational symmetry leads to eqn [18], where $\mathbf{r} \equiv (x, y)$ and $r = |\mathbf{r}|$, which is the angular momentum about the origin.

$$\mathcal{M} = -\iint \frac{1}{2}r^2\omega\,\mathrm{d}x\,\mathrm{d}y = \iint \hat{\mathbf{z}}\cdot(\mathbf{r}\times\mathbf{v})\,\mathrm{d}x\,\mathrm{d}y \qquad [18]$$

The discussion has so far neglected any contribution from boundary terms. They are easily included. In the presence of rigid lateral boundaries, for a complete mathematical specification of the problem, eqn [11] must be supplemented with the conditions [19] on each connected portion of the boundary.

$$\mathbf{v}\cdot\hat{\mathbf{n}} = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\oint \mathbf{v}\cdot\mathrm{d}\mathbf{s} = 0 \qquad [19]$$

Here $\hat{\mathbf{n}}$ is the outward-pointing normal, and $\mathbf{s}$ is the vector arc length along the boundary. The second of eqns [19] represents conservation of circulation, which follows from the underlying momentum equations but must be included as a separate condition in the vorticity formulation of the dynamics. Although the circulation integrals along each connected portion of the boundary are constants in time, they are independent dynamical variables and are needed to determine $\mathbf{v}$ from ω. The Hamiltonian formulation of eqn [13] may easily be extended to include the circulation integrals in addition to ω as dynamical variables. The Casimir invariants then include functions of these circulation integrals. With regard to the momentum invariants, of course, the rigid boundaries must respect the same symmetries; a zonal channel flow with walls at constant y breaks the translational symmetry in y and the rotational symmetry, leaving only the zonal impulse of eqn [16] as an invariant. The final equality of eqn [16] is then no longer strictly true, but the impulse and momentum differ only by terms involving the circulations along the channel walls, which are Casimirs. Since symmetry-based invariants are only defined to within a Casimir in any case, the impulse and momentum are essentially equivalent.

A simplified model of barotropic dynamics is the point-vortex model, where the vorticity is concentrated in Dirac delta functions. The point-vortex model has been used to study two-dimensional turbulence and certain kinds of atmospheric flow structures. It also turns out to be Hamiltonian, and is in fact a canonical system: the Casimirs are built into the model as parameters through the choice of the point-vortex strengths.

Other Balanced Models

The barotropic vorticity equation has a mathematical structure that is analogous to that of many models of

balanced, or potential-vorticity-driven, flow (*see* **Dynamic Meteorology:** Balanced Flows) and the results derived above extend in an obvious way to such systems. Inclusion of the beta effect means simply a change from ω to the potential vorticity $q = \omega + \beta y$. Since $\delta y = 0$ (recalling that the coordinate y is like an index), $\delta q = \delta\omega$ and eqns [11]–[14] go through unchanged with q in place of ω. However the beta effect breaks translational symmetry in y and rotational symmetry, leaving only the translational symmetry in x represented by the zonal impulse invariant of eqn [16]. Strictly speaking the latter should be written with q in place of ω, but the integrals differ by a constant and so represent the same invariant. Inclusion of topography is no more difficult; one simply includes an additional topographic term $h(x, y)$ in the definition of q. This will generally break all spatial symmetries, leaving only the energy $\mathcal{H}$ and Casimirs $\mathcal{C}$ as invariants. This illustrates a general and important point, namely, that symmetry-based invariants are fragile: a slight change in the conditions of the problem destroys their conservation properties. In contrast, the energy and the Casimirs are robust invariants (robust within the conservative context, of course) that survive such perturbations.

Stratification is most easily introduced in the context of the quasi-geostrophic (QG) model (*see* **Quasi-geostrophic Theory**). Layered QG models are completely trivial extensions of the barotropic system: their evolution is determined by the potential vorticity $q_i(x, y, t)$ in each layer i, governed by eqn [11] with q_i in place of ω, together with conservation of circulation along any rigid lateral boundaries that may be present. These are then the dynamical variables. The energy now includes available potential as well as kinetic energy, but, apart from some geometric factors representing the layer depths, one still recovers $\delta\mathcal{H}/\delta q_i = -\psi_i$ in each layer as well as eqn [13] with q_i in place of ω. The various invariants follow in the obvious way with the spatial integrals summed over the different layers. The same considerations, incidentally, apply to layered non-QG 'intermediate' models that still have the form of eqn [11] – namely, nondivergent horizontal advection of the potential vorticity q_i within each layer, with the flow in each layer driven by the potential vorticity in all layers (as described by the particular definition of q_i).

With continuous stratification and with upper and lower boundaries (at $z = 1$ and $z = 0$, say), there is an additional effect. It is well known that the temperature distribution along the upper and lower boundaries is equivalent to potential vorticity (*see* **Baroclinic Instability**), and independent evolution equations for these temperature distributions are required to fully specify the continuously stratified QG system, in addition to the equation for the interior potential vorticity (the latter being eqn [11], with q in place of ω, applied at every value of z; thus, the advection of q remains purely horizontal). The Eady model is an extreme case where the interior potential vorticity is uniform and the flow is driven entirely by the temperature distributions on the upper and lower boundaries; the dynamical structures driven from each boundary are known as Eady edge waves. Since these temperature distributions also evolve according to eqn [11], with the QG temperature ψ_z in place of ω, it is not surprising that the same kind of Hamiltonian structure also applies to this model. The energy is given by eqn [20].

$$\mathcal{H} = \iiint \frac{\rho_s}{2}\left\{|\nabla\psi|^2 + \frac{1}{S}\psi_z^2\right\}\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \qquad [20]$$

In eqn [20], the reference-state density $\rho_s(z)$ and stratification function $S(z) = N^2/f^2$ are both prescribed, with $N(z)$ the buoyancy frequency and f the Coriolis parameter, and where ∇ is still just the horizontal gradient operator. With the potential vorticity given by eqn [21], where f and β are constants, eqn [22] follows.

$$q(x, y, z, t) = \psi_{xx} + \psi_{yy} + \frac{1}{\rho_s}\left(\frac{\rho_s}{S}\psi_z\right)_z + f + \beta y \qquad [21]$$

$$\delta\mathcal{H} = \left[\iint \frac{\rho_s}{S}\psi\delta\psi_z\,\mathrm{d}x\,\mathrm{d}y\right]_{z=0}^{z=1} + \iiint \{\nabla\cdot(\rho_s\psi\delta\nabla\psi) - \rho_s\psi\delta q\}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \qquad [22]$$

This is like eqn [12], but with an additional term involving the temperature variations $\delta\psi_z$ at the upper and lower boundaries. Including these as independent dynamical variables, in addition to q (and possibly also circulation terms), the governing equations can be cast in the symplectic form of eqn [13]. The Casimirs now involve integrals of arbitrary functions of the temperature on the upper and lower boundaries, in addition to integrals of arbitrary functions of potential vorticity in the interior (eqn [23]).

$$\mathcal{C} = \iiint C(q; z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z + \iint C_0(\psi_z)\,\mathrm{d}x\,\mathrm{d}y\bigg|_{z=0} + \iint C_1(\psi_z)\,\mathrm{d}x\,\mathrm{d}y\bigg|_{z=1} \qquad [23]$$

The momentum invariants similarly extend in obvious ways: for example, the zonal impulse

invariant is given by eqn [24].

$$\mathcal{M} = \iiint \rho_s y q \, dx \, dy \, dz + \iint \frac{\rho_s}{S} y \psi_z \, dx \, dy \bigg|_{z=0} - \iint \frac{\rho_s}{S} y \psi_z \, dx \, dy \bigg|_{z=1} \quad [24]$$

The semi-geostrophic (SG) model (*see* **Frontogenesis**) is widely used in mesoscale dynamics because of its ability to represent realistic frontal structures. It turns out that the SG model can also be cast in the form of eqn [11], and hence in the symplectic form of eqn [13], provided the equations are written in isentropic–geostrophic coordinates. However, in these coordinates rigid boundaries appear to move in time. The SG equations, in contrast to the QG equations, make no geometrical distinction between horizontal and vertical boundaries – this is why they are also useful for the study of coastal dynamics in physical oceanography – and the same kind of independent dynamical degrees of freedom encountered in the QG system on upper and lower boundaries also appear on lateral boundaries. In the special case of channel walls, these degrees of freedom correspond to coastal Kelvin waves and are analogous in some respects to the Eady edge waves represented by both the QG and SG systems. They must be taken into account in the variational calculations, and enter into many of the resulting expressions.

Unbalanced Models

Balanced models are controlled by the advection of potential vorticity (perhaps augmented by the advection of isentropic surfaces on rigid boundaries), so for such models it is natural to seek a Hamiltonian description analogous to eqn [13]. However, models that include a representation of gravity waves or other high-frequency oscillations, called unbalanced models, do not fit within this framework. They necessarily have additional degrees of freedom. For such models, a description in terms of the velocity field is a more natural way to reflect the Hamiltonian structure. For example, the rotating shallow-water equations [25] with $\mathbf{v}(x, y, t) = (u, v)$ the horizontal velocity, $h(x, y, t)$ the fluid depth, g the gravitational acceleration, and with constant f, conserve the energy (eqn [26]).

$$\frac{\partial \mathbf{v}}{\partial t} + (f\hat{\mathbf{z}} + \nabla \times \mathbf{v}) \times \mathbf{v} + \nabla\left(\frac{1}{2}|\mathbf{v}|^2\right) = -g\nabla h,$$

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{v}) = 0 \quad [25]$$

$$\mathcal{H} = \iint \frac{1}{2}\{h|\mathbf{v}|^2 + gh^2\} \, dx \, dy \quad [26]$$

The dynamical variables are $\mathbf{v}$ and h, for which eqns [27] hold.

$$\frac{\delta \mathcal{H}}{\delta \mathbf{v}} = h\mathbf{v}, \qquad \frac{\delta \mathcal{H}}{\delta h} = \frac{1}{2}|\mathbf{v}|^2 + gh \quad [27]$$

Note that no integration by parts is necessary in this case; this is characteristic of velocity-based representations of the dynamics. It can easily be verified that eqns [25] may be cast in the symplectic form $(\partial \mathbf{u}/\partial t) = J(\delta\mathcal{H}/\delta\mathbf{u})$ with $\mathbf{u} = (u, v, h)$ with J given by eqn [28], where $q = (f + \hat{\mathbf{z}} \cdot \nabla \times \mathbf{v})/h$ is the potential vorticity of the shallow-water system.

$$J = \begin{pmatrix} 0 & q & -\partial_x \\ -q & 0 & -\partial_y \\ -\partial_x & -\partial_y & 0 \end{pmatrix} \quad [28]$$

The matrix (28) is evidently skew-symmetric; the signs on the derivative terms are indeed correct, since first-order differential operators are themselves skew-symmetric, as with the J in eqn [13]. The zonal (absolute) momentum invariant is given as expected by eqn [29], for which it is easy to verify that $J(\delta\mathcal{M}/\delta\mathbf{u}) = -\partial\mathbf{u}/\partial x$ in line with Noether's theorem, and the other momentum invariants follow similarly.

$$\mathcal{M} = \iint h(u - fy) \, dx \, dy \quad [29]$$

The Casimirs are given by eqn [30] for arbitrary functions $C(\cdot)$.

$$\mathcal{C} = \iint hC(q) \, dx \, dy \quad [30]$$

Thus, potential vorticity still plays a crucial role in the Hamiltonian description of the dynamics. Special cases of Casimirs are total mass ($C = 1$) and total circulation ($C = q$).

Stratification is easily incorporated. The hydrostatic primitive equations can be cast in Hamiltonian form isomorphic to that of eqn [28] when expressed in isentropic coordinates. Even the fully compressible stratified Euler equations, which form the most general system imaginable for (dry) dynamical meteorology, can be cast in an analogous form, although there are now additional dynamical variables associated with compressibility. The Casimirs are in this case given by eqn [31], where $\rho(x, y, z, t)$ is the density, $\theta(x, y, z, t)$ is the potential temperature, and $q = [(f\hat{\mathbf{z}} + \nabla \times \mathbf{v}) \cdot \nabla\theta]/\rho$ is the Ertel potential vorticity, with $\mathbf{v}$ and ∇ now acting in all three space dimensions.

$$\mathcal{C} = \iiint \rho C(q, \theta) \, dx \, dy \, dz \quad [31]$$

The invariance of the Casimirs is of course evident directly from the dynamical equations (eqn [32]) and reflects the material invariance of q and θ.

$$\frac{\partial q}{\partial t} + \mathbf{v}\cdot\nabla q = 0, \qquad \frac{\partial \theta}{\partial t} + \mathbf{v}\cdot\nabla\theta = 0,$$

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0 \qquad [32]$$

The fully compressible stratified Euler equations are, in fact, a straightforward expression of Newton's second law, without constraints such as hydrostatic balance, provided they are expressed in Lagrangian coordinates (*see* **Lagrangian Dynamics**). In Lagrangian coordinates, the dynamical variables are the positions and momenta of fluid elements, which are natural canonical variables. The thermodynamic fields can be expressed in terms of these variables: ρ can be written in terms of the Jacobian of particle positions (which describes the compression of the fluid), while θ can just be chosen as one of the Lagrangian coordinates. In this way, the fully compressible stratified Euler equations represent a canonical Hamiltonian system. But there are six dynamical variables in the Lagrangian description, compared with only five in the Eulerian description; in transforming to Eulerian coordinates, a reduction of the phase space takes place. This is where the potential vorticity comes in. In Lagrangian coordinates, the potential vorticity is still materially conserved; but what symmetry does it correspond to? The answer is a particle-relabeling symmetry: if one rearranges fluid elements while preserving the same Eulerian fields, then the dynamics is unchanged. There is just enough freedom to do this, because there is one more Lagrangian than Eulerian variable. Upon reduction to the Eulerian description, this additional degree of freedom disappears, and the particle-relabeling symmetry becomes invisible. That is why potential vorticity conservation then appears in the form of a Casimir invariant.

Disturbance Invariants

Probably the most powerful application of Hamiltonian dynamics to dynamical meteorology arises in the context of studying the properties of disturbances to basic states. In fluid dynamics, the question of how to define the energy of a wave has often been a point of confusion if not contention. For example, in the case of a basic flow, if the wave energy is defined as the energy in the frame of reference moving with the basic flow, then it is positive definite but not conserved. On the other hand, if it is defined as the difference energy relative to the basic-flow energy, then it is conserved but not positive definite. One would like both properties in order to define normal modes, spectra, etc. Another problem, at first sight unrelated, arises with momentum. The momentum of a wave would appear to be zero (the average of a sinusoid is zero), yet waves can certainly transfer momentum; this is what drives the quasi-biennial oscillation in the tropical stratosphere, for example (*see* **Middle Atmosphere:** Quasi-Biennial Oscillation). How is one to describe this wave momentum?

In canonical Hamiltonian mechanics, the disturbance energy about an equilibrium is always quadratic; from this one assesses stability and defines normal modes. There is no ambiguity. So why are things not equally clear for fluid dynamics? The answer lies in the noncanonical Hamiltonian structure of virtually every fluid dynamical system in the Eulerian representation. If $\mathbf{u} = \mathbf{U}$ is a steady solution of a Hamiltonian system, then eqn [33] holds.

$$J\left.\frac{\delta\mathcal{H}}{\delta\mathbf{u}}\right|_{\mathbf{u}=\mathbf{U}} = 0 \qquad [33]$$

For a canonical system, the invertibility of J then implies that $\delta\mathcal{H}/\delta\mathbf{u} = 0$ at $\mathbf{u} = \mathbf{U}$. This means that $\mathbf{U}$ is a conditional extremum of $\mathcal{H}$, and $\mathcal{H}[\mathbf{u}] - \mathcal{H}[\mathbf{U}]$ is quadratic in the disturbance. However, for a noncanonical system none of this follows and the disturbance energy is generally linear in the disturbance.

Pseudoenergy

Hamiltonian structure provides the solution to this quandary. Equation [33] is locally the same as the equation defining the Casimirs, which means that $\delta\mathcal{H}/\delta\mathbf{u}$ is locally parallel to $\delta\mathcal{C}/\delta\mathbf{u}$ for some $\mathcal{C}$ (a different $\mathcal{C}$ for each choice of $\mathbf{U}$). In other words, there exists a Casimir $\mathcal{C}$ such that eqn [34] holds.

$$\left.\frac{\delta\mathcal{H}}{\delta\mathbf{u}}\right|_{\mathbf{u}=\mathbf{U}} = -\left.\frac{\delta\mathcal{C}}{\delta\mathbf{u}}\right|_{\mathbf{u}=\mathbf{U}} \qquad [34]$$

Now, both $\mathcal{H}$ and $\mathcal{C}$ are invariants, and the combined invariant $\mathcal{H}+\mathcal{C}$ satisfies the extremal condition $\delta(\mathcal{H}+\mathcal{C}) = 0$ at $\mathbf{u} = \mathbf{U}$. We have thus constructed what we wanted, namely a disturbance quantity that is conserved and is locally quadratic in the disturbance (eqn [35]).

$$\mathcal{A} = (\mathcal{H}+\mathcal{C})[\mathbf{u}] - (\mathcal{H}+\mathcal{C})[\mathbf{U}] \qquad [35]$$

This quantity is known as the pseudoenergy. Provided one has a complete set of Casimirs, eqn [34] can always be solved for a Hamiltonian system and the pseudoenergy can always be constructed according to eqn [35]. This is one of the great attractions of Hamiltonian dynamics: it provides systematic recipes

in abstract terms, which can be worked out for any particular application.

A particularly illuminating application is the subject of available potential energy, highly useful in energy budget analyses. We demonstrate the method in the case of the three-dimensional stratified Boussinesq equations. The energy is given by eqn [36].

$$\mathcal{H} = \iint \left\{ \frac{1}{2}\rho_s |\mathbf{v}|^2 + \rho g z \right\} \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \qquad [36]$$

Here ρ_s is the constant reference-state density, and the dynamical variables are $\mathbf{v}$ and ρ, for which eqns [37] hold.

$$\frac{\delta\mathcal{H}}{\delta\mathbf{v}} = \rho_s \mathbf{v}, \qquad \frac{\delta\mathcal{H}}{\delta\rho} = gz \qquad [37]$$

The term $\rho g z$ in eqn [36] is the gravitational potential energy, and is linear in the dynamical variables. Now consider disturbances to a stably stratified, resting basic state $\mathbf{v} = 0,\ \rho = \rho_0(z)$. Although the Casimirs of this system include functions of the potential vorticity, because the basic state is at rest, $\delta\mathcal{H}/\delta\mathbf{v} = 0$ at $\mathbf{v} = 0$ and this dependence is unnecessary, so we may consider Casimirs of the form of eqn [38].

$$\mathcal{C} = \iiint \mathrm{C}(\rho)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \quad \text{with} \quad \frac{\delta\mathcal{C}}{\delta\rho} = \mathrm{C}'(\rho) \qquad [38]$$

Equation [34] then leads to the condition $\mathrm{C}'(\rho_0) = -gz$. This is the defining relation for the function $\mathrm{C}(\cdot)$. Thus, one has to express gz in terms of the same argument ρ_0. This can be done by inverting the functional dependence $\rho_0(z)$ to obtain $Z(\rho_0)$, where $Z(\rho_0(z)) = z$. This is always possible provided $\rho_0(z)$ is monotonic, which is the case for a stably stratified basic state. One then has eqn [39].

$$\mathrm{C}(\rho) = -\int^{\rho} gZ(\tilde{\rho})\,\mathrm{d}\tilde{\rho} \qquad [39]$$

From this the pseudoenergy of eqn [35] takes the form [40].

$$\mathcal{A} = \iiint \left\{ \frac{\rho_s}{2}|\mathbf{v}|^2 + (\rho - \rho_0)gz - \int_{\rho_0}^{\rho} gZ(\tilde{\rho})\,\mathrm{d}\tilde{\rho} \right\} \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \qquad [40]$$

The first term in the spatial integrand is the kinetic energy and is positive definite; the last two terms can be rewritten as in eqn [41].

$$-\int_0^{\rho-\rho_0} g[Z(\rho_0 + \tilde{\rho}) - Z(\rho_0)]\,\mathrm{d}\tilde{\rho} \qquad [41]$$

This is self-evidently positive definite for $\mathrm{d}\rho_0/\mathrm{d}z < 0$ and has the small-amplitude quadratic approximation [42].

$$-\frac{g(\rho - \rho_0)^2}{2(\mathrm{d}\rho_0/\mathrm{d}z)} \qquad [42]$$

Equation [41] is the exact, finite-amplitude expression for the available potential energy (*see* **General Circulation:** Energy Cycle) of disturbances to a stably stratified, resting basic state $\rho_0(z)$, while eqn [42] is its more familiar small-amplitude counterpart, widely used in the theory of internal gravity waves (*see* **Buoyancy and Buoyancy Waves:** Theory). Similar constructions can be performed to define the available potential energy of any stratified fluid system.

Although the small-amplitude expression of eqn [42] appears to be singular in regions where $\mathrm{d}\rho_0/\mathrm{d}z = 0$, the finite-amplitude expression of eqn [41] remains perfectly well-defined in such regions.

Pseudomomentum

The same kind of reasoning can be applied for disturbances to zonally symmetric (x-invariant) basic states, assuming that the underlying system possesses the same symmetry. For such states, with $\partial\mathbf{U}/\partial x = 0$, Noether's theorem implies that the zonal impulse or momentum invariant satisfies eqn [43].

$$J\frac{\delta\mathcal{M}}{\delta\mathbf{u}}\bigg|_{\mathbf{u}=\mathbf{U}} = 0 \qquad [43]$$

But just as with eqn [33], there is a Casimir $\mathcal{C}$ such that $\delta(\mathcal{M} + \mathcal{C}) = 0$ at $\mathbf{u} = \mathbf{U}$; with this $\mathcal{C}$, one may immediately construct the invariant [44], which is quadratic to leading order in the disturbance.

$$\mathcal{A} = (\mathcal{M} + \mathcal{C})[\mathbf{u}] - (\mathcal{M} + \mathcal{C})[\mathbf{U}] \qquad [44]$$

This quantity is known as the pseudomomentum.

We calculate the pseudomomentum for the case of barotropic flow on the beta-plane. Suppose we are given a monotonic basic state $q_0(y)$. From eqns [14] and [16], with q in place of ω, we have eqn [45].

$$\frac{\delta\mathcal{M}}{\delta q} = y, \qquad \frac{\delta\mathcal{C}}{\delta q} = \mathrm{C}'(q) \qquad [45]$$

The extremal condition $\delta(\mathcal{M} + \mathcal{C}) = 0$ at $q = q_0$ then leads to $\mathrm{C}'(q_0) = -y$. This is now isomorphic to the construction of the available potential energy, replacing gz with y and ρ with q. If we define the function $Y(\cdot)$ by $Y(q_0(y)) = y$, then evidently eqn [46] holds.

$$\mathcal{A} = \iint \left\{ -\int_0^{q-q_0} [Y(q_0 + \tilde{q}) - Y(q_0)]\,\mathrm{d}\tilde{q} \right\} \mathrm{d}x\,\mathrm{d}y \qquad [46]$$

The small-amplitude approximation to the spatial integrand is given by eqn [47].

$$-\frac{(q-q_0)^2}{2(\mathrm{d}q_0/\mathrm{d}y)} \qquad [47]$$

Equations [46] and [47] are evidently negative definite for $\mathrm{d}q_0/\mathrm{d}y > 0$, which is the case when q_0 is dominated by βy. These rather peculiar expressions have no obvious relation to zonal momentum at first sight, but they nevertheless explain why it is that Rossby waves always exert an eastward (positive) force when they leave a source region, and a westward (negative) force when they dissipate and deposit their momentum in a sink region: they carry negative pseudomomentum.

The general nature of the derivation ensures that exactly the same expressions hold for any balanced model having the basic form of eqn [13]. If the basic state q_0 is chosen to be the zonal mean $\bar{q}$, then the zonal mean of eqn [47] becomes eqn [48], where $q' = q - \bar{q}$.

$$-\frac{\overline{q'^2}}{2\bar{q}_y} \qquad [48]$$

In the case of stratified QG dynamics, the negative of eqn [48] is known as the Eliassen–Palm wave activity (*see* **Middle Atmosphere:** Zonal Mean Climatology), which has been widely used in dynamical meteorology to assess the effect of Rossby waves on the zonal mean flow. It is such an effective diagnostic precisely because it represents negative pseudomomentum. Moreover, and importantly, its use is not restricted to waves. The exact, finite-amplitude expression of eqn [46] ensures that the concept of pseudomomentum applies to fully nonlinear, even turbulent disturbances.

The robust negative definiteness of the pseudomomentum of balanced disturbances explains a great deal about the general circulation of the atmosphere. Propagation of synoptic-scale Rossby waves away from their source region in the baroclinic storm tracks implies an eastward force in the storm track regions, accounting for the maintenance of the westerlies (*see* **General Circulation:** Momentum Budget). The westward momentum deposition associated with breaking planetary-scale Rossby waves in the stratosphere drives the poleward Brewer–Dobson circulation (*see* **Middle Atmosphere:** Zonal Mean Climatology), which is responsible for the observed distribution of ozone and other chemical species in the stratosphere.

Stability Theorems

The pseudoenergy and pseudomomentum are, by their construction, conserved quantities that are quadratic to leading order in the disturbance quantities. In fact, their quadratic approximations are exactly conserved by the linearized dynamics. (The quadratic approximation to the pseudoenergy is the Hamiltonian of the linearized dynamics.) When either of these quantities is sign-definite for a given basic state, it follows that that basic state is stable to normal-mode instabilities. Indeed, in order to reconcile exponentially growing disturbances with conservation of pseudoenergy and pseudomomentum, the latter quantities must vanish for such disturbances. This fact provides a useful constraint on the structure of normal-mode instabilities, as well as a powerful unifying framework between different models.

This simple framework accounts for virtually every known stability theorem in dynamical meteorology. For resting, stratified basic states in unbalanced models, with pseudoenergy like eqn [40] for the Boussinesq model, the condition of positive definite pseudoenergy is the statement of static stability (*see* **Static Stability**). For basic flows in axisymmetric or symmetric stratified unbalanced models, the same condition is the statement of symmetric stability (*see* **Instability:** Symmetric Stability), which reduces to Rayleigh's centrifugal stability theorem in the special case of axisymmetric homogeneous flow (*see* **Instability:** Inertial Instability). These stability theorems are all quite analogous to static stability. A different situation arises for balanced models. There, the pseudoenergy can take either sign depending on the basic flow. The positive-definite and negative-definite cases correspond respectively to Arnol'd's first and second stability theorems. (They are analogous to the stability of a rigid body rotating about an axis of symmetry corresponding respectively to a maximum or minimum moment of inertia.) In the special case of a parallel basic flow, Arnol'd's first theorem states that the flow is stable if $u_0/(\mathrm{d}q_0/\mathrm{d}y) < 0$, which is the Fjørtoft–Pedlosky theorem.

With regard to pseudomomentum for balanced models, eqn [46] is sign-definite whenever $\mathrm{d}q_0/\mathrm{d}y$ is sign-definite. For barotropic flow with $q = \omega$, this corresponds to Rayleigh's inflection-point theorem; on the beta-plane with $q = \omega + \beta y$, to the Rayleigh–Kuo theorem; and for stratified QG flow with q given either by its multilevel forms q_i or by eqn [21] in the continuously stratified case, to the Charney–Stern theorem. For stratified QG dynamics in the presence of a lower boundary, the second terms of eqns [23] and [24] become relevant and there is an additional contribution to the pseudomomentum involving the temperature distribution on the lower boundary; it is isomorphic to the interior eqns [46]–[48], replacing q with ψ_z. Since the climatological temperature gradient along the Earth's surface is towards the Equator, the

pseudomomentum associated with surface disturbances is generally positive. In this case the Charney–Stern stability criterion is not satisfied for observed flows; on the other hand, normal-mode instabilities are generally required to involve both temperature disturbances on the lower boundary and potential-vorticity disturbances in the interior, in order to create a disturbance with zero total pseudomomentum. The Charney model of baroclinic instability (*see* **Baroclinic Instability**) is the best-known example of this. In the presence of an upper boundary, there is a further contribution to the pseudomomentum, with opposite sign to the lower contribution in accord with eqn [24]. Thus in the Eady model of baroclinic instability, where the potential vorticity is uniform and the interior contribution to the pseudomomentum disappears, the instability can arise from the interaction of disturbances on the upper and lower boundaries that together add up to zero total pseudomomentum.

These statements all concern normal-mode stability. But what can be said about stability goes much further than this. The existence of finite-amplitude disturbance invariants suggests the possibility of nonlinear, or Liapunov stability: namely, that small disturbances stay small for all time, where small is defined in terms of some disturbance norm. Mathematically, we say that a basic state $\mathbf{U}$ is Liapunov stable to disturbances $\mathbf{u}'$ in a given norm $||\mathbf{u}'||$ if for all $\varepsilon > 0$ there exists a $\delta(\varepsilon) > 0$ such that eqns [49] hold.

$$||\mathbf{u}'(0)|| < \delta \Rightarrow ||\mathbf{u}'(t)|| < \varepsilon \quad \text{for all } t \qquad [49]$$

Let us see how this applies to static stability for the Boussinesq model considered earlier. Suppose that the basic state has $\mathrm{d}\rho_0/\mathrm{d}z < 0$ and that furthermore the basic-state density gradients are bounded according to [50] for some constants c_1, c_2.

$$0 < c_1 \leq -g\frac{\mathrm{d}Z}{\mathrm{d}\rho_0} = -\frac{g}{\mathrm{d}\rho_0/\mathrm{d}z} \leq c_2 < \infty \qquad [50]$$

Then eqn [41] for the available potential energy is bounded from above and below according to eqn [51].

$$\frac{1}{2}c_1(\rho - \rho_0)^2 \leq [41] \leq \frac{1}{2}c_2(\rho - \rho_0)^2 \qquad [51]$$

Define the disturbance norm by eqn [52], with $c_1 \leq \lambda \leq c_2$.

$$||(\mathbf{v}, \rho - \rho_0)||^2 = \iiint \frac{1}{2}\{\rho_s|\mathbf{v}|^2 + \lambda(\rho - \rho_0)^2\}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \qquad [52]$$

Then using eqn [51] we obtain the chain [53] of inequalities, valid for any time t, involving the pseudoenergy $\mathcal{A}$ of eqn [40].

$$||(\mathbf{v}, \rho - \rho_0)(t)||^2 \leq \frac{\lambda}{c_1}\mathcal{A}(t) = \frac{\lambda}{c_1}\mathcal{A}(0) \leq \frac{c_2}{c_1}||(\mathbf{v}, \rho - \rho_0)(0)||^2 \qquad [53]$$

With the choice $\delta = \sqrt{c_1/c_2}\,\varepsilon$, eqn [53] establishes Liapunov stability in the norm defined by eqn [52]. Conservation of pseudoenergy is clearly central to the proof.

The finite-amplitude stability of stably stratified flow is not too surprising; it corresponds to physical intuition, and indeed motivates the very concept of available potential energy, which has a long pedigree. What is perhaps more surprising is that exactly the same kinds of constructions can be made for all of the stability theorems mentioned above, and for virtually any model within the same family. They can also be used to obtain rigorous upper bounds on the saturation of normal-mode instabilities, by considering the initial unstable flow (plus infinitesimal disturbance) to be a finite-amplitude disturbance to a stable basic state.

See also

Baroclinic Instability. **Buoyancy and Buoyancy Waves:** Theory. **Dynamic Meteorology:** Balanced Flows; Potential Vorticity. **Frontogenesis. General Circulation:** Energy Cycle; Momentum Budget. **Instability:** Inertial Instability; Symmetric Stability. **Lagrangian Dynamics**. **Middle Atmosphere:** Quasi-Biennial Oscillation; Zonal Mean Climatology. **Quasi-geostrophic Theory**. **Static Stability**.

Further Reading

Arnol'd VI (1989) *Mathematical Methods of Classical Mechanics*, 2nd edn. New York: Springer-Verlag.

Benjamin TB (1984) Impulse, flow force and variational principles. *IMA Journal of Applied Mathematics* 32: 3–68.

Holm DD, Marsden JE, Ratiu T and Weinstein A (1985) Nonlinear stability of fluid and plasma equilibria. *Physics Reports* 123: 1–116.

Landau LD and Lifshitz EM (1976) *Mechanics*, 3rd edn. New York: Pergamon Press.

Morrison PJ (1998) Hamiltonian description of the ideal fluid. *Reviews of Modern Physics* 70: 467–521.

Salmon R (1988) Hamiltonian fluid mechanics. *Annual Review of Fluid Mechanics* 20: 225–256.

Salmon R (1998) *Lectures on Geophysical Fluid Dynamics*. New York: Oxford University Press.

Shepherd TG (1990) Symmetries, conservation laws, and Hamiltonian structure in geophysical fluid dynamics. *Advances in Geophysics* 32: 287–338.

Shepherd TG (1993) A unified theory of available potential energy. *Atmosphere–Ocean* 31: 1–26.

HUMIDITY VARIABLES

J Curry, Georgia Institute of Technology, Atlanta, GA, USA

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

Atmospheric humidity is the amount of water vapor present in the atmosphere. Both the rate of evaporation and the time and place that condensation of cloud water occurs are controlled by humidity. Determination of the atmospheric humidity is important for determining surface evaporation, atmospheric radiative transfer, and certain chemical reactions in the atmosphere.

Vapor Pressure

The partial pressure of water vapor, e, is used as the fundamental measure of water vapor content in the atmosphere. Using Dalton's law of partial pressures, we can use the ideal gas law to write the following expression for water vapor pressure:

$$e = \rho_v R_v T \qquad [1]$$

where ρ_v is the water vapor density (often referred to as the absolute humidity), R_v is the specific gas constant for water, and T is the atmospheric temperature. The atmospheric relative humidity is defined as the ratio of the atmospheric vapor pressure to the equilibrium (or saturation) vapor pressure at the temperature of the air, e_s.

To understand what is meant by the saturation vapor pressure, we must first examine the equilibria between phases. Consider a closed container half full of pure water and overlain by dry air. As the water begins to evaporate from the water surface, a small increase in pressure is detected in the air above, resulting from the motion of the water vapor molecules that are added to the air through evaporation. As more and more molecules escape from the water surface in the closed container, the steadily increasing vapor pressure in the air above forces more and more of these molecules to return to the liquid. Eventually an equilibrium is reached where the number of water vapor molecules returning to the surface balances the number leaving. At that point the air is said to be saturated with water vapor, and the partial pressure of the water vapor is equal to the saturation vapor pressure. The relative humidity is equal to unity (or 100%). If the temperature of the water in the closed container were increased, more water would evaporate before a balance was reached.

A rigorous derivation of the saturation vapor pressure can be accomplished from the second law of thermodynamics. For the liquid–vapor equilibrium, we can write the Clausius–Clapeyron equation as follows,

$$\frac{de_s}{dT} = \frac{L_{lv} e_s}{R_v T^2} \qquad [2]$$

where L_{lv} is the latent heat of vaporization (energy per unit mass). At any given temperature, eqn [2] states that there is one and only one pressure at which water vapor is in equilibrium with liquid water, with the saturation vapor pressure increasing approximately exponentially with increasing temperature.

Integration of eqn [2] is made difficult owing to the variation of the latent heat of vaporization with temperature. Additionally, application of the Clausius–Clapeyron equation to determining the saturation vapor pressure in the atmosphere is not strictly valid because of the presence of other gases. Hence empirical values of the saturation vapor pressure are typically used (shown in **Table 1**).

These empirical values of saturation vapor pressure can be represented by a sixth-order polynomial:

$$e_s = a_1 + \sum_{n=2}^{7} a_n (T - T_{tr})^{n-1} \qquad [3]$$

where T_{tr}, the so-called 'triple point' of water, is 273.15 K, and the coefficients for the saturation vapor pressure over water and over ice are as given in **Table 2**. Note that, at a given temperature, the saturation vapor pressure over ice is smaller than the saturation vapor pressure over liquid water; this is because the latent heat of sublimation (associated with the phase transition between vapor and ice) is larger than the latent heat of vaporization.

Humidity Variables

Values of the saturation vapor pressure are used in the determination of some of the commonly used atmospheric humidity variables. A wide variety of different humidity variables are used in atmospheric science. The reasons for this are partly historical and partly related to the different methods by which atmospheric humidity is measured. Given information about the ambient atmospheric temperature and pressure, one humidity variable can be used to determine each of the other humidity variables.

Table 1 Saturation pressures over pure liquid water and pure ice as a function of temperature

T (°C)	e_w (hPa)	e_i (hPa)	T (°C)	e_w (hPa)	e_i (hPa)	T (°C)	e_w (hPa)	T (°C)	e_w (hPa)
−50	0.0635	0.0393	−24	0.8826	0.6983	1	6.565	26	33.606
−49	0.0712	0.0445	−23	0.9647	0.7708	2	7.054	27	35.646
−48	0.0797	0.0502	−22	1.0536	0.8501	3	7.574	28	37.793
−47	0.0892	0.0567	−21	1.1498	0.9366	4	8.128	29	40.052
−46	0.0996	0.0639	−20	1.2538	1.032	5	8.718	30	42.427
−45	0.1111	0.0720	−19	1.3661	1.135	6	9.345	31	44.924
−44	0.1230	0.0810	−18	1.4874	1.248	7	10.012	32	47.548
−43	0.1379	0.0910	−17	1.6183	1.371	8	10.720	33	50.303
−42	0.1533	0.1021	−16	1.7594	1.505	9	11.473	34	53.197
−41	0.1704	0.1145	−15	1.9114	1.651	10	12.271	35	56.233
−40	0.1891	0.1283	−14	2.0751	1.810	11	13.118	36	59.418
−39	0.2097	0.1436	−13	2.2512	1.983	12	14.016	37	62.759
−38	0.2322	0.1606	−12	2.4405	2.171	13	14.967	38	66.260
−37	0.2570	0.1794	−11	2.6438	2.375	14	15.975	39	69.930
−36	0.2841	0.2002	−10	2.8622	2.597	15	17.042	40	73.773
−35	0.3138	0.2232	−9	3.0965	2.837	16	18.171	41	77.798
−34	0.3463	0.2487	−8	3.3478	3.097	17	19.365	42	82.011
−33	0.3817	0.2768	−7	3.6171	3.379	18	20.628	43	86.419
−32	0.4204	0.3078	−6	3.9055	3.684	19	21.962	44	91.029
−31	0.4627	0.3420	−5	4.2142	4.014	20	23.371	45	95.850
−30	0.5087	0.3797	−4	4.5444	4.371	21	24.858	46	100.89
−29	0.5588	0.4212	−3	4.8974	4.756	22	26.428	47	106.15
−28	0.6133	0.4668	−2	5.2745	5.173	23	28.083	48	111.65
−27	0.6726	0.5169	−1	5.6772	5.622	24	29.829	49	117.40
−26	0.7369	0.5719	−0	6.1070	6.106	25	31.668	50	123.39
−25	0.8068	0.6322	—	—	—	—	—	—	—

The relative humidity, H, is defined as

$$H = \frac{e}{e_s} \qquad [4]$$

and H_i, the relative humidity with respect to ice saturation, is defined as

$$H_i = \frac{e}{e_{si}} \qquad [5]$$

where e_{si} is the saturation vapor pressure over ice. The relative humidity is the ratio of the actual partial pressure of water vapor in the air to the saturation vapor pressure, and is a function only of e and T. It is commonly multiplied by 100 and expressed as a percentage. Relative humidity can be changed if moisture is added or removed (which changes e), or if the air temperature changes (which changes e_s). At temperatures below 0°C, it is necessary to specify whether the relative humidity is being evaluated relative to the saturation vapor pressure over liquid water or over ice. **Table 3** shows that an atmosphere saturated with respect to liquid water is supersaturated with respect to ice, and that the degree of supersaturation increases with the supercooling.

Table 2 Coefficients of the sixth-order polynomial fits to saturation vapor pressure for the temperature range −50° to 50°C for both liquid water and ice

Coefficient	*Liquid water*	*Ice*
a_1	6.111 767 50	6.109 526 65
a_2	0.443 986 062	0.501 948 366
a_3	$0.143\,053\,301 \times 10^{-1}$	$0.186\,288\,989 \times 10^{-1}$
a_4	$0.265\,027\,242 \times 10^{-3}$	$0.403\,488\,906 \times 10^{-3}$
a_5	$0.302\,246\,994 \times 10^{-5}$	$0.539\,797\,852 \times 10^{-5}$
a_6	$0.203\,886\,313 \times 10^{-7}$	$0.420\,713\,632 \times 10^{-7}$
a_7	$0.638\,780\,966 \times 10^{-10}$	$0.147\,271\,071 \times 10^{-9}$

After Flatau PJ, Walko RL, Cotton WR (1992) Polynomial fits to saturation vapor pressure. *Journal of Applied Meteorology* 31: 1507–1513.

The water vapor mass mixing ratio, w_v, is the ratio of the mass of water vapor (m_v) present to the mass of dry air (m_d). It is thus defined, after substituting from the ideal gas law [1], as

$$w_v = \frac{m_v}{m_d} = \frac{\rho_v}{\rho_d} = \varepsilon \frac{e}{p - e} \qquad [6]$$

Table 3 Variation of H_i with T for constant $H = 1$

T (°C)	H	H_i
0	1.0	1.0
−10	1.0	1.10
−20	1.0	1.22
−30	1.0	1.34
−40	1.0	1.47

where $\varepsilon = M_v/M_d = 0.622$ is the ratio of the molecular weight of water to the molecular weight of the mixture of dry air gases, ρ_d is the density of dry air, and p is total atmospheric pressure.

A value of the saturation mass mixing ratio, w_s, is given by

$$w_s = \varepsilon \frac{e_s}{p - e_s} \quad [7]$$

Expressions for e and e_s from eqns [6] and [7] can be substituted into eqn [4]. Since $p \gg e$ and $p \ll e_s$,

$$H \approx \frac{w_v}{w_s} \quad [8]$$

is an approximate definition of the relative humidity.

The specific humidity is defined as the ratio of the mass of water vapor to the total mass of atmospheric gases. The water vapor mixing ratio can be related to the specific humidity, q_v, as

$$q_v = \frac{m_v}{m_d + m_v} = \varepsilon \frac{e}{p - (1-\varepsilon)e} = \frac{w_v}{1 + w_v} \quad [9]$$

Since both w_v and q_v are always smaller than 0.04 in the Earth's atmosphere, $q_v \approx w_v$.

The total mass of water vapor in a column of unit cross-sectional area extending from the surface to the top of the atmosphere is called the precipitable water, W_v:

$$W_v = \int_0^\infty \rho_v \, dz \quad [10]$$

where z is height above the Earth's surface. The term 'precipitable water' is used because, if all the vapor in the column were to be condensed into a pool of liquid at the base of the column, the depth of the pool would be equal to W_v/ρ_l, where ρ_l is the density of liquid water. To obtain a relationship between precipitable water and specific humidity, we can write eqn [10] in terms of pressure by incorporating the hydrostatic equation:

$$W_v = \frac{1}{g} \int_p^{p_0} \frac{\rho_v}{\rho_a} dp = \frac{1}{g} \int_p^{p_0} q_v \, dp \quad [11]$$

where p_0 is the surface pressure, corresponding to $z = 0$ and ρ_a is the air density.

The dew point temperature, denoted by T_D, is defined as the temperature to which moist air must be cooled, the atmospheric pressure and water vapor mixing ratio remaining constant, such that it becomes just saturated with respect to water. Obviously, at T_D the mixing ratio of the air becomes its saturation mixing ratio

$$w_v = w_s(T_D) \quad [12]$$

or equivalently

$$e = e_s(T_D) \quad [13]$$

We can determine the dew point temperature by inverting either eqn [12] or eqn [13], which can be done using eqns [3] and [6]. Although the dew point temperature is expressed in Kelvin, the dew point temperature is a measure not of temperature, but of atmospheric humidity. The term $T - T_D$ is called the dew point depression. Dew point depression is inversely proportional to relative humidity; a relative humidity of 100% corresponds to a dew point depression of zero.

Analogously to the dew point temperature, we define the frost point temperature as the temperature at which ice saturation occurs. The frost point temperature, T_F, is thus defined as

$$e = e_{si}(T_F) \quad [14]$$

or equivalently as

$$w_v = w_{si}(T_F) \quad [15]$$

where w_{si} is the saturation mass mixing ratio over ice. Note that for a given atmospheric mixing ratio, $T_F < T_D$, since $e_{si}(T) < e_s(T)$.

Example Calculations

Air has temperature of 30°C, a relative humidity of 50%, and a pressure of 1000 hPa. This information can be used to determine each of the humidity variables for these conditions:

partial pressure of water vapor, $e = 21.2$ hPa
water vapor mixing ratio, $w_v = 0.0135$
specific humidity, $q_v = 0.0133$
dew point temperature, $T_D = 291.8$ K

See also

Climate Variability: Glacial, Interglacial Variations. **Satellite Remote Sensing:** Water Vapor. **Thermodynamics:** Moist (Unsaturated) Air; Saturated Adiabatic Processes.

Further Reading

Curry JA and Webster PJ (1999) *Thermodynamics of Atmospheres and Oceans*. San Diego: Academic Press.

Flatau PJ, Walko RL and Cotton WR (1992) Polynomial fits to saturation vapor pressure. *Journal of Applied Meteorology* 31: 1507–1513.

HURRICANES

F D Marks, Hurricane Research Division, Miami, FL, USA

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

'Hurricane' is the term used in the Western Hemisphere for one of the general class of strong tropical cyclones, including western Pacific typhoons and similar systems, that are known simply as cyclones in the Indian and southern Pacific Oceans. A tropical cyclone is a low-pressure system which derives its energy primarily from evaporation from the sea in the presence of 1-minute sustained surface wind speeds $>17\,\mathrm{m\,s^{-1}}$ and the associated condensation in convective clouds concentrated near its center. In contrast, midlatitude storms (low-pressure systems with associated fronts) get their energy primarily from the horizontal temperature gradients that exist in the atmosphere. Structurally, the strongest winds in tropical cyclones are near the Earth's surface (a consequence of being 'warm-core' in the troposphere), while the strongest winds in midlatitude storms are near the tropopause (a consequence of being 'warm-core' in the stratosphere and 'cold-core' in the troposphere). 'Warm-core' refers to being warmer than the environment at the same pressure surface.

A tropical cyclone with the highest sustained wind speeds between 17 and $32\,\mathrm{m\,s^{-1}}$ is referred to as a tropical storm, whereas a tropical cyclone with sustained wind speeds $\geq 33\,\mathrm{m\,s^{-1}}$ is referred to as a hurricane or typhoon. Once a tropical cyclone has sustained winds $\geq 50\,\mathrm{m\,s^{-1}}$ it is referred to as a major hurricane or super typhoon. In the Atlantic and eastern Pacific Oceans hurricanes are also classified by the damage they can cause using the Saffir–Simpson scale (**Table 1**).

The Saffir–Simpson scale categorizes hurricanes on a scale from 1 to 5, with 1 the weakest and 5 the most intense. Major hurricanes correspond to categories 3 and higher. The reasons that some disturbances intensify to a hurricane, while others do not, are not well understood. Neither is it clear why some tropical cyclones become major hurricanes, while others do not. Major hurricanes produce 80–90% of the United States hurricane-caused damage despite accounting for only one-fifth of all landfalling tropical cyclones. Only two category 5 hurricanes made landfall on the mainland United States (Florida Keys 1935 and Camille 1969). Recent major hurricanes to make landfall on the United States were Hurricanes Bonnie and Georges in 1998, and Bret and Floyd in 1999.

As with large-scale extratropical weather systems, the structure and evolution of a tropical cyclone is dominated by the fundamental contradiction that while the airflow within a tropical cyclone represents an approximate balance among forces affecting each air parcel, slight departures from balance are essential for vertical motions and resulting clouds and precipitation, as well as changes in tropical cyclone intensity. As in extratropical weather systems, the basic vertical balance of forces in a tropical cyclone is hydrostatic except in the eyewall, where convection is superimposed on the hydrostatic motions. However, unlike in extratropical weather systems, the basic horizontal balance in a tropical cyclone above the boundary layer is between the sum of the Coriolis 'acceleration' and the centripetal 'acceleration', balanced by the horizontal pressure gradient force. This balance is referred to as gradient balance, where the Coriolis 'acceleration' is defined as the horizontal velocity of an air parcel, v, times the Coriolis parameter, f. (f is the Coriolis parameter ($f = 2\Omega \sin\phi$), where Ω is the angular velocity of the Earth ($7.292 \times 10^{-5}\,\mathrm{s^{-1}}$) and ϕ is latitude. The Coriolis parameter is zero at the equator and 2Ω at the pole.) Centripetal 'force' is defined as the acceleration on a parcel of air moving in a curved path, directed toward the center of curvature of the path, with magnitude v^2/r, where v is the horizontal velocity of the parcel and r the radius of curvature of the path. The centripetal force alters the original two-force geostrophic balance and creates a nongeostrophic gradient wind.

Table 1 Saffir–Simpson scale of hurricane intensity

Category	*Pressure (hPa)*	*Wind ($\mathrm{m\,s^{-1}}$)*	*Storm surge (m)*	*Damage*
1	>980	33–42	1.0–1.7	Minimal
2	979–965	43–49	1.8–2.6	Moderate
3	964–945	50–58	2.7–3.8	Extensive
4	944–920	59–69	3.9–5.6	Extreme
5	<920	≥ 70	≥ 5.7	Catastrophic

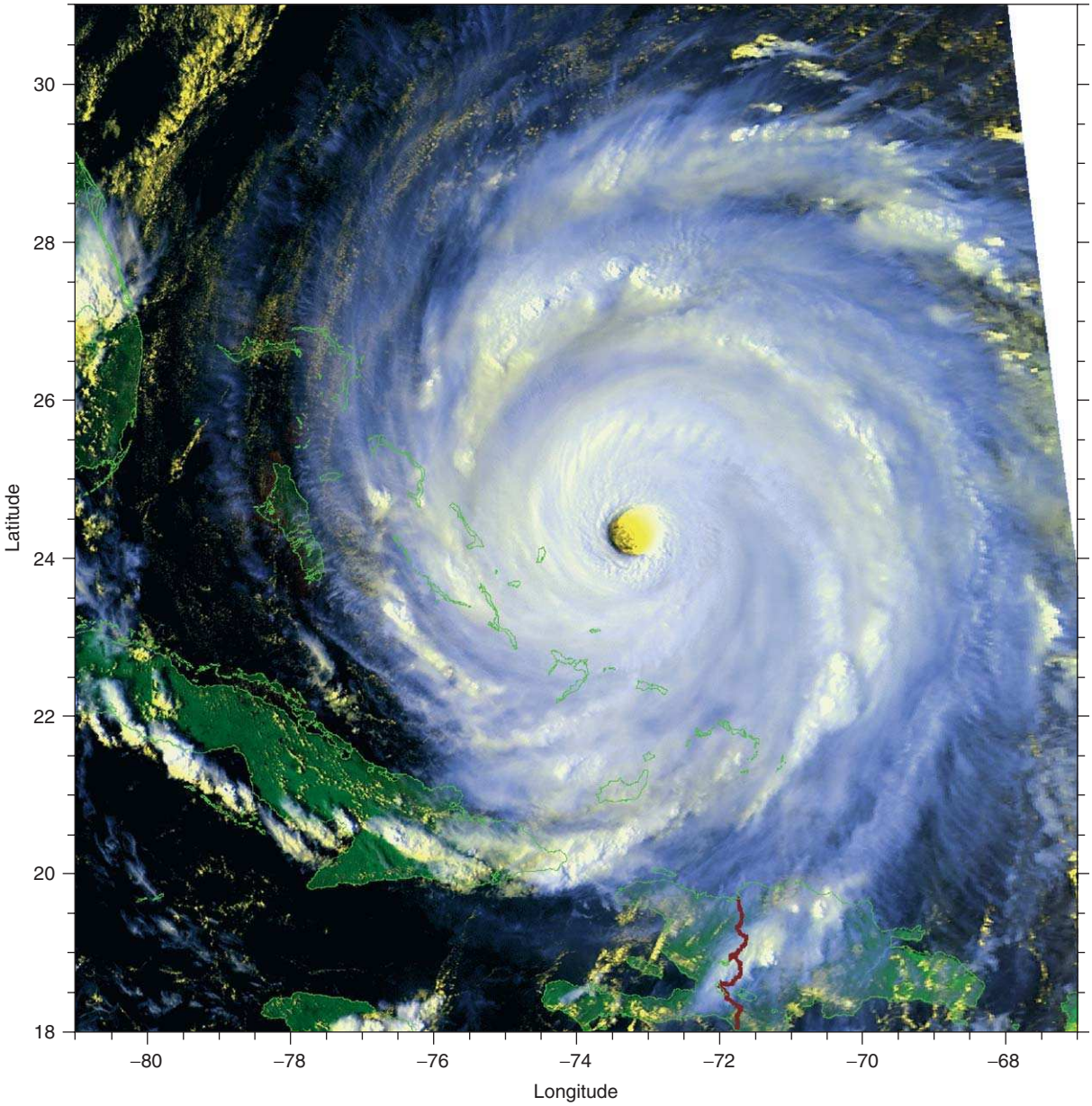

Figure 1 NOAA-14 AVHRR multispectral false color image of Hurricane Floyd at 2041 UTC, 13 September 1999 about 800 km east of southern Florida. (Photo courtesy of NOAA Operationally Significant Event Imagery website: http://www.osei.noaa.gov/.)

The inner region of the tropical storm, termed the cyclone 'core', contains the spiral bands of precipitation, the eyewall, and the eye that characterize tropical cyclones in radar and satellite imagery (**Figure 1**). The primary circulation – the tangential or swirling wind – in the core becomes strongly axisymmetric as the cyclone matures. The strong winds in the core, which occupies only 1–4% of the cyclone's area, threaten human activities and make the cyclone's dynamics unique. In the core, the local Rossby number is always >1 and may be as high as 100. The Rossby number indicates the relative magnitude of centrifugal (v/r) and Coriolis (f) accelerations, $Ro = V/fr$, where V is the axial wind velocity, f the Coriolis parameter, and r the radius from the storm center. An approximate breakdown of regimes is: $Ro < 1$, geostrophic flow; $Ro > 1$, gradient flow; and $Ro > 50$, cyclostrophic flow. When the Rossby number significantly exceeds unity, the balance in the core becomes more cyclostrophic, where the pressure gradient force is almost completely balanced by the centrifugal 'force'. The time scales are such that air swirling around the center completes an orbit in much less than a pendulum day (defined as $1/f$).

When the atmosphere is in approximate horizontal and vertical balance, the wind and mass fields are

tightly interconnected. The distribution of a single mass or momentum variable may be used as a starting point to infer the distribution of all other such variables. One such variable is potential vorticity (PV), approximately equal to the vorticity times the thermal stratification, which is related to the three-dimensional mass and momentum fields through an inverse second-order Laplacian-like operator. The benefit of such a relationship is that PV variations in a single location are diagnostically related to variations in mass and wind fields at a distance. Areas of high PV correspond locally to low mass, or cyclones, while areas of low PV correspond to anticyclones.

Typical extratropical weather systems contain high PV values around $0.5 \times 10^{-6}\,\mathrm{m^2\,s^{-1}\,K\,kg^{-1}}$ (0.5 PVU) to 5 PVU, whereas typical values in the tropical cyclone core are ≥ 10 PVU. **Figure 2** shows the wind and mass fields associated with an idealized axially symmetric tropical cyclone PV anomaly with the PV concentrated near the surface rather than in a vertical column. The cyclonic anomaly (positive in the Northern Hemisphere) is associated with a cyclonic circulation that is strongest at the level of the PV anomaly near the surface, and decreases upward. Temperatures are anomalously warm above the PV anomaly (isentropic surfaces are deflected downward). While consistent with the simple PV distribution, the wind and mass fields are also in horizontal and vertical balance. The tropical cyclone being a warm-core vortex, the PV inversion dictates that the winds that swirl about the center decrease with increasing height, but they typically fill the depth of the troposphere. If the PV reaches values ≥ 10 PVU, the inner region winds can become intense, as in Hurricane Gloria (**Figure 3**). Gloria had PV values exceeding 500 PVU just inside the radius of maximum winds of 15 km where the axisymmetric mean tangential winds exceeded $65\,\mathrm{m\,s^{-1}}$.

Many features in the core, however, persist with little change for (pendulum) days (mean life span of a tropical cyclone is about 5–10 days). Because these long lifetimes represent tens or hundreds of orbital periods (~ 1 h), the flow is nearly balanced. Moreover, at winds $> 35\,\mathrm{m\,s^{-1}}$, the local Rossby radius of deformation is reduced from its normal $\sim 10^3$ km to a value comparable with the eye radius. The Rossby radius of deformation is the ratio of the speed of the relevant gravity wave mode and the local vorticity, or, equivalently, the ratios of the Brunt–Väisälä and inertial frequencies. This scale indicates the amount of energy that goes into gravity waves compared with inertial acceleration of the wind. In very intense tropical cyclones, the eye radius may approach the

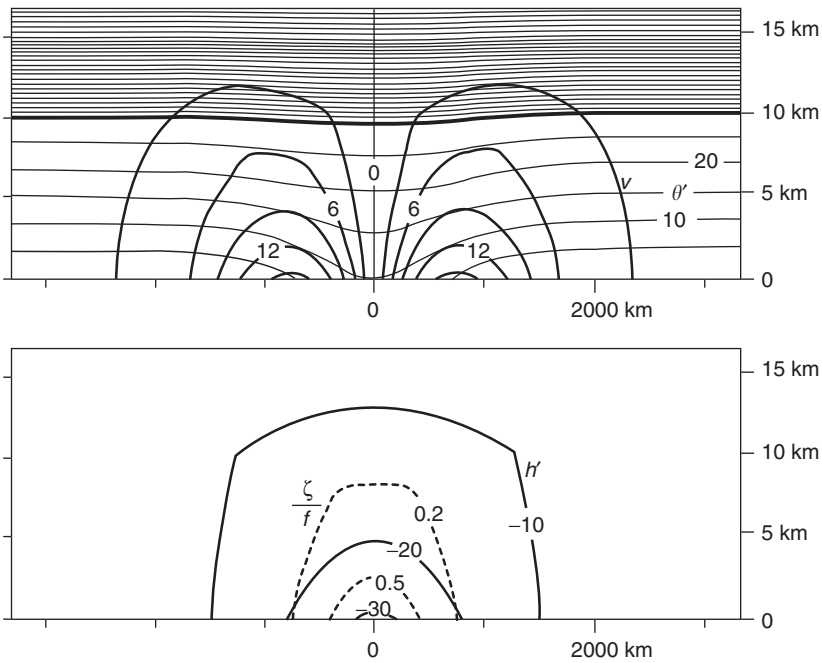

Figure 2 v (gradient wind) $(\mathrm{m\,s^{-1}})$ and θ' (perturbation potential temperature) (K, top panel); and h' (geopotential height perturbation) (dm) and ζ/f (bottom panel) for a warm core, lower cyclone. The tropopause location is denoted by the bold solid line, and the label 0 on the horizontal axis indicates the core (and axis of symmetry) of the disturbance. The equivalent pressure deviation at the surface in the center of the vortex is -31 hPa. (Reproduced with permission from Thorpe AJ (1986) Synoptic scale disturbances with circular symmetry. *Monthly Weather Review* 114: 1384–1389; © American Meteorological Society.)

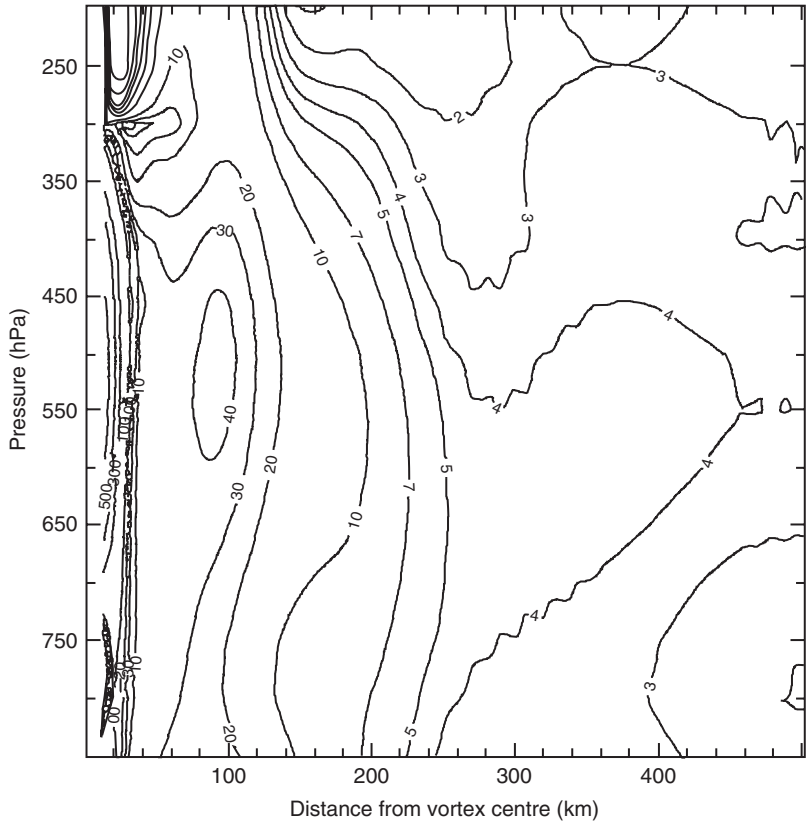

Figure 3 Radial-height cross-section of symmetric potential vorticity for Hurricane Gloria, 24 September 1985. Contours are 0.1 PVU. Values in the data-sparse region, within 13 km of vortex center, are not displayed. (Reproduced with permission from Shapiro LJ and Franklin JL (1995) Potential vorticity in Hurricane Gloria. *Monthly Weather Review* 123: 1465–1475; © American Meteorological Society.)

depth of the troposphere (15 km), making the aspect ratio unity. Thus, the dynamics near the center of a tropical cyclone are so exotic that conditions in the core differ from the Earth's day-to-day weather as much as the atmosphere of another planet does.

Climatology

There are 80–90 tropical cyclones worldwide per year, with the Northern Hemisphere having more tropical cyclones than the Southern Hemisphere. **Table 2** shows that of the 80–90 tropical cyclones, 45–50 reach hurricane or typhoon strength and 20 reach major hurricane or super typhoon strength. The western North Pacific (27 tropical cyclones), eastern North Pacific (17 tropical cyclones), south-west Indian Ocean (10 tropical cyclones), Australia/south-west Pacific (10 tropical cyclones), and North Atlantic (10 tropical cyclones) are the major tropical cyclone regions. There are also regional differences in the tropical cyclone activity by month with the majority of the activity in the summer season for each basin. Hence, in the Pacific, Atlantic, and North Indian Ocean the maximum numbers of tropical cyclones occur in August through October, while in the South Pacific and Australia regions the maxima are in February and March. In the South Indian Ocean, the peak activity occurs in June. In the western North Pacific, Bay of Bengal, and South Indian Ocean regions tropical cyclones may occur in any month, while the other regions at least one tropical cyclone-free month occurs per year. For example, in the North Atlantic, there has never been tropical cyclone activity in January.

Some general conclusions can be drawn from the global distribution of tropical cyclone locations (**Figure 4A**). Tropical cyclone formation is confined to a region approximately 30° N and 30° S, with 87% of them located within 20° of the Equator. There is a lack of tropical cyclones near the Equator, as well as in the eastern South Pacific and South Atlantic basins. From these observations there appear to be at least five necessary conditions for tropical cyclone development.

- *Warm sea surface temperature (SST) and large mixed-layer depth* (i.e., the thickness of the mixed

Table 2 Mean annual frequency, standard deviation (σ) and percentage of global total of the number of tropical storms (winds $\geq 17\,\mathrm{m\,s^{-1}}$), hurricane-force tropical cyclone (winds $\geq 33\,\mathrm{m\,s^{-1}}$), and major hurricane-force tropical cyclone (winds $\geq 50\,\mathrm{m\,s^{-1}}$). Dates in parentheses provide the nominal years for which accurate records are currently available

Tropical cyclone basin	*Tropical storm annual frequency (σ)*	*% of total*	*Hurricane annual frequency (σ)*	*% of total*	*Major hurricane annual frequency (σ)*	*% of total*
Atlantic (1944–00)	9.8 (3.0)	11.4	5.7 (2.2)	12.1	2.2 (1.5)	10.9
NE Pacific (1970–00)	17.0 (4.4)	19.7	9.8 (3.1)	20.7	4.6 (2.5)	22.9
NW Pacific (1970–00)	26.9 (4.1)	32.1	16.8 (3.6)	35.5	8.3 (3.2)	41.3
N Indian (1970–00)	5.4 (2.2)	6.3	2.2 (1.8)	4.6	0.3 (0.5)	1.5
SW Indian (30–100° E) (1969–00)	10.3 (2.9)	12.0	4.9 (2.4)	10.4	1.8 (1.9)	9.0
Australian/SE Indian (100–142° E) (1969–00)	6.5 (2.6)	7.5	3.3 (1.9)	7.0	1.2 (1.4)	6.0
Australian/SW Pacific (142° E) (1969–00)	10.2 (3.1)	11.8	4.6 (2.4)	9.7	1.7 (1.9)	8.5
Global (1970–00)	86.1 (8.0)		47.3 (6.5)		20.1 (5.7)	

layer, defined as the depth of the sharp temperature inversion (also referred to as the thermocline) between the cooler bottom water and the warmer near surface water). Numerous studies suggest a minimum SST criterion of 26°C for development. The warm water must also have sufficient depth (i.e., 50 m). Comparison of **Figures 4A** and **4B**, the annual mean global SST, shows the strong correlation between regions where the SST is >26°C and annual tropical cyclone activity. An SST of >26°C is sufficient but not necessary for tropical cyclone activity, as is evidenced by the regions with tropical cyclone activity when the SST <26°C. Some of the discrepancy exists because storms that form over regions where the SST is >26°C are advected poleward during their life cycle. However, tropical cyclones are observed to originate over regions where the SST <26°C. These occurrences are not many, but the fact that they exist suggests that other factors are important.

- *Background Earth vorticity.* Tropical cyclones do not form within 3° of the Equator. The Coriolis parameter vanishes at the Equator and increases to extremes at the poles. Hence, a threshold value of Earth vorticity (f) must exist for a tropical cyclone to form. However, the likelihood of formation does not increase with increasing f. Thus, nonzero Earth vorticity is necessary, but not sufficient to produce tropical cyclones.
- *Low vertical shear of the horizontal wind.* In order for tropical cyclones to develop, the latent heat generated by the convection must be kept near the center of the storm. Historically, shear was thought to 'ventilate' the core of the cyclone by advecting the warm anomaly away. The ventilation argument suggests that if the storm travels at nearly the same speed as the environmental flow in which it is embedded then its heating remains over the disturbance center. However, if it is moving slower than the mean wind at upper levels then the heating in the upper troposphere is carried away by the mean flow. Recent analysis suggests that the effect of shear is to force the convection into an asymmetric pattern such that the convective latent heat release forces flow asymmetry and irregular motion rather than intensification of the symmetric vortex. Thus, if the vertical shear is too strong ($>16\,\mathrm{m\,s^{-1}}$) then existing tropical cyclones are ripped apart and new ones cannot form.
- *Low atmospheric static stability.* Static stability is the ability of a fluid to become turbulent (unstable) or laminar (stable) due to the effects of buoyancy neglecting all other inertial effects of motion. The troposphere must be potentially unstable to sustain convection for an extended period. Typically measured as the difference between the equivalent potential temperature, θ_e, at the surface and 500 hPa, instability must typically be >10 K for convection to occur. This value is usually satisfied over tropical oceans.
- *Tropospheric humidity.* The higher the midlevel humidity, the longer a parcel of air can remain saturated as it entrains the surrounding air during its ascent. Vigorous convection occurs if the parcel remains saturated throughout its ascent. A relative humidity of 50–60% at lower to midlevels (700–500 hPa) is often sufficient to keep a parcel saturated during ascent. This condition is regularly evident over tropical oceans.

These conditions are usually satisfied in the summer and fall seasons for each tropical cyclone basin.

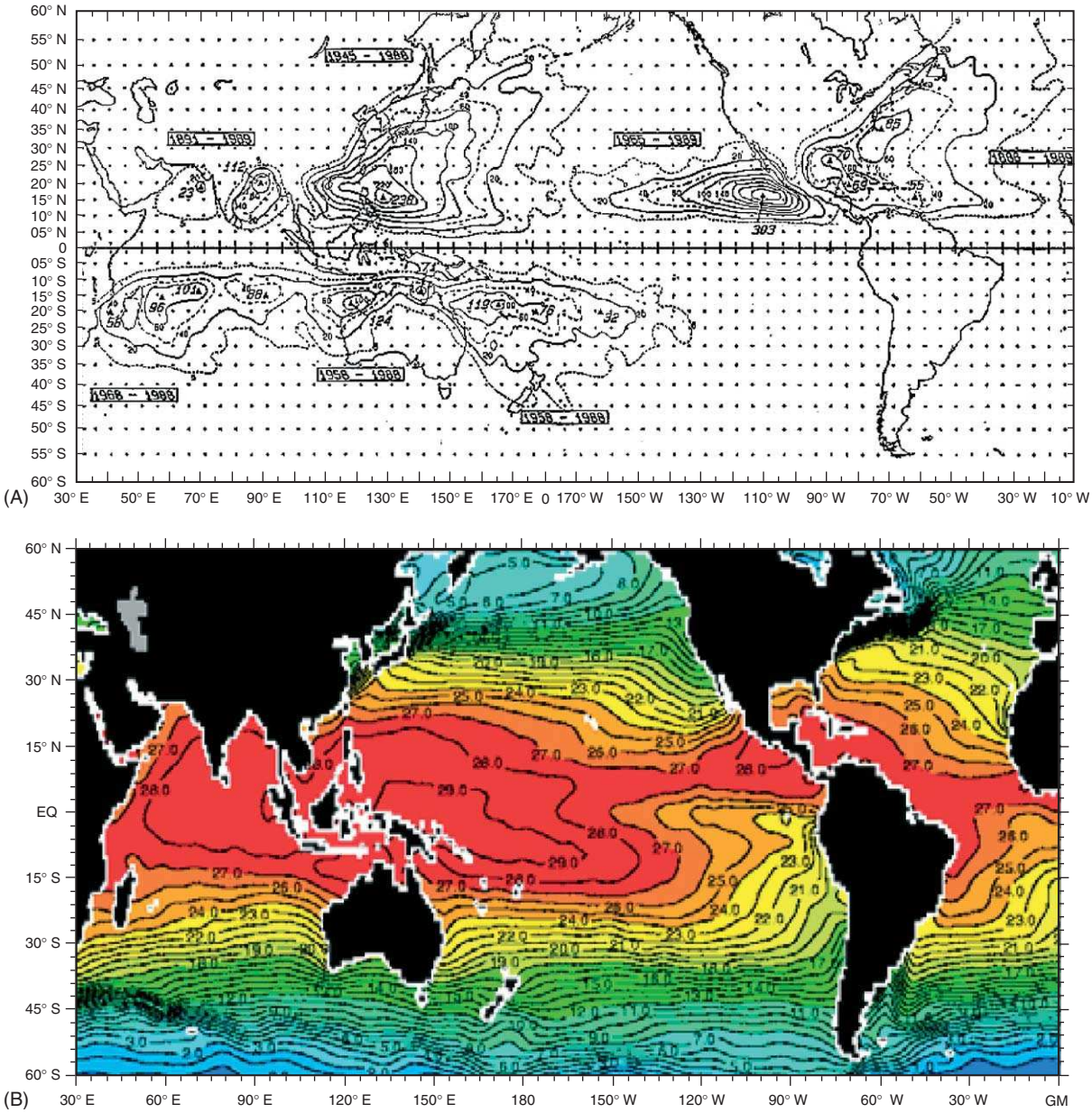

Figure 4 (A) Frequency of tropical cyclones per 100 years within 140 km of any point. Solid triangles indicate maxima, with values shown. Period of record is shown in boxes for each basin. (B) Annual sea surface temperature distribution (°C).

However, even when all of the above conditions are favorable, tropical cyclones don't necessarily form. In fact, there is growing evidence for significant interannual variability in tropical cyclone activity, where numerous tropical cyclones form in a given basin over a week to 10 days, followed by 2–3 weeks with little or no tropical cyclone activity. **Figure 5** shows just such an active period in the Atlantic basin in mid-September 1999, where two hurricanes (Floyd and Gert), both major, and an unnamed tropical depression formed within a few days of each other. During these active phases almost every disturbance makes at least tropical storm strength, whereas in the inactive phase practically none of the disturbances intensify. The two hurricanes and unnamed depression in **Figure 5** represented the second 10-day active period during the summer of 1999. An earlier period in mid-August also resulted in the development of three hurricanes (Brett, Cindy, and Dennis), two of which were major, as well as a tropical storm (Emily). There is speculation that the variability is related to the propagation of a global wave. Because the SST, static stability, and Earth vorticity don't vary that much during the season, the interannual variability is most likely related

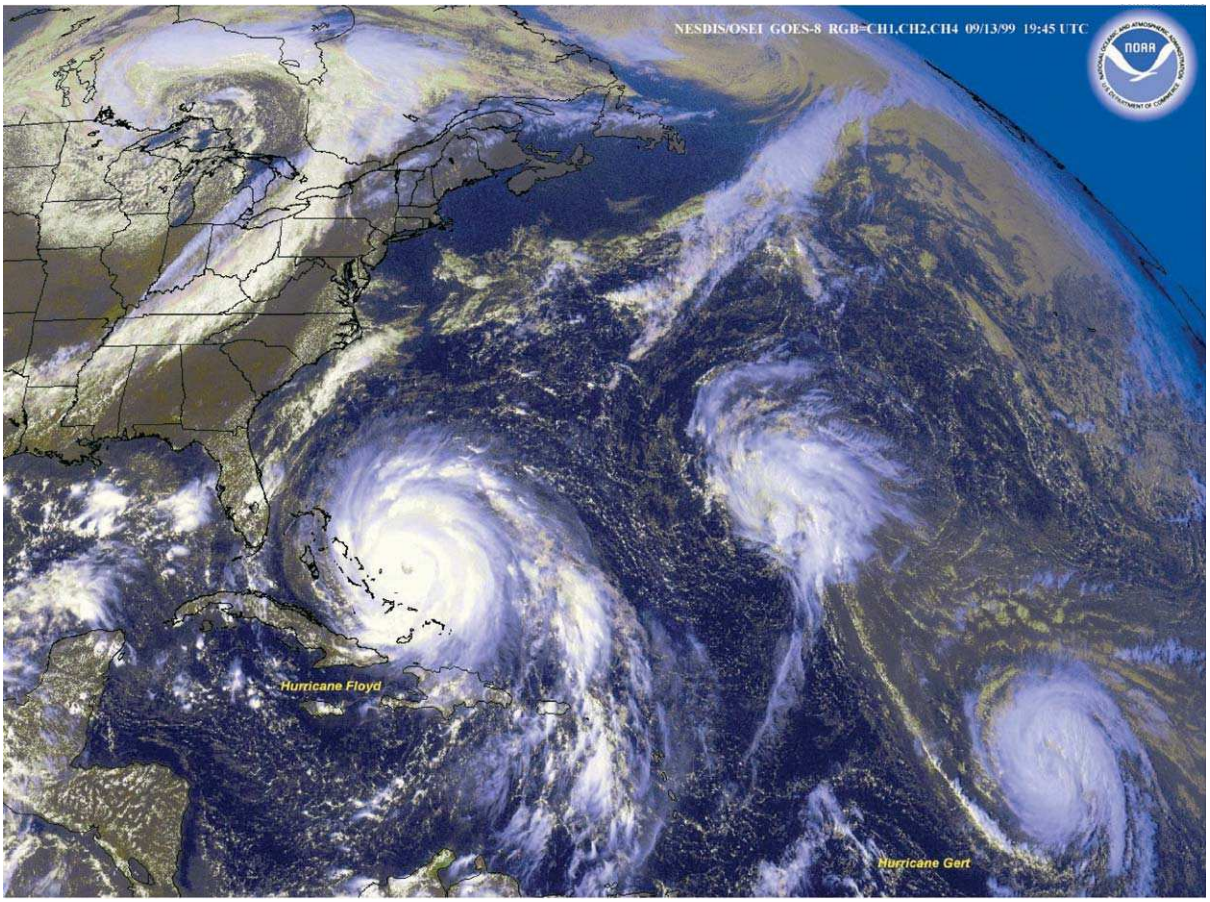

Figure 5 GOES multispectral false color image of Hurricanes Floyd and Gert and an unnamed tropical depression at 1935 UTC, 13 September 1999. (Photo courtesy of NOAA Operationally Significant Event Imagery website: http://www.osei.noaa.gov/.)

to variations in tropospheric relative humidity and vertical wind shear.

It has long been recognized that the number of tropical cyclones in a given region varies from year to year. The exact causes of this remain largely speculative. The large-scale global variations in atmospheric phenomena such as the El Niño Southern Oscillation (ENSO) and the Quasi-Biennial Oscillation (QBO) appear to be related to annual changes in the frequency of tropical cyclone formation, particularly in the Atlantic Ocean. The ENSO phenomenon is characterized by warmer SSTs in the eastern South Pacific and anomalous winds over much of the equatorial Pacific. It influences tropical cyclone formation in the western North Pacific, South Pacific, and even the North Atlantic.

During the peak phase of the ENSO, often referred to as El Niño (which usually occurs during the months July–October), anomalous westerly winds near the Equator extend to the dateline in the western North Pacific acting to enhance the intertropical convergence zone (ITCZ) in this area, making it more favorable for formation of tropical cyclones. Another effect of the El Niño circulation is warmer SST in the eastern South Pacific. During such years, tropical cyclones form closer to the Equator and farther east. Regions such as French Polynesia, which are typically unfavorable for tropical cyclones owing to a strong upper-level trough, experience numerous tropical cyclones. The eastern North Pacific is also affected by the El Niño through a displacement of the ITCZ south to near 5° N. Additionally, the warm ocean anomaly of El Niño extends to near 20° N, which enhances the possibility of tropical cyclone formation. The result is an average increase of two tropical cyclones during El Niño years. Cyclones also develop closer to the Equator and farther west than during a normal year.

The QBO is a roughly 2-year oscillation of the equatorial stratosphere (30–50 hPa) winds from easterly to westerly and back. The phase and magnitude of QBO are associated with the frequency of tropical cyclones in the Atlantic. Hurricane activity is more frequent when the 30-hPa stratospheric winds are

westerly. The exact mechanism by which the QBO affects tropical cyclones in the troposphere is not clear; however, there are more North Atlantic tropical cyclones when the QBO is in the westerly phase than when it is in the easterly.

Tropical Cyclogenesis

Enthalpy is a thermodynamic state function defined for an ideal gas as the temperature times the specific heat at constant pressure plus a constant. For a system like the atmosphere which consists of a mixture of components the total enthalpy is the mass-weighted sum of the enthalpies of each component. Thus, the total enthalpy for a system consisting of a mixture of dry air, water vapor, and liquid water is defined as a constant plus the temperature times the sum of the specific heats at constant pressure for each component times the masses of each component, respectively. In an adiabatic, reversible process, the total enthalpy is conserved, although the component enthalpies may not be due to the exchange of enthalpy between components in phase changes. Most of the energy needed for tropical cyclones to form and maintain themselves is realized through the difference in enthalpy between the warm near-surface waters of the tropical ocean and the tropospheric column. The process of bringing the late-summer tropical troposphere into thermodynamic equilibrium with the sea surface at 28–30°C, mainly through the irreversible energy transfer from the ocean to the air by evaporation, can produce hydrostatic pressures as low as the minimum sea-level pressures of the most intense tropical cyclones. Thus, much of the tropical oceans contain enough moist enthalpy to support a major hurricane.

Throughout most of the Trade Wind regions, gradual subsidence causes an inversion that traps water vapor in the lowest kilometer. Sporadic convection (often in squall lines) that breaks through the inversion exhausts the moist enthalpy stored in the near-surface boundary layer quickly, leaving a wake of cool, relatively dry air. This air comes from just above the inversion and is brought to the surface by downdrafts driven by the weight of hydrometeors and cooling due to their evaporation. If the squall line does not keep moving it quickly runs out of energy. A day, or even several days, may pass before normal fair-weather evaporation can restore the preexistent moist enthalpy behind the squall. The reasons why squall line convection generally fails to produce hurricanes lie in the limited amount of enthalpy that can be stored in the sub-inversion layer and the slow rate of evaporation under normal wind speeds in the trades.

For a tropical cyclone to occur, evaporation must speed up and the equilibrium enthalpy at the sea surface temperature must rise through a lowering of the surface pressure. Tropical cyclones are thus finite-amplitude phenomena. They do not grow by some linear process from infinitesimal amplitude. The normal paradigm of searching for the most rapidly growing unstable linear mode used to study midlatitude cyclogenesis through baroclinic instability fails here. The surface wind has to exceed roughly 20 m s^{-1} before evaporation can prevail against downdraft cooling.

How then do tropical cyclones reach the required finite amplitude? The answer seems to lie in the structure of tropical convection. As explained previously, behind a squall line the lower troposphere (below the 0°C isotherm at ~5 km) is dominated by precipitation-driven downdrafts which lie under the 'anvil' of nimbostratus and cirrostratus that spreads behind the active convection. Above 5 km, a combination of differential radiative fluxes at the top and bottom of the anvil and residual condensational heating from the main updraft maintains weak rising motion. This updrafts-over-downdrafts arrangement requires horizontal convergence centered near 5 km altitude to maintain mass continuity. The important kinematic consequence is formation of patchy shallow vortices near the altitude of the 0°C isotherm. The typical horizontal scales of these 'mesovortices' are tens to hundreds of kilometers. If they were at the surface or if their influence could be extended downward to the surface then they would be the means to get the system to the required finite amplitude.

The foregoing reasoning defines the important unanswered questions: (1) how do the midlevel mesovortices extend their influence to the surface, and (2) what are the detailed thermodynamics at the air–sea interface during this process? Leading hypotheses for (1) are related to processes that can increase the surface vorticity through changes in static stability and momentum mixing, both horizontally and vertically. However, the answers to these questions await new measurements that are just becoming available through improved observational tools.

Basic Structure

Primary and Secondary Circulations

Inner core dynamics have received a lot of attention over the last 40 years through aircraft observations of the inner core structure. These observations show that the tropical cyclone inner core dynamics are dominated by interactions between 'primary' (horizontal axisymmetric), 'secondary' (radial and vertical)

circulations, and a wavenumber one asymmetry caused by the storm motion. The primary circulation is so strong in the cyclone core that it is possible to consider axisymmetric motions separately, if account is taken of forcing by the asymmetric motions. The primary circulation is in near-gradient balance, and evolves when heat and angular momentum sources (often due to asymmetric motions) force secondary circulations, which in turn redistribute heat and angular momentum.

Figure 6 shows that the primary circulation is sustained by the secondary circulation that consists of frictional inflow that loses angular momentum to the sea as it gains moist enthalpy. (Angular momentum $M = Vr + fr^2/2$, where V is the tangential wind velocity, f the Coriolis parameter, and r the radius from the storm center.) The inflow picks up latent heat through evaporation, and exchanges sensible heat with the underlying ocean, as it spirals into lower levels of the storm under influence of friction. The evaporation of sea spray adds moisture to the air, while at the same time cooling it. This process is important in determining the intensity of a tropical cyclone. Near the vortex center, the inflow turns upward and brings the latent heat it acquires in the boundary layer into the free atmosphere. Across the top of the boundary layer, turbulent eddies cause significant downward flux of sensible heat from the free atmosphere to the boundary layer. The energy source for the turbulent eddies is mechanical mixing caused by the strong winds. The eddies are also responsible for downward mixing of angular momentum. Hence, these turbulent eddy fluxes fuel the storm.

As the air converges towards the eye and is lifted in convective clouds that surround the clear eye, it ascends to the tropopause (the top of troposphere, where temperature stops decreasing with height). As shown in **Figure 6**, the convective updrafts in the eyewall turn the latent heat into sensible heat through the latent heat of condensation to provide the buoyancy needed to loft air from the surface to tropopause level. The updraft entrains midlevel air, promoting mass and angular momentum convergence into the core. It is the midlevel inflow that supplies the excess angular momentum needed to spin up the vortex. The thermodynamics of a storm can be modeled as an idealized heat engine, running between a warm heat reservoir, the sea, at around 300 K, and a cold reservoir, 15–18 km up in the tropical troposphere, at about 200 K. The net energy realized in the whole process is proportional to the difference in temperature between the ocean and the upper troposphere. Storm-induced upwelling of cooler water reduces ocean SST by a few degrees, which has a considerable effect on the storm's intensity.

As shown in **Figure 7**, the secondary circulation also controls the distribution of hydrometeors and radar reflectivity. It is much weaker than the primary circulation except in the anticyclonic outflow, where the vortex is also much more asymmetric. Precipitation-driven convective updrafts form as hydrometeors fall from the outward sloping updraft. Condensation in the anvil causes a mesoscale updraft above the 0°C isotherm and precipitation loading by snow falling from the overhanging anvil causes a mesoscale downdraft below 0°C isotherm. The melting level itself

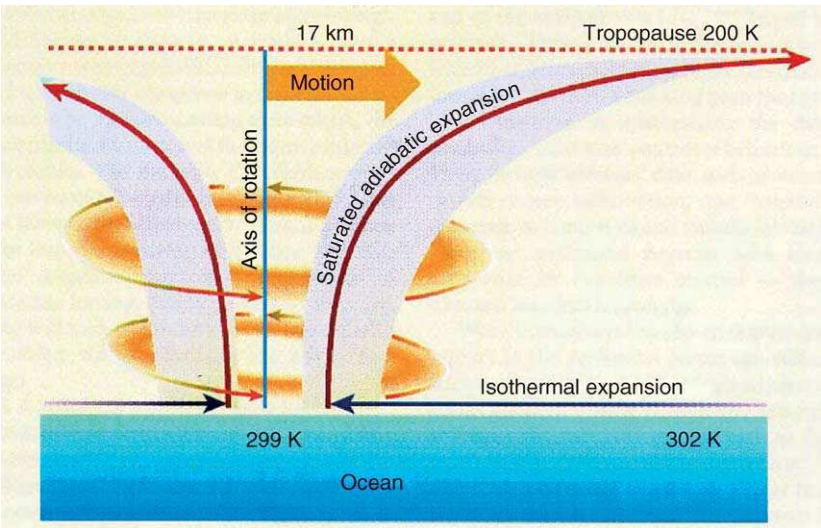

Figure 6 Schematic of the secondary circulation thermodynamics. (Reproduced with permission from Willoughby HE (1999) Hurricane heat engines. *Nature* 401: 649–650; © Macmillan Magazines Ltd.)

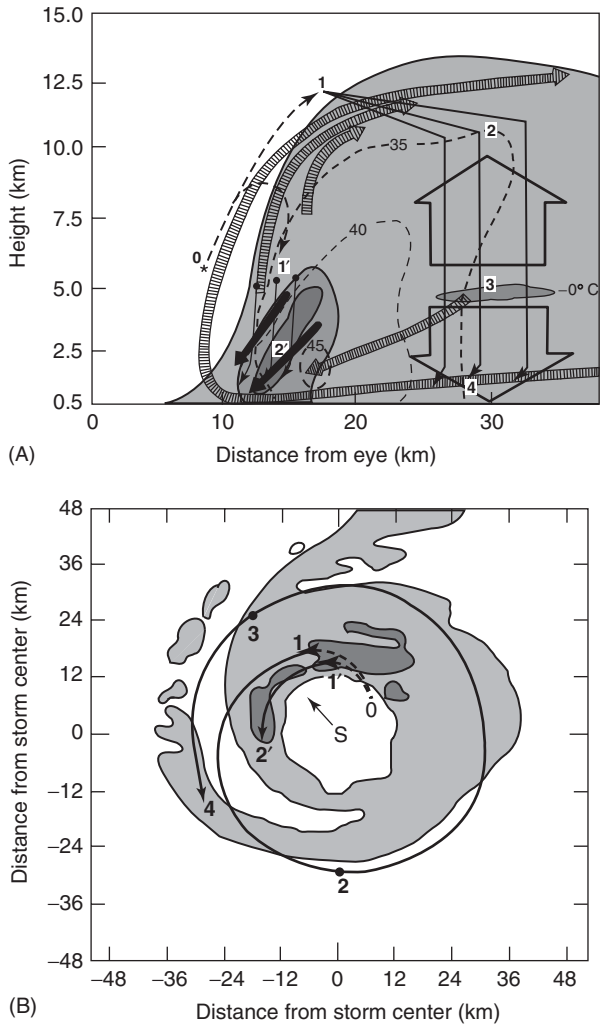

Figure 7 (A) Schematic of the radius–height circulation of the inner core of Hurricane Alicia (1983). Shading depicts the reflectivity field, with contours of 5, 30, and 35 dBZ. The primary circulation ($m\,s^{-1}$) is depicted by dashed lines and the secondary circulation by the wide hatched streamlines. The convective downdrafts are denoted by the thick solid arrows, while the mesoscale up- and downdrafts are shown by the broad arrows. (B) Schematic plan view of the low-level reflectivity field in the inner core of Hurricane Alicia superimposed with the middle of the three hydrometeor trajectories in (A). Reflectivity contours in (b) are 20 and 35 dBZ. The storm center and direction are also shown. In (A) and (B) the hydrometeor trajectories are denoted by dashed and solid lines labeled 0-1-2-3-4 and 0′-1′-2′. (Reproduced with permission from Marks FD and Houze RA (1987) Inner core structure of Hurricane Alicia from airborne Doppler radar observations. *Journal of the Atmospheric Sciences* 44: 1296–1317; ©American Meteorological Society.)

marks the height of maximum mass convergence. Inside the eye, dynamically driven descent and momentum mixing leads to substantial pressure falls.

In order for the primary circulation to intensify, the flow cannot be in exact balance. Vertical gradients of angular momentum due to vertical shears of the primary circulation cause updrafts to pass through the convective heat sources, because the path of least resistance for the warmed air lies along constant angular momentum surfaces. Similarly, horizontal temperature gradients due to vertical shears cause the horizontal flow to pass through momentum sources, because the path of least resistance lies along isentropes (potential temperature or θ surfaces). Although the flow lies generally along the angular momentum or isentropic surfaces, it has a small component across them. The advection by this component, not the direct forcing, is the mechanism by which the primary circulation evolves.

Some of the most intense tropical cyclones exhibit 'concentric' eyewalls, i.e., two or more eyewall structures centered at the circulation center of the storm. In much the same way as the inner eyewall forms, convection surrounding the eyewall can become organized into distinct rings. Eventually, the inner eye begins to feel the effects of the subsidence resulting from the outer eyewall, and the inner eyewall weakens, to be replaced by the outer eyewall. The pressure rises due to the destruction of the inner eyewall are usually more rapid than the pressure falls, due to the intensification of the outer eyewall, and the cyclone itself weakens for a short period of time. This mechanism, referred to as the eyewall replacement cycle, often accompanies dramatic changes in storm intensity. The intensity changes are often associated with the development of secondary wind maxima outside the storm core.

A good example of contracting rings of convection effecting the intensification of a hurricane is shown in **Figure 8** for Hurricane Gilbert on 14 September 1988. Two convective rings, denoted by intense radar reflectivity, are evident in **Figure 8A**. The outer ring is located near 80–90 km radius and the inner one at 10–12 km radius. **Figure 8B** shows that both are associated with maxima in tangential wind and vorticity. **Figure 9** shows that in the ensuing 12–24 h the storm filled dramatically. However, it is not clear how much of the filling was caused by the storm moving over land and how much by the contracting outer ring and decaying inner ring of convective activity.

A process has been proposed whereby: (1) nonlinear balanced adjustment of the vortex to eddy heat and angular momentum sources generated by some environmental interaction in the storm's periphery produces an enhanced secondary circulation; (2) a secondary wind maximum develops in response; and (3) the wind maximum contracts as a result of differential adiabatic warming associated with the convective diabatic heating in the presence of a inward radial gradient of inertial stability. Under these

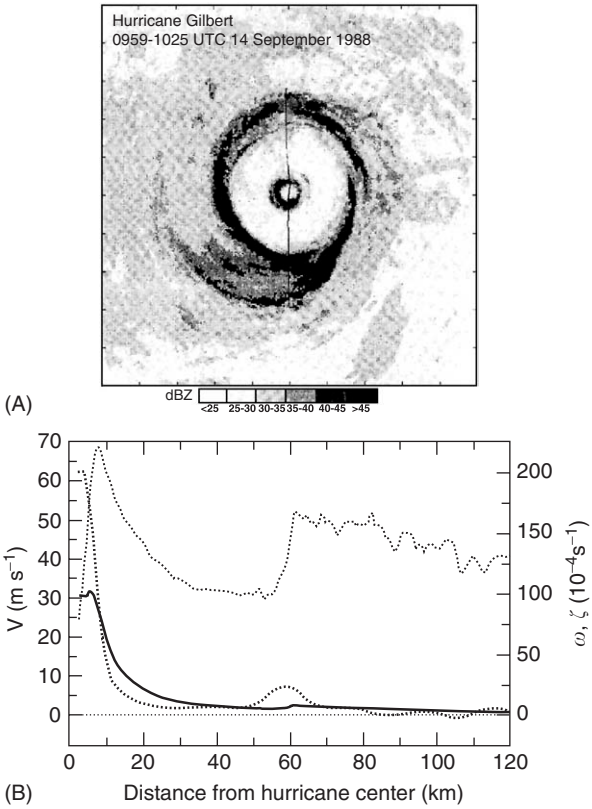

Figure 8 (A) Composite horizontal radar reflectivity of Hurricane Gilbert for 0959–1025 UTC, 14 September 1988; the domain is 360 km × 360 km, marked every 36 km. The line through the center is the WP-3D aircraft flight track. (B) Profiles of flight-level angular velocity (ω, solid) tangential wind (short dash), and smoothed relative vorticity (ζ, long dash) along the southern leg of the flight track shown in (A). (Reproduced with permission from Kossin JP, Schubert WH, and Montgomery MT (2000) Unstable interactions between a hurricane's primary eyewall and a secondary ring of enhanced vorticity. *Journal of the Atmospheric Sciences* 57: 3893–3917; © American Meteorological Society.)

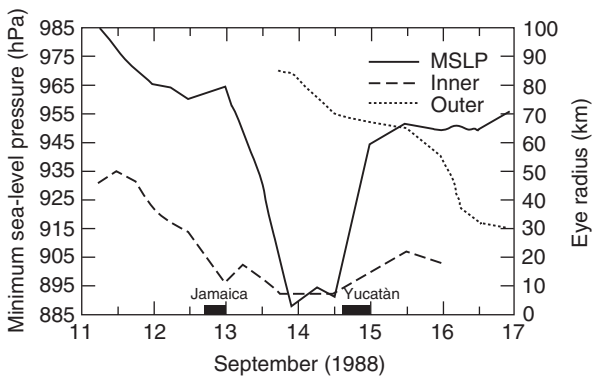

Figure 9 Hurricane Gilbert's minimum sea-level pressure (MSLP) and radii of the inner and outer eyewalls as a function of time, September 1988. Solid blocks at bottom indicate times over land. (Reproduced with permission from Black ML and Willoughby HE (1999) The concentric eyewall cycle of Hurricane Gilbert. *Monthly Weather Review* 120: 947–957; © American Meteorological Society.)

circumstances, understanding the intensification of the tropical cyclone reduces to determining what mechanisms can produce an enhanced secondary wind maximum.

Inner core – eyewall and eye The most recognizable feature found within a hurricane is the eye (**Figure 10**). It is found at the center and is typically between 20–50 km in diameter. The eye is the focus of the hurricane, the point about which the primary circulation rotates and where the lowest surface pressures are found in the storm. The eye is a roughly circular area of comparatively light winds and fair weather found at the center of strong tropical cyclones. Although the winds are calm at the axis of rotation, strong winds may extend well into the eye. As seen in **Figure 10**, there is little or no precipitation and sometimes blue sky or stars can be seen. The eye is the region of warmest temperatures aloft – the eye temperature may be $\geq 10°C$ warmer at an altitude of 12 km than the surrounding environment, but only 0–2°C warmer at the surface.

The eye is surrounded by the eyewall, the roughly circular area of deep convection associated with the up-branch of the secondary circulation and the highest surface winds. The eye is composed of air that is slowly sinking and the eyewall has a net upward flow because of many moderate – occasionally strong – updrafts and downdrafts. The eye's warm temperatures are due to warming by compression of the subsiding air. Most soundings taken within the eye are similar to that for Hurricane Hugo in **Figure 11**. They show a low-level layer which is relatively moist, with an inversion above, suggesting that the sinking in the eye typically does not reach the ocean surface, but instead gets only within 1–3 km of the surface. An eye is usually present only in hurricane-strength tropical cyclones.

The general mechanisms by which the eye and eyewall are formed are not fully understood, although observations shed some light on the problem. The calm eye of the tropical cyclone shares many qualitative characteristics with other vortical systems such as tornadoes, waterspouts, dust devils, and whirlpools. Given that many of these lack a change of phase of water (i.e., no clouds and diabatic heating are involved), it may be that the eye feature is a fundamental component to all rotating fluids. It has been hypothesized that supergradient wind flow (i.e., swirling winds generating stronger centrifugal 'force' than the local pressure gradient can support) present near the radius of maximum winds causes air to be centrifuged out of the eye into the eyewall, thus accounting for the subsidence in the eye. However, others found that the swirling winds within several tropical cyclones were within 1–4% of gradient

Figure 10 Eyewall of Hurricane Georges, 1945 UTC, 19 September 1998. (Photo courtesy of M. Black, NOAA/OAR/AOML Hurricane Research Division.)

balance. It may be thought that the amount of supergradient flow needed to cause such centrifuging of air is only on the order of a couple of percent and thus difficult to measure.

Another feature of tropical cyclones that probably plays a role in forming and maintaining the eye is the eyewall convection. As shown in **Figure 12**, convection in developing tropical cyclones is organized into

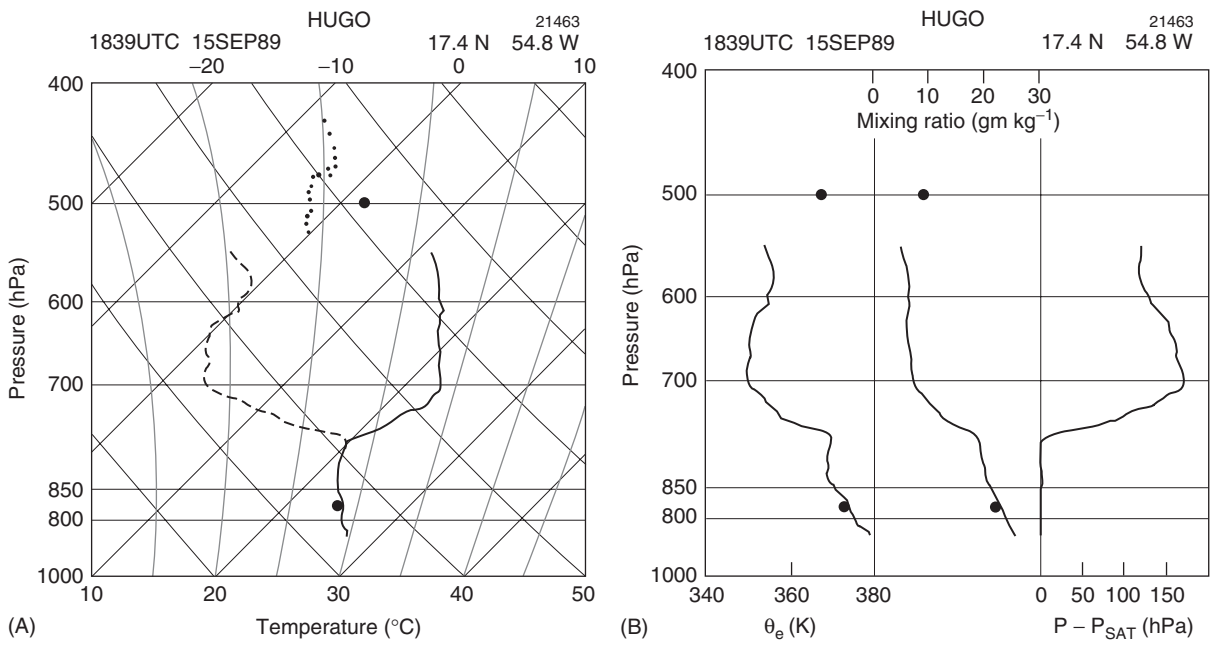

Figure 11 (A) Skew $T \lg p$ diagram of the eye sounding in Hurricane Hugo at 1839 UTC, 15 September 1989, 17.4° N, 54.8° W. Isotherms slope upward to the right; dry adiabats slope upward to the left; moist adiabats are nearly vertical curving to the left. Solid and dashed curves denote temperature and dew point, respectively. The smaller dots denote saturation points computed for the dry air above the inversion, and the two larger dots temperature observed at the innermost saturated point as the aircraft passed through the eyewall. (B) θ_e, water vapor mixing ratio, and saturation pressure difference, P-P_{SAT}, as functions of pressure at 2123 UTC. (Reproduced with permission from Willoughby HE (1998) Tropical cyclone eye thermodynamics. *Monthly Weather Review* 126: 3189–3211; © American Meteorological Society.)

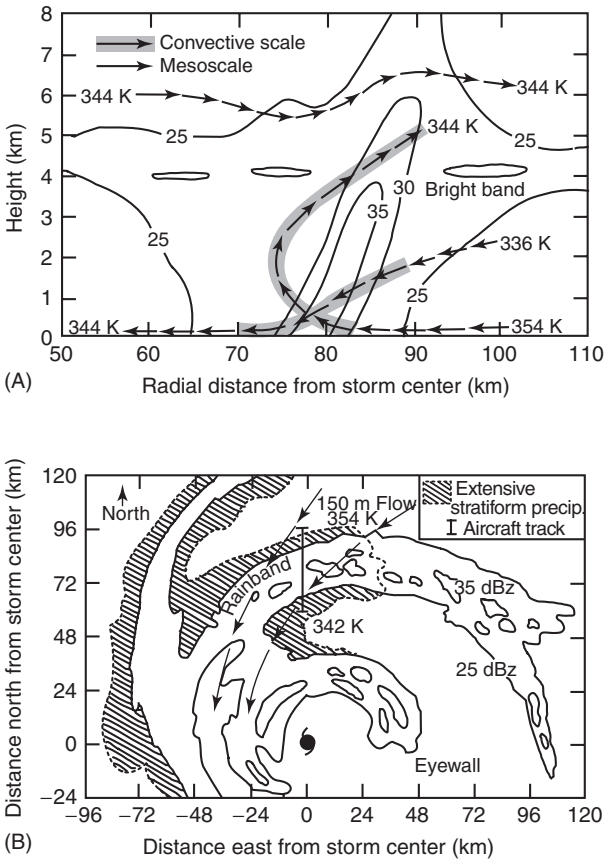

Figure 12 (A) Schematic of the rainband in radius–height coordinates. Reflectivity, θ_e, mesoscale (arrows), and convective scale motions are shown. (B) Plan view. Aircraft track, reflectivities, cells, stratiform precipitation, 150 m flow, and θ_e values are shown (Reproduced with permission from Barnes GM, Zipser EJ, Jorgensen DP, and Marks FD (1983) Mesoscale and convective structure of a hurricane rainband. *Journal of the Atmospheric Sciences* 40: 2125–2137; © American Meteorological Society.)

long, narrow rainbands that are oriented in the same direction as the horizontal wind. Because these bands seem to spiral into the center of a tropical cyclone, they are sometimes called spiral bands. The earliest radar observations of tropical cyclones detected these bands, which are typically 5–50 km wide and 100–300 km long. Along these bands, low-level convergence is a maximum, and therefore upper-level divergence is most pronounced. A direct circulation develops in which warm, moist air converges at the surface, ascends through these bands, diverges aloft, and descends on both sides of the bands. Subsidence is distributed over a wide area outside of the rainband, but is concentrated in the small inside area. As the air subsides, adiabatic warming takes place, and the air dries. Because subsidence is often concentrated on the inside of the band, the adiabatic warming is stronger inward from the band, causing a sharp contrast in pressure falls across the band since warm air is lighter than cold air. Because of the pressure falls on the inside, the tangential winds around the tropical cyclone increase, owing to an increased pressure gradient. Eventually, the band moves toward the center and encircles it and the eye and eyewall form.

The circulation in the eye is comparatively weak and, at least in the mature stage, thermally indirect (warm air descending), so it cannot play a direct role in the storm energy production. On the other hand, the temperature in the eye of many hurricanes exceeds that which can be attained by any conceivable moist adiabatic ascent from the sea surface, even accounting for the additional entropy (positive potential temperature, θ, anomaly) owing to the low surface pressure in the eye (the lower the pressure, the higher the θ at a given altitude and temperature). Thus, the observed low central pressure of the storm is not consistent with that calculated hydrostatically from the temperature distribution created when a sample of air is lifted from a state of saturation at sea surface temperature and pressure. The thermal wind balance restricts the amount of warming that can take place. In essence, the rotation of the eye at each level is imparted by the eyewall, and the pressure drop from the outer to the inner edge of the eye is simply that required by gradient balance.

Because the eyewall azimuthal velocity decreases with height, the radial pressure drop decreases with altitude, requiring, through the hydrostatic equation, a temperature maximum at the storm center. Thus, given the swirling velocity of the eyewall, the steady-state eye structure is largely determined. The central pressure, which is estimated by integrating the gradient balance equation inward from the radius of maximum winds, depends on the assumed radial profile of azimuthal wind in the eye.

In contrast, the eyewall is a region of rapid variation of thermodynamic variables. As shown in **Figure 13**, the transition from the eyewall cloud to the nearly cloud-free eye is often so abrupt that it has been described as a form of atmospheric front. Early studies were the first to recognize that the flow under the eyewall cloud is inherently frontogenetic. The eyewall is the upward branch of the secondary circulation and a region of rapid ascent that, together with slantwise convection, leads to the congruence of angular momentum and moist entropy (θ_e) surfaces. Hence, the three-dimensional vorticity vectors lie on θ_e surfaces, so that the moist PV vanishes. As the air is saturated, this in turn implies, through the invertibility principle applied to flow in gradient and hydrostatic balance, that the entire primary circulation may be deduced from the radial distribution of θ_e in the boundary layer and the distribution of vorticity at the tropopause.

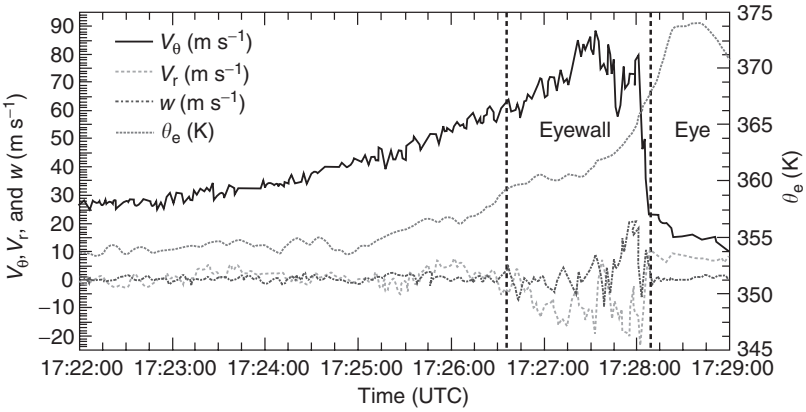

Figure 13 Time series plots of tangential wind (V_θ), radial wind (V_r), vertical velocity (w), and θ_e in Hurricane Hugo at 1721–1730 UTC, 15 September 1989. The aircraft flight track was at 450 m. Thick dashed vertical lines denote the width of the eyewall reflectivity maximum at low levels.

In the classic semigeostrophic theory of deformation-induced frontogenesis, the background geostrophic deformation flow provides the advection of temperature across surfaces of absolute momentum that drives the frontogenesis, whereas in the hurricane eyewall, surface friction provides the radial advection of entropy across angular momentum surfaces. Also note that the hurricane eyewall is not necessarily a front in surface temperature, but instead involves the θ_e distribution, which is related directly to density in saturated air.

There is likely a two-stage process in eye formation. The amplification of the primary circulation is strongly frontogenetic and results, in a comparatively short time, in frontal collapse at the inner edge of the eyewall. (Frontal collapse is an increase in the horizontal gradient of an airmass property, principally density, and the development of the accompanying features of the wind field through the secondary circulation that typify a front.) The frontal collapse leads to a dramatic transition in the storm dynamics. While the tropical cyclone inner core is dominated by axisymmetric motions, hydrodynamic instabilities are potential sources of asymmetric motions within the core. In intense tropical cyclones the wind profile inside the eye is often U-shaped, in the sense that the wind increases outwards more rapidly than linearly with radius (**Figure 13**). The strong cyclonic shear just inside the eyewall may result in a local maximum of absolute vorticity or angular momentum, so that the profile may actually become barotropically unstable. (This refers to the hydrodynamic instability arising from certain distributions of vorticity in a two-dimensional nondivergent flow. It is an inertial instability in that kinetic energy is the only form of energy transferred between the current and perturbation. A well-known necessary condition for barotropic instability is that the basic state vorticity gradient must have both signs in the domain of interest.) This instability leads to frontal collapse as a result of radial diffusion of momentum into the eye, and also may explain the 'polygonal eyewalls' where the eyewall appear on radar to be made up of a series of line segments rather than as a circle. It may also explain intense mesoscale vortices observed in the eyewalls of Hurricanes Hugo of 1989 and Andrew of 1992.

Once the radial turbulent diffusion of momentum driven by the instability of the primary circulation becomes important, it results in a mechanically induced, thermally indirect (warm air sinking) component of the secondary circulation in the eye and eyewall. Such a circulation raises the vertically averaged temperature of the eye beyond its value in the eyewall and allows for an amplification of the entropy distribution. Feedbacks with the surface fluxes then allow the boundary layer entropy to increase and result in a more rapid intensification of the swirling wind. Thus, the frontal collapse of the eyewall is an essential process in the evolution of tropical cyclones. Without it, amplification of the temperature distribution relies on external influences, and intensification of the wind field is slow. Once it has taken place, the mechanical spinup of the eye allows the temperature distribution to amplify without external influences and, through positive feedback with surface fluxes, allows the entropy field to amplify and the swirling velocity to increase somewhat more rapidly.

Outer structure and rainbands The axisymmetric core is characteristically surrounded by a less symmetric outer vortex that diminishes into the synoptic 'environment'. In the lower troposphere, the cyclonic circulation may extend more than 1000 km from the center. As evident in **Figure 14** the boundary between

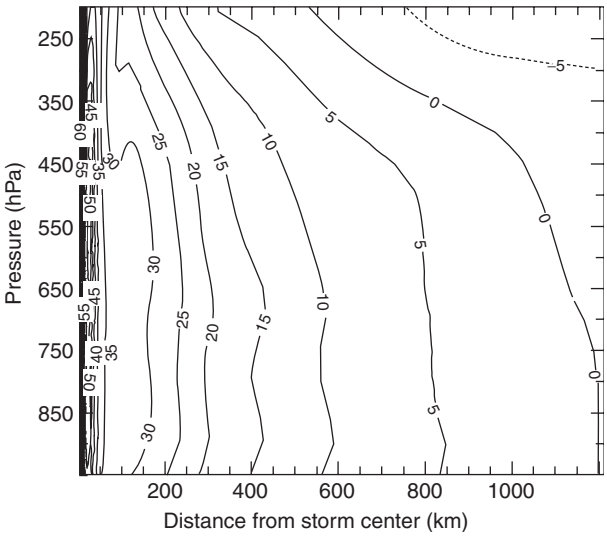

Figure 14 Vertical cross-section of the azimuthal mean tangential wind for Hurricane Gloria on 24 September 1985. Anticyclonic contours are dashed. (Reproduced with permission from Franklin JL, Lord SJ, Feuer SE, and Marks FD (1993) The kinematic structure of Hurricane Gloria (1985) determined from nested analyses of dropwindsonde and Doppler radar data. *Monthly Weather Review* 121: 2433–2451; © American Meteorological Society.)

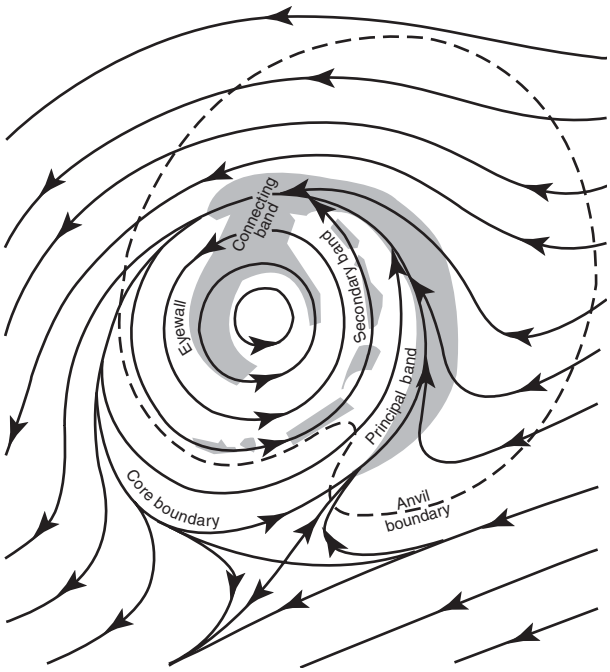

Figure 15 Schematic representation of the stationary band complex, the entities that compose it, and the flow in which it is embedded. (Reproduced with permission from Willoughby HE, Marks FD, and Feinberg RJ (1984) Stationary and moving convective bands in hurricanes. *Journal of the Atmospheric Sciences* 41: 3189–3211; © American Meteorological Society.)

cyclonic and anticyclonic circulation slopes inward with increasing height, so that the circulation in the upper troposphere is primarily anticyclonic, except near the core. In the outer vortex, there are no scale separations between the primary and the secondary circulations, the asymmetric motions, or the vortex translation, as they are roughly all of the same magnitude. The asymmetric flows in this region control the vortex motion and sustain an eddy convergence of angular momentum and moisture toward the center. Interactions between the symmetric motions of the inner core with the more asymmetric motions in the outer portion of the storm are the key to improved forecasts of tropical cyclone track and intensity.

Spiral bands of precipitation characterize radar and satellite images in this region of the storm (**Figures 1** and **15**). As seen in **Figures 8**, **12** and **15**, radar reflectivity patterns in tropical cyclones provide a good means for flow visualization, although they represent precipitation, not winds. Descending motion occupies precipitation-free areas, such as the eye. The axis of the cyclone's rotation lies near the center of the eye. The eyewall surrounds the eye. In intense hurricanes, it may contain reflectivities as high as 50 dBZ equivalent to rainfall rates of $74\,\mathrm{mm\,h^{-1}}$. (1 dBz $10\lg Z$, where Z is equivalent radar reflectivity factor ($\mathrm{mm^6\,m^{-3}}$).) Less extreme reflectivities, 40 dBZ ($13\,\mathrm{mm\,h^{-1}}$), characterize most convective rainfall in the eyewall and spiral bands. The vertical velocities (both up and downdrafts) in convection with highest reflectivity may reach $25\,\mathrm{m\,s^{-1}}$, but typical vertical velocities are $<5\,\mathrm{m\,s^{-1}}$. Such intense convection occupies less than 10% of the tropical cyclone's area. Outside convection, reflectivities are still weaker, 30 dBZ, equivalent to a $2.4\,\mathrm{mm\,h^{-1}}$ rain rate. This 'stratiform rain', denoted by a distinct reflectivity maximum or 'bright band' at the altitude of the 0°C isotherm, falls out of the anvil cloud that grows from the convection. The spiral bands tend to lie along the friction-layer wind that spirals inward toward the eyewall (**Figure 12**).

Many aspects of rainband formation, dynamics, and interaction with the symmetric vortex are still unresolved. The trailing spiral shape of bands and lanes arises because the angular velocity of the vortex increases inward and deforms them into equiangular spirals. In the vortex core, air remains in the circulation for many orbits of the center, while outside the core, the air passes through the circulation in less than the time required for a single orbit. As the tropical cyclone becomes more intense, the inward ends of the bands approach the center less steeply approximating arcs of circles. Some bands appear to move outward, while others maintain a fixed location relative to the translating center.

As shown in **Figure 15**, motion of the vortex through its surroundings may cause one stationary band, called the principal band, to lay along a convergent streamline asymptote that spirals into the core. A tropical cyclone advected by midlevel steering with westerly shear moves eastward through surrounding air at low levels. Thus, the principal band may be like a bow wave, caused by the displacement of the environmental air on the eastern side of the vortex. Its predominant azimuthal wavenumber is one.

Moving bands, and other convective features, are frequently associated with cycloidal motion of the tropical cyclone center, and intense asymmetric outbursts of convection are observed to displace the tropical cyclone center by tens of kilometers. The bands observed by radar are often considered manifestations of internal gravity waves, but these waves can exist only in a band of Doppler-shifted frequencies between the local inertia frequency (defined as the sum of the vertical component of the earth's inertial frequency, f, and the local angular velocity of the circulation, V/r) and the Brunt–Väisälä frequency (i.e., the natural gravity wave frequency, the square root of the static stability defined as $(g/\theta)\,\partial\theta/\partial z$). Only two classes of trailing-spiral, gravity wave solutions lie within this frequency band: (1) waves with any tangential wavenumber that move faster than the swirling wind, and (2) waves with tangential wavenumber ≥ 2 that move slower than the swirling wind.

Bands moving faster than the swirling wind with outward phase propagation are observed by radar. They are more like squall lines than linear gravity waves. Waves moving slower than the swirling wind propagate wave energy and anticyclonic angular momentum inward, grow at the expense of the mean-flow kinetic energy, and reach appreciable amplitude if they are excited at the periphery of the tropical cyclone. Alternate explanations for these inward-propagating bands involve filamentation of vorticity from the tropical cyclone environment, asymmetries in the radially shearing flow of the vortex, and high-order vortex Rossby waves. Detailed observations of the vortex-scale rainband structure and windfield are necessary to determine which mechanisms play a role in rainband development and maintenance.

While the evolution of the inner core is dominated by interactions between the primary, secondary, and track-induced wavenumber-one circulation, there is some indication that the local convective circulations in the rainbands may impact on intensity change. Although precipitation in some bands is largely stratiform, condensation in most bands tends to be concentrated in convective cells rather than spread over wide mesoscale areas. As shown in **Figure 12**, convective elements form, move through the bands, and dissipate as they move downwind. Doppler radar observations indicate that the roots of the updrafts lie in convergence between the low-level radial inflow and gust fronts produced by convective downdrafts. This convergence may occur on either side of the band. A 20 K decrease in low-level θ_e was observed in a rainband downdraft, suggesting that the draft acts as a barrier to inflow. This reduction in boundary layer energy may be advected near the center, inhibit convection, and thereby alter storm intensity.

Motion

Tropical cyclone motion is the result of a complex interaction between a number of internal and external influences. Environmental steering is typically the most prominent external influence on a tropical cyclone, accounting for as much as 70–90% of the motion. Theoretical studies show that in the absence of environmental steering, tropical cyclones move poleward and westward owing to internal influences.

Accurate determination of tropical cyclone motion requires accurate representation of interactions that occur throughout the depth of the troposphere on a variety of scales. Observations spurred improved understanding of how tropical cyclones move using simple barotropic and more complex baroclinic models. To first order, the storm moves with some layer average of the lower-tropospheric environmental flow: the translation of the vortex is roughly equal to the speed and direction of the basic 'steering' current. However, the observations show that tropical cyclone tracks deviate from this simple steering flow concept in a subtle and important manner. Several physical processes may cause such deviations. The approach in theoretical and modeling of tropical cyclones has been to isolate each process in a systematic manner to understand the magnitude and direction of the track deviation caused by each effect. The β effect opposes the advection of relative vorticity through the differential advection of the Earth's vorticity, f, that slows the advection of the disturbance. (The β effect is the asymmetric vorticity advection around the vortex caused by the latitudinal gradient of f; $\beta = 2\Omega\cos\phi$. β has a maximum value at the Equator (i.e., $2.289 \times 10^{-11}\,\mathrm{s}^{-1}$) and becomes zero at the pole.) Models that are more complete describe not only the movement of the vortex but also the accompanying wavenumber-one asymmetries that develop owing to the differential advection of f on the east and west side of the vortex. It was also discovered that the role of meridional and zonal gradients of the environmental flow could add greatly to the complexity even in the barotropic evolution of a vortex. Hence, the evolution

of the movement depends on not only the relative vorticity gradient and on shear of the environment but also the structure of the vortex itself.

Generally, the propagation vector of these model baroclinic vortices is very close to that expected from a barotropic model initialized with the vertically integrated environmental wind. An essential feature in baroclinic systems is the relative vorticity advection through the storm center, where the vertical structure of the tropical cyclone produces a tendency for the low-level vortex to move slower than the simple propagation of the vortex due to β. Vertical shear plays an important factor in determining what the relative flow is, though there is no unique relation between the shear and storm motion. Diabatic heating effects also alter this flow and change the propagation velocity. Thus, tropical cyclone motion is primarily governed by the dynamics of the low-level cyclonic circulation; however, the addition of observations of the upper-level structure may alter this finding.

Interaction with the Atmospheric Environment

A consensus exists that small vertical shear of the environmental wind and lateral eddy imports of angular momentum are favorable to tropical cyclone intensification. The inhibiting effect of vertical shear in the environment on tropical cyclone intensification is well known from climatology and forecasting experience. Exposure of the low-level circulation from under the tropical cyclones large area of cirrus (central dense overcast) in satellite imagery is universally recognized as a symptom of shear, and as an indication of a nonintensification or weakening. Nevertheless, the detailed dynamics of a vortex in shear has been the topic of surprisingly little study, probably because while the effect is a reliable basis for practical forecasting, it is difficult to measure and model.

In contrast, the positive effect of eddy momentum imports at upper levels has received extensive study. Modeling studies with composite initial conditions show that eddy momentum fluxes can intensify a tropical cyclone even when other conditions are neutral or unfavorable. It has been shown theoretically that momentum imports can form a tropical cyclone in an atmosphere with no buoyancy. Statistical analysis of tropical cyclones reveals a clear relationship between angular momentum convergence and intensification, but only after the effects of shear and SST variations are accounted for. Such interactions occur frequently (35% of the time, defined by eddy angular momentum flux convergence exceeding $10\,\mathrm{m\,s^{-1}\,day^{-1}}$), and likely represent the more common upper-boundary interaction for tropical cyclones. Frequently they are accompanied by eyewall cycles and dramatic intensity changes.

The environmental flows that favor intensification, and presumably inward eddy momentum fluxes, usually involve interaction with a synoptic-scale cyclonic feature, such as a midlatitude upper-level trough or PV anomaly. Given the interaction of an upper-level trough and the tropical cyclone, the exact mechanism for intensification is still uncertain. The secondary circulation response to momentum and heat sources is very different. Upper-tropospheric momentum sources can influence the core directly. As can be seen in **Figure 16**, large inertial stability in the lower troposphere protects the mature tropical cyclone core from direct influence by momentum sources (inertial stability is a measure of the resistance to horizontal displacements, based on the conservation of angular momentum for a vortex in gradient balance, and is defined as $(f+\zeta)(f+V^2/r)$, where ζ is the relative vorticity, V the axial wind velocity, f the Coriolis parameter, and r the radius from the storm center); however, the inertial stability in the upper troposphere is smaller and a momentum source can induce an outflow jet with large radial extent just below the tropopause. If the eyewall updraft links to the direct circulation at the entrance region of the jet, as shown in **Figures 17C** and **17D**, the exhaust outflow is unrestricted. The important difference between heat and momentum sources is that the roots of the diabatically induced updraft must be in the inertially

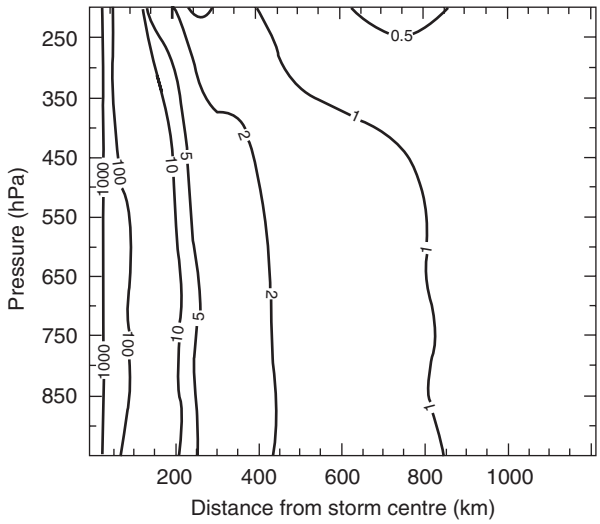

Figure 16 Axisymmetric mean inertial stability for Hurricane Gloria on 24 September 1985. Contours are shown as multiples of f^2 at the latitude of Gloria's center. (Reproduced with permission from Franklin JL, Lord SJ, Feuer SE, and Marks FD (1993) The kinematic structure of Hurricane Gloria (1985) determined from nested analyses of dropwindsonde and Doppler radar data. *Monthly Weather Review* 121: 2433–2451; © American Meteorological Society.)

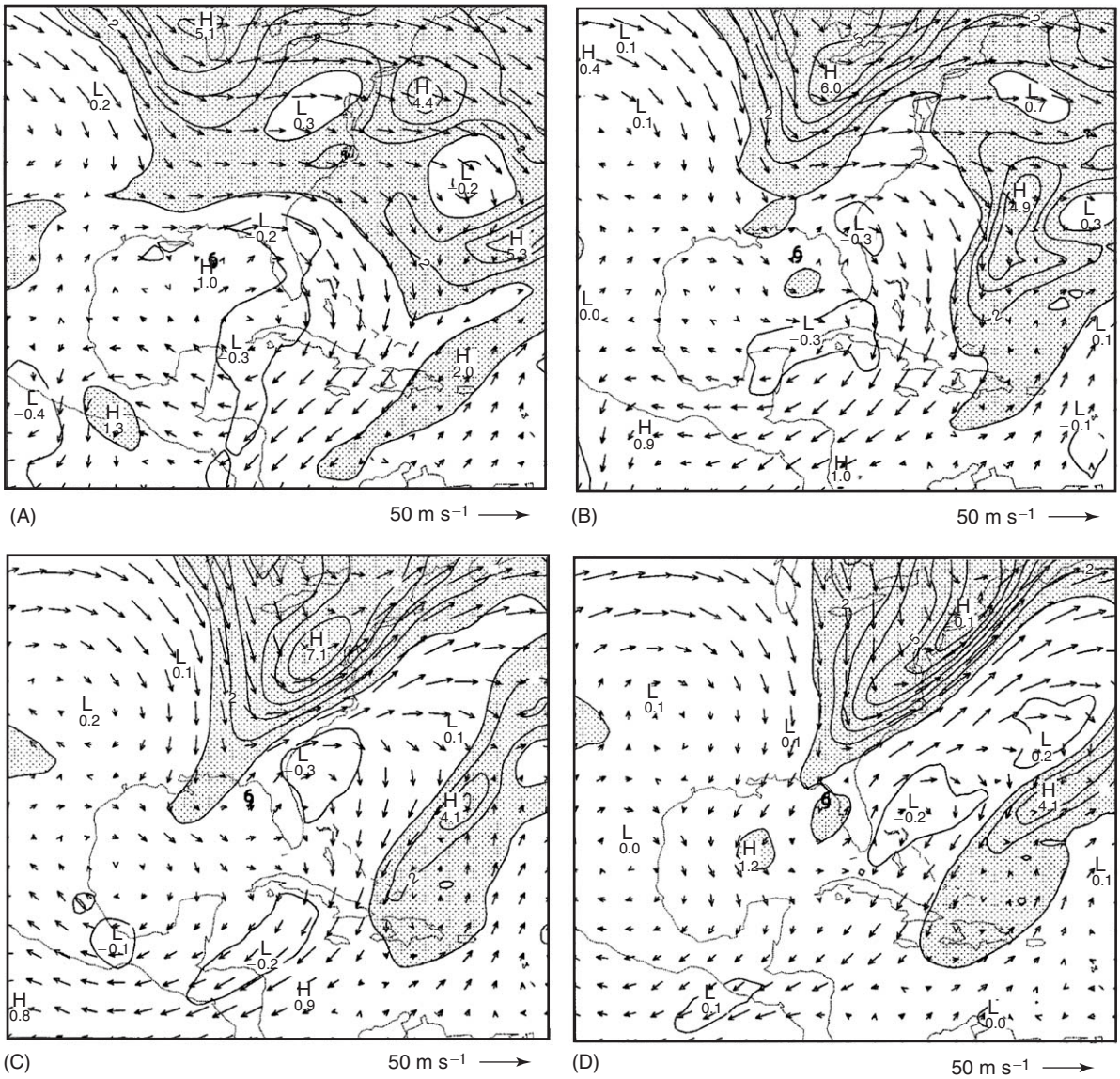

Figure 17 Wind vectors and potential vorticity on the 345 K isentropic surface at (A) 1200 UTC 30 August; (B) 0000 UTC 31 August; (C) 1200 UTC 31 August; and (D) 0000 UTC 1 September 1985. PV increments are 1 PVU and values >1 PVU are shaded. Wind vectors are plotted at 2.25° intervals. The 345 K surface is approximately 200 hPa in the hurricane environment and ranges from 240 to 280 hPa at the storm center. The hurricane symbol denotes the location of Hurricane Elena. (Reproduced with permission from Molinari J, Skubis S, and Vollaro D (1995) External influences on hurricane intensity. 3. Potential vorticity structure. *Journal of the Atmospheric Sciences* 52: 3593–3606; © American Meteorological Society.)

stiff lower troposphere, but the outflow jet due to a momentum–flux convergence can be confined to the inertially stable upper troposphere. Momentum forcing does not spin the vortex up directly. It makes the exhaust flow stronger and reduces local compensating subsidence in the core, thus cooling the upper troposphere and destabilizing the sounding. The cooler upper troposphere leads to less thermal-wind shear and a weaker upper anticyclone.

A two-dimensional balanced approach provides reasonable insight into the nature of the tropical cyclone intensification as a trough approaches. Isentropic PV analysis (**Figure 17**), which express the problem in terms of a quasi-conserved variable in three dimensions, are used to describe various processes in idealized tropical cyclones with considerable success. The eddy heat and angular momentum fluxes are related to changes in the isentropic PV through their contribution to the eddy flux of PV.

It has been suggested that outflow-layer asymmetries, as in **Figure 17**, and their associated circulations could create a mid- or lower-tropospheric PV

maximum outside the storm core, either by creating breaking PV waves on the mid-tropospheric radial PV gradient (**Figure 3**) or by diabatic heating. It has been shown that filamentation of any such PV maximum in the 'surf zone' outside the tropical cyclone core (the sharp radial PV gradient near 100 km radius) produces a feature much like a secondary wind maximum, which was apparent in the PV fields of Hurricane Gloria in **Figure 3**. These studies thus provide mechanisms by which outflow-layer asymmetries could bring about a secondary wind maximum.

An alternative argument has been proposed for storm reintensification as a 'constructive interference without phase locking', as shown in **Figure 18**. As the PV anomalies come within the Rossby radius of deformation, the pressure and wind perturbations associated with the combined anomalies are greater than when the anomalies are apart, even though the PV magnitudes are unchanged. The perturbation energy comes from the basic-state shear that brought the anomalies together. However, constructive interference without some additional diabatic component cannot account for intensification. It is possible that intensification represents a baroclinic initiation of a wind-induced surface heat exchange. By this mechanism, the constructive interference induces stronger surface wind anomalies, which produce larger surface moisture fluxes and thus higher surface moist enthalpy. This feeds back through the associated convective heating to produce a stronger secondary circulation and thus stronger surface winds. The small effective static stability in the saturated, nearly moist neutral storm core ensures a deep response, so that even a rather narrow upper trough can initiate this feedback process. The key to this mechanism is the direct influence of the constructive interference on the surface wind field, as that controls the surface flux of moist enthalpy.

Interaction with the Ocean

As pointed out in the climatology section, preexisting SSTs $>26^\circ$C are a necessary but insufficient condition for tropical cyclogenesis. Once the tropical cyclone develops and translates over the tropical oceans, statistical models suggest that warm SSTs describe a large fraction of the variance (40–70%) associated with the intensification phase of the storm. However, these predictive models do not account either for the oceanic mixed layers having temperatures of 0.5–1°C cooler than the relatively thin SST over the upper meter of the ocean or horizontal advective tendencies by basic-state ocean currents such as the Gulf Stream and warm core eddies. Thin layers of warm SST are well mixed with the underlying cooler mixed layer water well in advance of the storm where winds are a few meters per second, reducing SST as the storm approaches. However, strong oceanic baroclinic features advecting deep, warm oceanic mixed layers represent moving reservoirs of high-heat-content water available for the continued development and intensification phases of the tropical cyclone. Beyond a first-order description of the lower boundary providing the heat and moisture fluxes derived from low-level convergence, little is known about the complex boundary layer interactions between the two geophysical fluids.

One of the more apparent aspects of the atmospheric–oceanic interactions during tropical cyclone passage is the upper-ocean cooling as manifested in the SST (and mixed-layer temperature) decrease starting just in back of the eye. As seen in **Figure 19**, ocean mixed-layer temperature profiles acquired during the passage of several tropical cyclones revealed a crescent-shaped pattern of upper-ocean cooling and mixed-layer depth changes, which indicated a rightward bias in the mixed-layer temperature response with cooling by 1–5°C extending from the right rear quadrant of the storm into the wake regime. These SST decreases are observed through satellite-derived SST images, such as that of the post-Hurricane Bonnie SST (**Figure 20**), which are indicative of mixed-layer depth changes due to stress-induced turbulent mixing in the front of the storm and shear-induced mixing between the layer and thermocline in the rear half of the storm. The mixed-layer cooling represents the thermodynamic and dynamic response to the strong wind that typically accounts for 75–85% of the ocean heat loss, compared with the 15–25% caused by surface latent and sensible heat fluxes from the ocean to the atmosphere. Thus, the upper ocean's heat content for tropical cyclones is governed not solely by SST; rather, it is the mixed-layer depths and temperatures that are significantly affected along the lower boundary by the basic state and transient currents.

Recent observational data have shown that the horizontal advection of temperature gradients by basic state currents in a warm core ring affected the mixed-layer heat and mass balance, suggesting the importance of these warm oceanic baroclinic features. In addition to enhanced air–sea fluxes, warm temperatures ($>26^\circ$C) may extend to 80–100 m in warm core rings, significantly impacting the mixed-layer heat and momentum balance. That is, strong current regimes (1–2 m s^{-1}) advecting deep, warm upper-ocean layers not only represent deep reservoirs of high-heat-content water with an upward heat flux, but transport heat from the tropics to the subtropical and polar regions as part of the annual cycle. Thus, the basic state of the mixed layer and the subsequent response represent an evolving three-dimensional process with

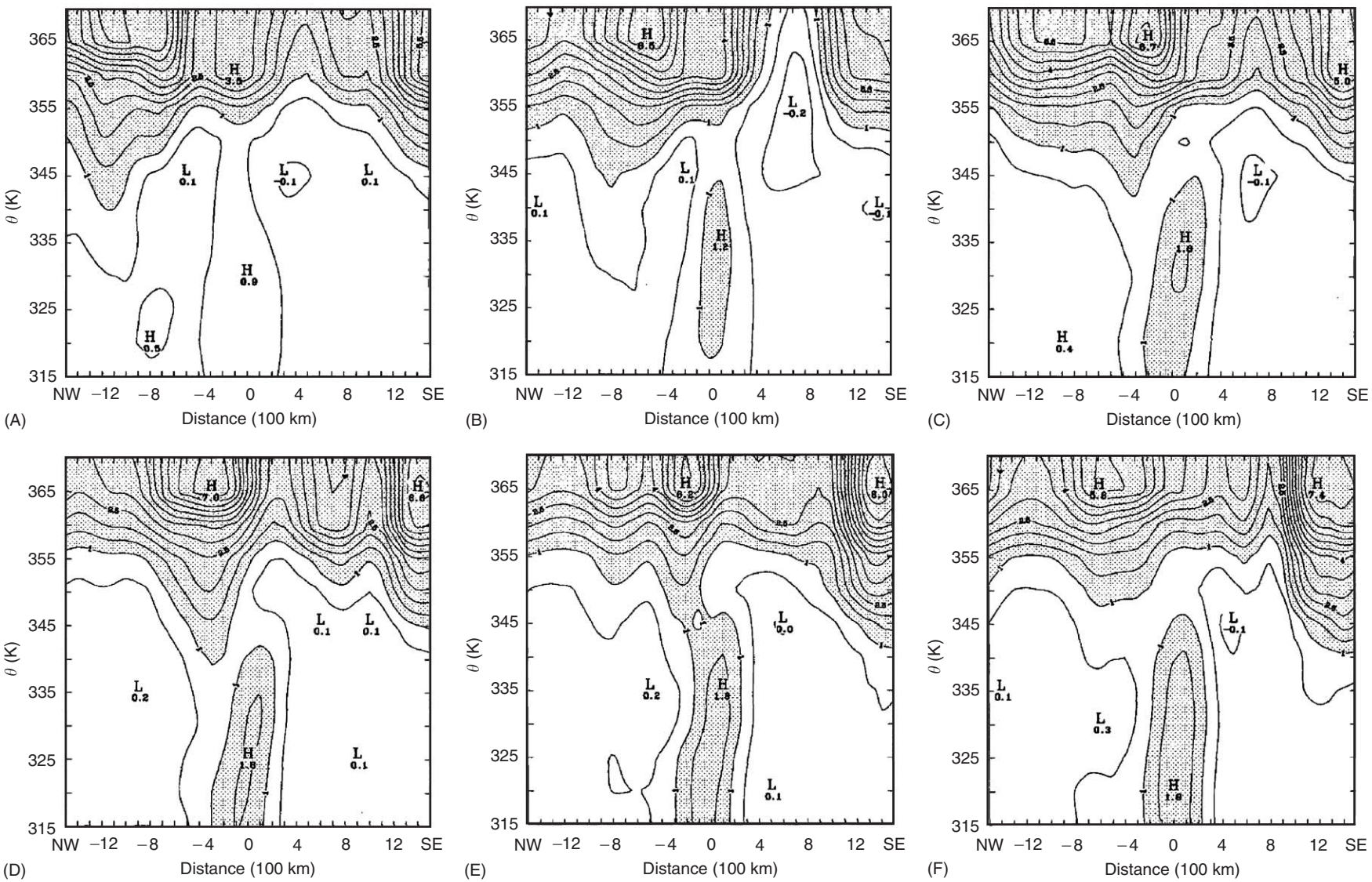

Figure 18 Cross-sections of potential vorticity from north-west (left) to south-east (right) through the observed center of Hurricane Elena at the same times as in **Figure 17**, plus (A) at 0000 UTC 30 August and (**F**) 1200 UTC 1 September. Increment is 0.5 PVU and shading above 1 PVU. (Reproduced with permission from Molinari J, Skubis S, and Vollaro D (1995) External influences on hurricane intensity. 3. Potential vorticity structure. *Journal of the Atmospheric Sciences* 52: 3593–3606; © American Meteorological Society.)

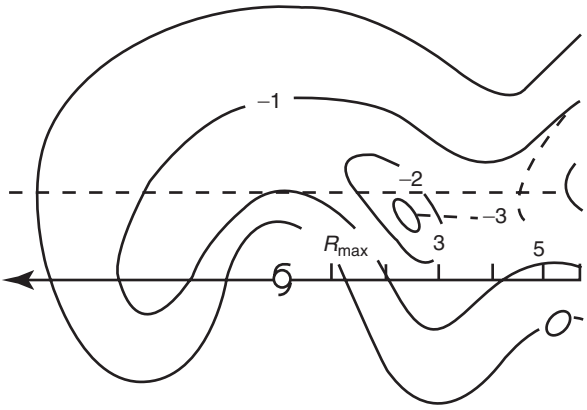

Figure 19 Schematic SST change (°C) induced by a hurricane. The distance scale is indicated in multiples of the radius of maximum wind. Storm motion is to the left. The horizontal dashed line is at 1.5 times the radius of maximum wind. (Reproduced with permission from Black PG, Elsberry RL, and Shay LK, (1988) Airborne surveys of ocean current and temperature perturbations induced by hurricanes. *Advances in Underwater Technology, Ocean Science and Offshore Engineering* 16: 51–58; © Society for Underwater Technology (Graham & Trotman).)

surface fluxes, vertical shear across the entrainment zone, and horizontal advection. Simultaneous observations in both fluids are lacking over these baroclinic features prior, during, and subsequent to tropical cyclone passage, and are crucially needed to improve our understanding of the role of lower boundary in intensity and structural changes to intensity change.

In addition, wave height measurements and current profiles revealed the highest waves and largest fetches to the right side of the storm where the maximum mixed-layer changes occurred. Mean wave-induced currents were in the same direction as the steady mixed-layer currents, modulating vertical current shears and mixed-layer turbulence. These processes feed back to the atmospheric boundary layer by altering the surface roughness and hence the drag coefficient. However, little is known about the role of strong surface waves on the mixed-layer dynamics, and their feedback to the atmospheric boundary layer under tropical cyclone force winds by altering the drag coefficient.

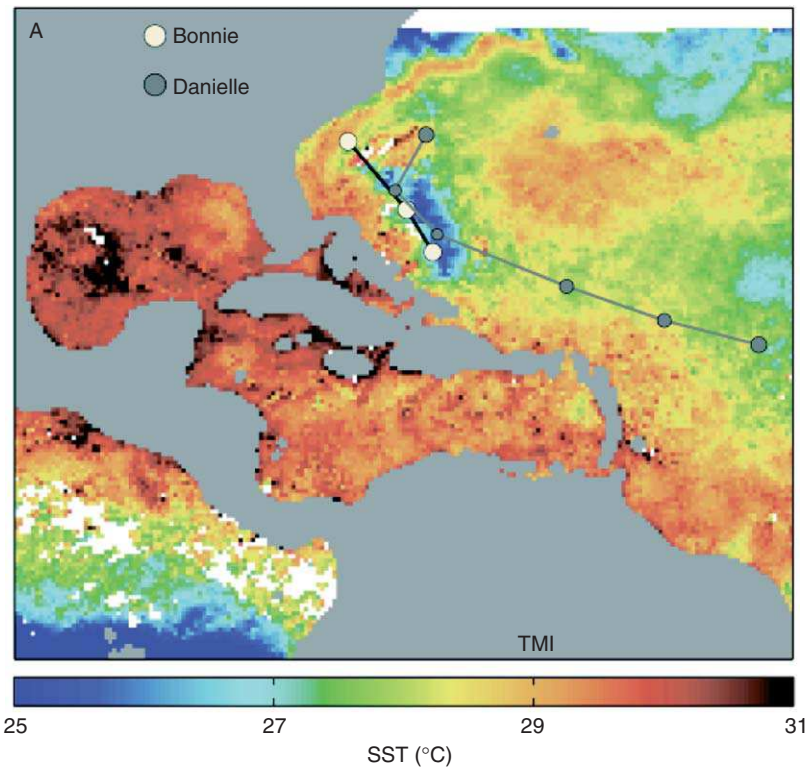

Figure 20 Cold wake produced by Hurricane Bonnie for 24–26 August 1998, as seen by the NASA TRMM satellite Microwave Imager (TMI). Small white patches are areas of persistent rain over the 3-day period. White dots show Hurricane Bonnie's daily position from 24 to 26 August. Gray dots show the later passage of Hurricane Danielle from 27 August to 1 September. Danielle crossed Bonnie's wake on 29 August and its intensity dropped. (Reproduced with permission from Wentz FJ, Gentemann C, Smith D, and Chelton D (2000) Satellite measurements of sea surface temperature through clouds. *Science* 288: 847–850; © owned by the American Geophysical Union (http:/ www.sciencemag.org).)

Tropical Cyclone Rainfall

Precipitation in tropical cyclones can be separated into either convective or stratiform regimes. Convective precipitation occurs primarily in the eyewall and rainbands, producing rains $>25\,\mathrm{mm\,h^{-1}}$ over small areas. However, observations suggest that only 10% of the total rain area is comprised of these convective rain cores. The average core is 4 km in radius (area of $50\,\mathrm{km^2}$), and has a relatively short lifetime, with only 10% lasting longer than 8 min (roughly the time a 1 mm diameter raindrop takes to fall from the mean height of the 0°C isotherm at terminal velocity). The short life cycle of the cores and the strong horizontal advection produce a well-mixed and less asymmetric precipitation pattern in time and space. Thus, over 24 h the inner core of a tropical cyclone as a whole produces 1–2 cm of precipitation over a relatively large area, and 10–20 cm in the core. After landfall, orographic forcing can anchor heavy precipitation to a local area for an extended time. Additionally, midlatitude interaction with a front or upper-level trough can enhance precipitation, producing a distortion of the typical azimuthally uniform precipitation distribution.

Energetics

Energetically, a tropical cyclone can be thought of, to a first approximation, as a heat engine; obtaining its heat input from the warm, humid air over the tropical ocean, and releasing this heat through the condensation of water vapor into water droplets in deep thunderstorms of the eyewall and rainbands, then giving off a cold exhaust in the upper levels of the troposphere ($\sim$12 km up). One can look at the energetics of a tropical cyclone in two ways: (1) the total amount of energy released by the condensation of water droplets or (2) the amount of kinetic energy generated to maintain the strong swirling winds of the hurricane. It turns out that the vast majority of the heat released in the condensation process is used to cause rising motions in the convection, and only a small portion drives the storm's horizontal winds.

Using the first approach, we assume an average tropical cyclone produces $1.5\,\mathrm{cm\,day^{-1}}$ of rain inside a circle of radius 665 km. Converting this to a volume of rain gives $2.1\times10^{16}\,\mathrm{cm^3\,day^{-1}}$ ($1\,\mathrm{cm^3}$ of rain weighs 1 g). The energy released through the latent heat of condensation to produce this amount of rain is $5.2\times10^{19}\,\mathrm{J\,day^{-1}}$ or $6.0\times10^{14}\,\mathrm{W}$, which is equivalent to 200 times the worldwide electrical generating capacity.

Under the second approach we assume that for a mature hurricane, the amount of kinetic energy generated is equal to that being dissipated due to friction. The dissipation rate per unit area is air density times the drag coefficient times the wind speed cubed. Assuming an average wind speed for the inner core of the hurricane of $40\,\mathrm{m\,s^{-1}}$ winds over a 60-km radius, the wind dissipation rate (wind generation rate) would be $1.5\times10^{12}\,\mathrm{W}$. This is equivalent to about half the worldwide electrical generating capacity.

Either method suggests hurricanes generate an enormous amount of energy. However, they also imply that only about 2.5% of the energy released in a hurricane by latent heat released in clouds actually goes to maintaining the hurricane's spiraling winds.

Tropical Cyclone-Related Hazards

In the coastal zone, extensive damage and loss of life are caused by the storm surge (a rapid, local rise in sea level associated with storm landfall), heavy rains, strong winds, and tropical cyclone-spawned severe weather (e.g. , tornadoes). The continental United States currently averages nearly $5 billion (in 1998 dollars) annually in tropical-cyclone-caused damage, and this is increasing, owing to growing population and wealth in the vulnerable coastal zones.

Before 1970, large loss of life stemmed mostly from storm surges. The height of storm surges varies from 1 to 2 m in weak systems to more than 6 m in major hurricanes that strike coastlines with shallow water offshore. The storm surge associated with Hurricane Andrew (1992) reached a height of about 5 m, the highest level recorded in south-east Florida. Hurricane Hugo's (1989) surge reached a peak height of nearly 6 m about 20 miles north-east of Charleston, South Carolina, and exceeded 3 m over a length of nearly 180 km of coastline. In recent decades, large loss of life due to storm surges in the United States has become less frequent because of improved forecasts, fast and reliable communications, timely evacuations, a better-educated public, and a close working relationship between the National Hurricane Center (NHC), local weather forecast offices, emergency managers, and the media. Luck has also played a role, as there have been comparatively few landfalls of intense storms in populous regions in the last few decades. The rapid growth of coastal populations and the complexity of evacuation raises concerns that another large storm surge disaster might occur along the eastern or Gulf Coast shores of the United States.

In regions with effectively enforced building codes designed for hurricane conditions, wind damage is typically not so lethal as the storm surge, but it affects a much larger area and can lead to large economic loss. For instance, Hurricane Andrew's winds produced over $25 billion in damage over southern Florida and Louisiana. Tornadoes, although they occur in many

hurricanes that strike the United States, generally account for only a small part of the total storm damage.

While tropical cyclones are most hazardous in coastal regions, the weakening, moisture-laden circulation can produce extensive, damaging floods hundreds of miles inland long after the winds have subsided below hurricane strength. In recent decades, many more fatalities in North America have occurred from tropical-cyclone-induced inland flash flooding than from the combination of storm surge and wind. For example, although in 1972 the deaths from storm surge and wind along the Florida coast from Hurricane Agnes were minimal, inland flash flooding caused more than 100 deaths over the north-eastern United States. More recently, rains from Hurricane Mitch in 1998 killed at least 10 000 people in Central America, the majority after the storm had weakened to tropical storm strength. An essential difference in the threat from flooding rains, compared with that from wind and surge, is that the rain amount is not tied to the strength of the storm's winds. Hence, any tropical disturbance, from depression to major hurricane, is a major rain threat.

Over each ocean basin that experiences tropical cyclones, the poleward movement of a tropical cyclone into the midlatitudes is normally associated with the decay stage of its life cycle. However, these systems can develop into fast-moving and rapidly developing extratropical cyclones that contain tropical cyclone-force winds. These transforming tropical cyclones may accelerate from forward speeds of $5\,\mathrm{m\,s^{-1}}$ in the tropics to $>20\,\mathrm{m\,s^{-1}}$ in the midlatitudes. They pose a serious threat and forecast problem to maritime activities and shore locations over wide geographic regions that do not normally experience such conditions. A common problem in all these regions is the difficult challenge of predicting accurately the track, intensity, and impacts of these rapidly changing systems after advisories have been discontinued by the tropical cyclone forecast center. Forecasters responsible for producing warnings and advisories during extratropical transition are faced with the potential for large amounts of precipitation, continued high wind speed, and generation of large ocean waves and swell common in tropical cyclones. However, the increased translation speed decreases the warning time. Over land, the impacts are related to the intensity of surface winds and precipitation as in Hurricane Hazel in 1954, which resulted in a rapid intensification and precipitation amounts >200 mm, leading to 83 deaths in the Toronto area of southern Ontario. Over the open ocean the increased storm motion combined with the continued high wind speeds produces extremely large surface waves, as in Hurricane Luis in 1995, which produced waves >30 m, causing extensive damage to the luxury liner *Queen Elizabeth II*.

Tropical Cyclone Forecasting

When a tropical cyclone threatens there are four questions that must be answered: (1) where will it hit, (2) when will it hit, (3) how strong will it be, and (4) what type of threat should be expected (i.e., wind, storm surge, heavy rain, severe weather). The answers to these four questions are the goal of tropical cyclone forecasters. Track is the most important forecast, as it determines the answers to the first two questions. Operational tropical cyclone track forecasting is a semiobjective process that combines conventional, satellite, and reconnaissance observations with input from objective prediction models. At 12–24-h forecast intervals, persistence of storm motion is a major component of the forecast. However, an error in the initial motion of $1\,\mathrm{m\,s^{-1}}$ will yield an 84.6 km error in the 24 h forecast position. Since the tropical cyclone motion processes are complex and nonlinear, track uncertainty increases with time. For example, in the United States, where track errors are the lowest globally, the mean 24-h track error over the last 10 years is 170 km. However, 5% of the 24-h track errors over the last 10 years are >370 km. To minimize the possibility that a coastal area may be struck without time to prepare, much larger areas are warned than will actually experience damaging winds. While specific track models have indicated up to 15% improvement over the past 2–3 years, the average length of coastline warned, 730 km (roughly a 4:1 ratio to the track error) has not decreased over the past decade. In fact, it has increased over the 30-year mean of 556 km in response to the emergency manger's desire for longer lead time.

In the United States, emergency managers require communities with limited escape routes to complete preparation and evacuation before $17\,\mathrm{m\,s^{-1}}$ winds arrive on the coast. Hence, the length of coastline warned is a combination of the forecast uncertainty in the track forecast and the uncertainty in the forecast of the radius of the $17\,\mathrm{m\,s^{-1}}$ winds. The mean 24-h error of the forecast $17\,\mathrm{m\,s^{-1}}$ wind radii is 71 km, which is about 30–35% of the actual radii, and represents a 4-h error in lead time for a typical storm motion of $5\,\mathrm{m\,s^{-1}}$. However, 5% of the $17\,\mathrm{m\,s^{-1}}$ wind radii forecasts exceed 255 km. The warning of 730 km of the coastline is justified as the sum of the 370 km uncertainty in the track forecast plus the 255 km uncertainty in the $17\,\mathrm{m\,s^{-1}}$ wind radii forecast to be more than 95% confident that a coastal region will not be struck without warning. Such overwarning is

costly! Our best estimates of average preparation costs for the warned coastline have increased roughly sixfold from \$50 million in 1989 to \$300 million in 1996 (\$410 000 km^{-1} warned).

Another factor in the warning equation is storm intensity, which addresses the third question. Preparations differ considerably for a major hurricane or super typhoon than for a lesser storm (e.g., storm surge inundation varies greatly with storm intensity). In the United States the mean 24-h intensity forecast error over the last 10 years is 5 $m\,s^{-1}$ (half a category on the Saffir–Simpson scale; **Table 1**), and 5% of the intensity errors are $>12.5\ m\,s^{-1}$ (over 1 category on the Saffir–Simpson scale). At 48 h, the mean intensity error is 8 $m\,s^{-1}$, and 5% are $>20\ m\,s^{-1}$ (two categories on the Saffir–Simpson scale). Because the forecast intensity of the storm at landfall is a key factor in who evacuates, greater accuracy can lead to increased public safety and reduced costs, although these savings have yet to be quantified.

Understanding and predicting intensity change is more complex than that for track as it requires knowledge of interactions throughout the depth of the troposphere over a broad spectrum of scales. Observations are sparse in upper troposphere, atmospheric boundary layer, and upper ocean, limiting knowledge of environmental interactions, angular momentum imports, boundary layer stress, and air–sea interactions. Moreover, sea surface temperature remains an important but incomplete measure of the ocean's influence on tropical cyclone intensity change. Crucial unanswered questions concerning tropical cyclone intensity change lie in the relative impact and interactions of three major components: (1) the structure of the upper-ocean circulations that control the oceanic mixed-layer heat content, (2) the storm's inner core dynamics, and (3) the structure of the synoptic-scale upper-tropospheric environment. A successful intensity forecast requires knowledge of the mechanisms that modulate tropical cyclone intensity within the envelope defined by these three components.

Once the 17 $m\,s^{-1}$ wind radius crosses the coast the important forecast issues relate to determining how much damage is likely from wind, ensuing surge, rain, and severe weather, the answer to the fourth question. These factors will determine the type and level of emergency management response, and dictate where the most resources for recovery are needed. To a first order, the areas impacted by wind, ensuing surge, rain, and severe weather are determined by the horizontal and vertical wind distribution in the boundary layer and the interaction of the wind field with coast and inland topography. Currently, only a little information is known about the wind structure and its distribution with height and radial distance near landfall because of the paucity of observations (research quality or otherwise). Hence, a concerted effort is needed to better understand what determines the distribution of wind with height and radial distance in a high-wind boundary layer. As with the intensity forecasts, greater accuracy in the specification of the three-dimensional wind structure at landfall should lead to increased public safety and reduced costs, although these savings have yet to be quantified.

Summary

'The eye of the storm' is a metaphor for calm within chaos. The core of a tropical cyclone, encompassing the eye and the inner 100–200 km of the cyclone's 1000–1500 km radial extent, is hardly tranquil. However, the rotational inertia of the swirling wind makes it a region of orderly, but intense, motion. It is dominated by a cyclonic primary circulation in balance with a nearly axisymmetric, warm-core, low-pressure anomaly. Superimposed on the primary circulation are weaker asymmetric motions and an axisymmetric secondary circulation. The asymmetries modulate precipitation and cloud into trailing spirals. Because of their semibalanced dynamics, the primary and secondary circulations are relatively simple and well understood. These dynamics are not valid in the upper troposphere, where the outflow is comparable to the swirling flow, nor do they apply to the asymmetric motions. Since the synoptic-scale environment appears to interact with the vortex core in the upper troposphere by means of the asymmetric motions, future research should emphasize this aspect of the tropical cyclone dynamics and their influence on the track and intensity of the storm.

Improved track forecasts, particularly the location and time when a tropical cyclone crosses the coast, are achievable with more accurate specification of the initial conditions of the large-scale environment and the tropical cyclone wind fields. Unfortunately, observations are sparse in the upper troposphere, atmospheric boundary layer, and upper ocean, limiting knowledge of environmental interactions, angular momentum imports, boundary layer stress, and air–sea interactions. In addition to the track, an accurate forecast of the storm intensity is needed because it is the primary determinant of localized wind damage, severe weather, storm surge, ocean wave runup, and even precipitation during landfall. A successful intensity forecast requires knowledge of the mechanisms that modulate tropical cyclone intensity through the relative impact and interactions of three major components: (1) the structure of the upper ocean circulations that control the mixed-layer heat content, (2) the

storm's inner core dynamics, and (3) the structure of the synoptic-scale upper-tropospheric environment. Even if we could make a good forecast of the landfall position and intensity, our knowledge of how a tropical cyclone's structure changes as it makes landfall is in its infancy, because few hard data survive the harsh condition. To improve forecasts, developments to improve our understanding through observations, theory, and modeling need to be advanced together.

See also

Convective Storms: Overview. **Cyclogenesis**. **Dynamic Meteorology:** Balanced Flows; Overview; Potential Vorticity. **El Niño and the Southern Oscillation:** Observation; Theory. **Middle Atmosphere:** Quasi-Biennial Oscillation. **Severe Storms**. **Tropical Meteorology:** Inter Tropical Convergence Zones (ITCZ).

Further Reading

Elsberry R (ed.) (1995) *Global Perspectives of Tropical Cyclones*. World Meteorological Organization Report No. TCP-38. Geneva: WMO.

Emanuel KA (1986) An air–sea interaction theory for tropical cyclones. 1. Steady-state maintenance. *Journal of the Atmospheric Sciences* 43: 585–604.

Ooyama KV (1982) Conceptual evolution of the theory and modeling of the tropical cyclone. *Journal of the Meteorological Society of Japan* 60: 369–380.

HYDRAULIC FLOW

R B Smith, Yale University, New Haven, CT, USA

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

The study of hydraulic flow is one branch of a broader field of fluid mechanics dealing with the dynamics of density stratified flow under the influence of a gravity field. It has a natural application to the stratified atmosphere and ocean. The field of hydraulics is distinguishable from other studies of stratified flow by its emphasis on layered flow and the use of the hydrostatic or longwave approximation. Typically, in hydraulic flow formulations, the fluid system is composed of one or more homogeneous fluid layers, separated by sharp interfaces with density discontinuities. This formulation, together with the hydrostatic assumption, insures that the velocity is nearly uniform with height within each layer. In this way, a continuous problem is reduced to a problem with one or more discrete layers; this results in a vast reduction in the number of degrees of freedom. The possibilities for mathematical analysis, numerical computation, and physical conceptualization are greatly enhanced by the simple formulation of hydraulic theory.

Historically, the field of hydraulics arose out of, and is still largely involved in, the study of natural river flow and engineering problems related to water flow in channels. Its application to atmosphere and ocean dynamics is more recent. Beginning in the 1950s, a growing number of atmospheric applications have been suggested. On large scales, C. G. Rossby, G. Benton, and N. A. Phillips developed two-layer mathematical models of the midlatitude atmosphere including the Coriolis force. On smaller scales, following the pioneering work of R. R. Long and M. Tepper, a variety of atmospheric phenomena have been treated with hydraulic models. Cool outflowing air from thunderstorms, sea breeze fronts, and the leading edges of cold fronts all behave like gravity currents. Existing cool layers beneath marine inversions and frontal layers behave hydraulically in mountainous areas, causing barrier jets, gap jets, hydraulic jumps, severe downslope winds, and wake eddies. Cold high terrain can generate layered cold air avalanches and katabatic winds.

In oceanography too, hydraulic theory has found wide application. Basin to basin exchange of water masses is limited by hydraulic control at sills and straits. The propagation of tidal currents and tsunamis is controlled by the long-wave speed. Turbidity currents slump into the deep ocean according to gravity current dynamics. Coastally trapped currents obey a modified set of hydraulic equations. Even large-scale wind-driven ocean currents are often modeled as two layers, defined by the thermocline, with wind stress and the Coriolis force playing dominant roles.

The theory of hydraulic flow is based on a few fundamental definitions and concepts. These are: reduced gravity, the long-wave speed, Froude number, hydraulic control, conjugate states, the hydraulic jump, and gravity or density current. Reduced gravity (g') is a measure of the effective magnitude of gravity acting on layers of different density. It defined as the product of the acceleration of gravity

($g = 9.81\,\mathrm{m\,s^{-2}}$) times the relative density difference ($\Delta\rho/\rho$) between the two superposed layers; thus $g' = (\Delta\rho/\rho)g$. In the compressible atmosphere, the relative density difference is approximately the difference in potential temperature (Θ), so that $g' = (\Delta\Theta/\Theta)g$. For example, if the air above an inversion is 3° warmer than the air below, and the average potential temperature is 300 K, the effective gravity is $g' = (3/300)g = 0.0981\,\mathrm{m\,s^{-2}}$. The long-wave speed for a single layer is given by $C = (g'H)^{1/2}$, where H is the depth of the layer. If a cool marine layer of air has an effective gravity of $g' = 0.1$ and a depth of 1000 m, long gravity waves will propagate along it at a speed of $C = (0.1\times1000)^{1/2} = 10\,\mathrm{m\,s^{-1}}$.

The Froude number plays a central role in hydraulic theory. It is defined as the ratio of flow speed (u) to long-wave speed (C), i.e., $Fr = u/C$. It is related to the ratio of kinetic to potential energy in a layer, but is more useful as a measure of whether waves can move upstream against the current. Flows are categorized as subcritical, critical, or supercritical according to whether the Froude number is less than, equal to, or greater than unity. In supercritical flow ($Fr > 1$), long waves cannot move upstream or even stand steady, against the fluid flow. As the long waves are usually the fastest waves in the system, information carried by waves cannot then be felt upstream of a disturbance. The nature of a fluid response to any disturbance is highly sensitive to the Froude number.

Hydraulic control occurs whenever a layered flow is forced to transition from subcritical to supercritical flow by the narrowing of a channel or valley, the rising of a sill or mountain, or the alteration of some other geometric or external parameter (e.g., Coriolis force, coastal slope, etc.). Downstream of the control point, i.e., the point where the Froude number is unity, supercritical flow prevents information from propagating upstream. As a result, the amount of flow through the channel cannot be altered from downstream. Only the upstream conditions and control point characteristics have an influence.

Conjugate states are defined as multiple states of flow, defined by fluid speed and layer depth, with identical mass and momentum flux. They can be computed easily in the hydraulic formalism. If a flow has a conjugate state, it can, in principle, jump spontaneously to its other state, without the loss or gain of mass or momentum. Most commonly, this occurs in a hydraulic jump (i.e., an abrupt thickening of a layer), where energy may be dissipated by turbulence, but mass and momentum are conserved. Jumps are commonplace events in rivers, and related phenomena have been identified in the atmosphere and ocean.

A so-called gravity current or density current occurs when a new fluid pushes its way over or under an existing denser or less dense fluid, under the influence of gravity. While it resembles a hydraulic jump, a density current is not a sudden thickening of a pre-existing layer, but the introduction of a new fluid layer.

The literature is not clear on whether all layered formulations of stratified fluid mechanics should be classified as 'hydraulics'. When friction or Coriolis forces dominate, the term hydraulics is less often used.

See also

Convective Storms: Convective Initiation. **Density Currents**. **Downslope Winds**. **Dynamic Meteorology:** Overview. **Gust Fronts**. **Katabatic Winds**. **Land–Sea Breeze**. **Lee Waves and Mountain Waves**. **Mesoscale Meteorology:** Overview. **Microbursts**. **Mountain Meteorology**. **Solitary Waves**. **Static Stability**. **Valley Winds**.

Further Reading

Baines PG (1995) *Topographic Effects in Stratified Flows*. Cambridge: Cambridge University Press.

Hughes RL (1989) The hydraulics of local separation in a coastal current with application the the Kuroshio meander. *Journal of Physical Oceanography* 19: 1809–1820.

Jackson PL and Steyn DG (1994) Gap winds in a fjord. 2. Hydraulic analog. *Monthly Weather Review* 122: 2666–2676.

Pratt LJ and Lundberg PA (1991) Hydraulics of rotating strait and sill flow. *Annual Review of Fluid Mechanics* 23: 81–106.

Seitter KL (1987) Numerical study of atmospheric density current motion including the effects of condensation. *Journal of the Atmospheric Sciences* 43: 3068–3076.

Smith RB (1985) On severe downslope winds. *Journal of the Atmospheric Sciences* 42: 2597–2603.

Smith RB and Smith DF (1995) Pseudoinviscid wake formation by mountains in shallow-water flow with a drifting vortex. *Journal of the Atmospheric Sciences* 52: 436–454.

Yih C-S (1965) *Dynamics of Nonhomogeneous Fluids*. New York: Macmillan.

HYDROLOGY

Contents

Overview
Ground and Surface Water
Modeling and Prediction
Soil Moisture

Overview

R C Bales, University of Arizona, Tucson, AZ, USA

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

Hydrology is the science that encompasses the study of water on the Earth's surface and beneath the surface of the Earth, the occurrence and movement of water, the physical and chemical properties of water, and its relationship with the living and material components of the environment. Ultimately, many hydrologic questions involve the transport of solutes, nutrients, energy, sediment, or contaminants, as well as the fluxes of water itself.

As a science, hydrology has both basic and applied aspects. The first relates to questions about the Earth system, and specifically about the role of water in natural processes, particularly as related to the Earth's biosphere. The second relates to using scientific knowledge to provide a sound basis for wise usage of water resources. The development of hydrologic science in recent years is based on both of these aspects, which are equally important and intimately linked.

Water is central to most natural processes. Water weathers, then transports sediment and solutes to lakes and oceans, thereby shaping the landscape. The land-based part of the Earth's water cycle is important for transporting carbon from the continents to the ocean. The high capacity of water for storing thermal energy and the large amount of heat required to change between solid, liquid, and vapor forms of water strongly influence the global energy balance. The distribution of atmospheric water and its regulation by oceanic and land-surface processes make it a central aspect of climate. Water vapor is the most important greenhouse gas. In short, life depends on water.

Hydrologic science has an important place in the field of water resources, especially freshwater resources, which are the subject of intense concern and study. In arid and semi-arid regions, the fair allocation and wise use of water are significant societal challenges, affecting relations between nations, states, cities, and individual users. As a global resource, water appears abundant. Locally and regionally it is often taken for granted. However, the twentieth century has witnessed a tremendous growth in the use of water, resulting in limits on both its availability, due to human exploitation, and its quality, owing to contamination.

Water Cycle

A fundamental concept of hydrology is the hydrologic cycle, which can be described at many different scales of space and time. At the global scale, the hydrologic cycle is the endless recirculatory process linking water in the atmosphere, on the continents, and in the oceans. We can think of this recirculatory process in terms of reservoirs or compartments that store water (e.g., oceans, atmosphere, glaciers, ice sheets, ground water) and the movement of water between them. Movement of water from one compartment to another can occur in any of the three phases. For example, the movement of water between the land surface and the atmosphere occurs in the vapor phase (evaporation and condensation), liquid phase (rain), and solid phase (snowfall).

Solar energy and gravity are the main forces driving the hydrologic cycle. The dynamic processes of water vapor formation and transport of vapor and liquid in the atmosphere are driven largely by solar energy. Precipitation and the flow of water on and beneath the Earth's surface are driven primarily by gravity. Within partially dry soil, gravitational pressure and capillary forces are responsible for the movement of water.

The hydrologic cycle can be considered to start anywhere, but it is often convenient to consider

atmospheric water first. The dominant hydrologic process involving atmospheric water is precipitation of water to the land surface. Condensation of water from the atmosphere to the land surface (e.g., dew, frost) and fog deposition can be important locally, in the absence of precipitation. Precipitation that reaches the land surface as snow or ice can be retained on vegetation and subsequently evaporate or fall to the ground, accumulate in seasonal snowpacks, and later melt or evaporate or accumulate in glaciers, ice caps, or ice sheets. Water is also lost from glaciers and ice caps by evaporation and melting; pieces of ice can also break off from the edges of glaciers and ice sheets (this is known as calving) and be returned directly to the ocean, in the form of icebergs. A portion of rain or snowfall can be retained temporarily on vegetation surfaces, and subsequently evaporate or fall to the ground. Rain or snowmelt can also collect in surface depressions, enter into the soil (infiltration), or flow as runoff over the land surface into small rivulets and ultimately into larger streams, lakes, and rivers. Water that infiltrates into the soil can also follow different paths. Some returns to the atmosphere by evaporating from the soil or being transpired by plants (transpiration), the sum of which is termed evapotranspiration. The remaining water continues to move downward through the soil and recharges the saturated portion of the subsurface, becoming groundwater. Ground water discharges into streams and rivers, or directly to the ocean. Water evaporates from the surface of the oceans and thereby replenishes the water in the atmosphere.

Much effort in hydrology goes toward estimating the amounts of water in the various storage compartments and the magnitudes of the various flows to and from these compartments at local, regional, and global scales. Nearly 97% of all water on the Earth is stored in the oceans, while only about 0.001% is stored in the atmosphere. Fresh water accounts for about 2.5% of the total storage, 70% of which is contained in the two polar ice sheets and 30% is found in ground water. Only about 0.4% is found in glaciers and ice caps. The fresh water in lakes, streams, rivers, and marshes represents only 0.26% of all fresh water and 0.008% of all water on Earth. That is, if the Earth's ocean were represented as a 16 l (4 gallon) bucket, the fresh water fraction would be equal to a little over 1 ml ($\frac{1}{4}$ teaspoonful).

Another useful concept for enumerating reservoirs and the flows of water through them is residence time, which is a measure of how long, on average, a molecule of water spends in that reservoir before moving on to another reservoir of the hydrologic cycle. For a system at steady state, i.e. with inflow and outflow the same, residence time is equal to the size of the reservoir (e.g., in m^3) divided by the flow through the system (e.g., in $m^3\,yr^{-1}$). Water in the oceans has a residence time approaching 3000 years, less than half of the residence time for ice sheets, while in the atmosphere it has a residence time of only 0.02 years or about 8 days; the residence time in rivers is 0.05 years or about 17 days.

Most of hydrology deals with scales smaller than global; however, the same concepts of fluxes and reservoirs apply. A catchment is an area in which water falling on or flowing across the land surface drains into a particular stream or river and flows ultimately through a single point or outlet. Thus a catchment is defined relative to a specific location and includes all of the land area that drains to that point; i.e., it can be considered to catch the water that flows past that point. Clearly, then, any number of catchments can be defined for a particular river (corresponding to any location along the river). Some special locations of interest for defining catchments are points corresponding to dams, stream gauges, cities, and a river's mouth. Ground water reservoirs (aquifers), on the other hand, are defined by subsurface geologic structure. Aquifers, or water-bearing formations, are bounded by material of low permeability, i.e., material with a very small ability to transmit water.

Precipitation

Precipitation is the deposition of liquid water droplets and ice particles that have formed in the atmosphere and grown to a size sufficient to fall to the Earth's surface by gravity. Precipitation is classified according to the phase it is in when it reaches the ground, i.e., solid (snow, sleet, and hail) or liquid (rain and freezing rain). Other deposition processes (e.g., direct deposition of dew and fog), though generally small, can however be important in terms of chemical fluxes (e.g., acidic fog).

Most of the precipitation falling over North America originates from the bordering oceans, even in the interior of the continent. However, over the Amazon basin, a significant fraction of the precipitation is derived from within-basin evapotranspiration. In other continental basins, local evapotranspiration does have some influence on local precipitation, but most of the precipitated water must be transported significant distances across the continents from the oceans.

Average precipitation over the continents is extremely variable geographically, reflecting the influence of a number of important physiographic factors. First, precipitation increases with elevation owing to orographic cooling. Second, precipitation is typically

higher on windward than on leeward sides of mountain ranges. Third, precipitation tends to drop off as air masses move further inland, away from the ocean. Fourth, the temperature differences between adjacent land and ocean influence moisture transport. Fifth, prevailing wind direction has local to regional effects.

Global average precipitation is about 1000 mm yr^{-1}. In the continental US, average annual precipitation ranges from about 40 mm yr^{-1} at Death Valley, California (in the Mojave Desert), to over 3000 mm yr^{-1} in parts of the Pacific Northwest. In the Atacama Desert of northern Chile rainfall is infrequent, averaging under 1 mm yr^{-1}.

Rainfall and snowfall are measured at a point by collectors of very simple construction. Essentially, any receptacle with a reasonable opening can serve to estimate the precipitation per unit area. In the US, the standard gauge has a 20 cm diameter opening. Wind is probably the single most important factor in rain gauge accuracy. Updrafts resulting from air moving up and around the instrument reduce the catch, which has led to the development of elaborate shields to mitigate wind effects and detailed correction procedures.

In a few areas, radar is also used for precipitation measurement, although rain gauges are generally required for calibration. Radar emits electromagnetic energy in narrow bands, which upon hitting precipitation is partially absorbed, scattered, and reflected. Some of the reflected energy returns to the transmitter, and the attenuation of the signal indicates the intensity of rainfall. In principle, the same method could be used for snowfall.

Snow accumulation at a point is more accurately measured, either *in situ* by snow pillows equipped with pressure transducers or manually. Snow accumulation is reported in terms of equivalent water depth rather than snow depth, the proportionality factor being the density. Snow accumulation can also be measured from aircraft in terms of the extinction of gamma or other types of radiation. Satellites are used for estimating snowpack depth and areal coverage. A combination of visible and infrared signals are used to extract snow-covered area. Over flat terrain, passive microwave signals indicate snow water equivalent. Because much snow falls in heterogeneous terrain and is thus not uniformly distributed, accurate measurements of snow water equivalent remain a major challenge. Snow measurement networks in the western US consist of index sites, for which correlations with streamflow have been developed. They do a relatively poor job of providing direct estimates of snow volume.

Snowmelt plays a major role in the hydrology of midlatitudes, as many rivers originate in mountains where snow dominates the precipitation. In contrast to rainfall, snowfall has a delayed effect on river flow and hydrology. Accumulation occurring during winter months becomes all-important during spring runoff, which provides much of the streamflow, ground water recharge, and soil moisture for wide areas of the Earth. Melt waters can also cause serious floods, particularly when compounded with spring rainfall.

Catchment

Much effort in hydrology focuses on water balances at the scale of a catchment, be it a 1 km^2 headwater basin or a 100 000 km^2 regional basin. Hydrologists often think of a catchment as functioning like a group of reservoirs that store and release water. Precipitation and snowmelt are the inputs to the catchment reservoir, and evapotranspiration and runoff are the outputs. Integrating the various processes that affect runoff involves quantitative partioning of the different fluxes and reservoirs into downstream fluxes and reservoirs. For example, a fraction of precipitation is intercepted by vegetation, with the remainder falling to the ground. Intercepted rain and snow then either evaporates or eventually falls to the ground. Snow on the ground partitions between snowmelt and sublimation. Rainfall or snowmelt reaching the ground either infiltrates or runs off, depending on the infiltration capacity of the soil. A portion may flow downstream in the subsurface and reemerge at a lower point in the catchment. Much infiltrating water may eventually be transpired by plants or evaporated, though some will travel downward sufficiently far to enter, or recharge, a ground water aquifer. Of all these quantities, surface runoff in streams and rivers, or discharge, is most often measured.

Discharge is estimated by continuously measuring the stage, or height of a stream at a point using a pressure transducer or a mechanical float. This point measurement is then related to the volumetric discharge through a calibration, or rating, curve, which is developed by measuring the volumetric discharge at the gauged point for many different stages. A time series graph of discharge is commonly referred to as a hydrograph.

A streamflow hydrograph is often defined in terms of two components, quickflow and baseflow. Separation of a hydrograph into two components suggests that water is being routed through two different storage reservoirs. During and after rainfall and snowmelt events, water moves through the catchment into the stream channel and the discharge increases (quickflow). The resulting peak in the hydrograph is generally defined as a flood, regardless of whether the river actually leaves its banks and causes damage.

Background discharge between floods (baseflow) is supplied by inflow of ground water, which may lag the occurrence of precipitation by days, weeks, or even years.

Water flowing across the ground surface is termed overland flow. For this to occur, water must accumulate at the surface rather than infiltrate into the soil. This happens for three main reasons: (1) the catchment surface may be nearly impermeable owing to the presence of exposed bedrock or pavement; (2) the instantaneous rate of infiltration through the pervious surface may be exceeded by the instantaneous rate of rainfall (or snowmelt) onto the catchment surface, causing ponding of water at the surface, and (3) the catchment soil upon which the rainfall is precipitated may be saturated to the soil surface, causing ponding because the precipitated water cannot infiltrate into an already saturated soil. Overland flow in catchments is one of the most rapid paths that water can follow to the stream channel.

Water that has infiltrated the soil surface continues to be influenced by gravity, so that it percolates downward through the soil profile. In general, the ability of the soils and rocks of a catchment to conduct water (hydraulic conductivity) decreases with depth; water percolating downward has thus been observed to back up, causing local areas of saturation in the soil. In these instances, water may move laterally toward a stream by a process known as shallow subsurface stormflow. Some of the water in subsurface stormflow moves at a relatively slow pace through the soil and contributes to the baseflow of streams, particularly during wetter winter and spring periods. Subsurface stormflow also may occur quite rapidly along preferred flow pathways or macropores (e.g., soil cracks, old animal burrows, and decayed root channels).

Vadose

In most areas, the water table is some distance below the ground surface. Between the ground surface and the water table is a region in which the pore spaces of the rock or soil may be partly filled with air and partly with water. This region is referred to as the unsaturated zone or vadose zone, and water in this zone is referred to as soil moisture. Hydrologists want to be able to describe the amount of water in the unsaturated zone and fluxes through the zone for two main reasons. First, ground water recharge occurs through this zone. Second, most terrestrial plants extract water from the vadose zone. Plants wilt when soils become too dry because the tension forces holding the water in the soil are too great to allow the plants access to the water. Related to the water balance of plants is the practice of irrigation in agriculture, which accounts for about two-thirds of global water use. Understanding the movement of soil water, its uptake by plants, and its loss through evapotranspiration and recharge to the groundwater system is essential in this regard.

Hydrologists have traditionally recognized three divisions within the unsaturated zone: the capillary fringe, the intermediate belt, and the belt of soil water. The capillary fringe is a zone in which the pressure is less than atmospheric, overlying the zone of saturation and containing capillary interstices, some or all of which are filled with water that is continuous with the water in the zone of saturation but is held above that zone by capillarity forces acting against gravity. That is, the capillary fringe is a saturated zone above the water table where water is affected by capillary forces. Above that is a zone of soil water from which water is discharged to the atmosphere by the action of plants or by evaporation. For the most part, plants extract water from a portion of the soil near the surface (the 'root zone'). Depending on the depth of the vadose zone and the plant, roots may lie only in the upper part of the vadose zone, or extend into the water table. For example, most grasses have roots extending only a few centimeters to tens of centimeters, whereas some trees in semi-arid regions have roots that extend through vadose zones that are tens of meters thick and reach the water table. The volumetric moisture content (volume of water per bulk volume of soil sample) in the capillary fringe is the saturation value. In other words, the pores are completely filled with water. As water drains or is withdrawn by plants, soil moisture content decreases from saturation to a fairly constant value, termed the field capacity.

Rates of removal of water from the unsaturated zone by evapotranspiration are controlled by a number of factors, including the wetness of the soil itself. If a vegetated surface is supplied with plenty of water (e.g., a well-watered lawn), evapotranspiration will be controlled by atmospheric conditions, e.g. solar radiation, wind speed, and humidity. That is, evapotranspiration will proceed at the maximum rate (potential evapotranspiration). As a soil dries, evapotranspiration will proceed at the potential rate for some time, but ultimately the rate will drop. As water is pulled from the soil near a plant root, the moisture content in the soil surrounding the root decreases. In order to maintain a steady flow of water to the plant root, the plant must exert ever greater suction (ever greater negative capillary pressure heads). At some point, the plant cannot sustain this battle with a drying soil and the transpiration rate falls below the potential rate.

Most plants have openings (stomata) on their leaves to allow them to take up carbon dioxide from the

atmosphere. When the stomata are open, plants transpire water. Unlike evaporation, transpiration is not controlled solely by physical conditions because plants regulate the rate at which water is released in transpiration in a manner that varies by plant type and ecological conditioning. Of the water taken up by plant roots, most is transpired through the stomata. A few percent is concerted to biomass through photosynthesis. Hence, to first order, the water taken up by the roots is converted to vapor and lost to the atmosphere. The degree of stomatal restriction varies considerably across plant species and even throughout the year for a given species.

Ground Water

The largest accessible fresh water reservoir for human exploitation is ground water, or water that is present in the fractures and interstitial spaces in subsurface geologic materials. In contrast to the vadose zone, in ground water the void spaces are completely filled with water. An aquifer is a saturated geological formation that contains and transmits significant quantities of water under normal field conditions. 'Significant' is a vague term, but the implication is that aquifers are formations that can be used for water supply. Many aquifers are unconsolidated materials, mainly gravel and sand. Examples of this type of aquifer include those in coastal plains and intermontane valleys. Limestones, partially cemented sandstones and conglomerates, and permeable volcanic and igneous rocks are also important as aquifers.

An aquitard is a formation of relatively low permeability, and includes both formations that contain water but do not transmit significant quantities (e.g., clays and shales) and formations that neither contain nor transmit significant quantities of water (e.g., unfractured crystalline rocks).

Aquifers are classified according to hydraulic conditions as well as type of material. Ground water by definition refers to water in the saturated zone of the subsurface; one type of aquifer is an unconfined or water table aquifer. Deeper in the soil profile, saturated conditions prevail (the saturated zone). The water table is defined as a surface of zero (gauge) pressure within the subsurface, and separates the saturated and unsaturated zones. Water will flow into an excavation or well up to this level; the water table is equivalent to a free surface. An aquifer with the water table as the bounding surface at its top is an unconfined aquifer.

The second type of aquifer is a confined or artesian one. This is found when permeable material (the aquifer) is overlain by relatively impermeable material. The water in a confined aquifer is under pressure and, in a well penetrating the aquifer, will rise above the top of the aquifer. The height to which water rises in a well defines the piezometric surface, or pressure of water in the aquifer.

In areas where the water table is sufficiently close to the ground surface, ground water levels are influenced directly by transpiration. During the day, when transpiration is high, water movement is upward from the water table and the level declines. At night, transpiration is reduced, groundwater flows laterally from locations relatively unaffected by direct transpiration effects, and the water table recovers.

Recharge to aquifers can occur from direct infiltration of rainfall or snowmelt past the root zone and to the water table. Water can also seep from surface water bodies, such as rivers, ponds and lakes, into the ground. Artificial recharge (recharge induced by activities of people as opposed to that which occurs naturally) can be implemented by introducing water into recharge wells or by routing water into infiltration basins in permeable material.

The specific yield is a hydrologic parameter that determines the response of the water table to changes in inputs and outputs. In the case of an increase in evapotranspiration, the change in water table level may be fairly uniform over a given area, although variations will occur due to the lateral movement of groundwater and spatial variations in evaporation rate and vegetation. Pumping a well has a different effect. Pumping produces a decrease in hydraulic head at a point, which increases the hydraulic gradient toward the well. The change in water level in the pumping well, or in observation wells nearby, is referred to as drawdown. The amount of this drawdown will decrease as one moves away from the pumping well, and the pattern produced is called a cone of depression because of its characteristic shape.

Integration

The need for an integrated understanding of water cycles and linked chemical cycles is critical for a number of reasons. For example, a quantitative understanding of how ground water recharge changes with precipitation in a variable or changed climate requires an understanding of how catchment processes respond to translate precipitation into recharge. Understanding how evapotranspiration and ecosystem functioning respond to change in land use depends on how water and nutrient fluxes change. Hydrologic science is at the center of many pressing issues in other natural sciences.

See also

Climate Variability: Seasonal to Interannual Variability. **Hydraulic Flow**. **Hydrology:** Modeling and Prediction; Soil Moisture. **Land–Atmosphere Interactions:** Overview

Further Reading

Bras RL (1990) *Hydrology – An Introduction to Hydrologic Science*. Reading MA: Addison-Wesley.

Fetter CW (1988) *Applied Hydrogeology*. New York: Macmillan.

Freeze RA and Cherry JA (eds) (1979) *Groundwater*. Englewood Cliffs, NJ: Prentice-Hall.

Hornberger GM, Raffensperger JP, Wiberg PL and Eshleman KN (eds) (1988) *Elements of Physical Hydrology*. Baltimore, MD: Johns Hopkins University Press.

Maidment DR (1992) *Handbook of Hydrology*. New York: McGraw-Hill.

National Research Council (1991) *Opportunities in Hydrologic Sciences*. Washington, DC: National Academy Press.

Ground and Surface Water

S Ge, University of Colorado, Boulder, CO, USA

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

Water is one of the most precious and indispensable natural resources for civilization. As important components of the hydrological system on Earth, ground water and surface water impact numerous aspects of the Earth's processes and many facets of our daily lives. Water at the Earth's surface interacts directly with the atmosphere, and water in the subsurface continuously redistributes thermal energy and dissolved minerals in the Earth's crust at a variety of temporal and spatial scales. Hydrology encompasses the study of the occurrence and movement of water both at the land surface and in the subsurface environment. Water stored in the subsurface is known as ground water and is either in a moisture state in near-surface soils or in a liquid state in rock voids at greater depth. Some texts exclude moisture from the broad definition of ground water.

Focusing on the physical aspect of the dynamics of ground water and surface water, this article reviews our current knowledge of water as a resource, and explains the main concepts associated with ground water and surface water storage. The fundamental principles and material properties governing water movement are presented. Finally, the applied aspects of hydrology as it relates to water contamination, land subsidence, and geological processes are discussed.

Water as a Resource

While the presence of abundant liquid water makes the Earth a unique planet in the solar system, that abundance has been challenged throughout human history as numerous local and regional conflicts over water resources have erupted. Evidence of early attempts to harness water for human purpose has been documented by archaeologists. For example, clever water usage for irrigation can be traced as far back as 4500 years ago in the Middle East. Also, a remarkable water-collecting tunnel system dating from around 500 BC in Egypt has been unearthed. As world population grows at an unprecedented rate, the demand for water is expected to continue, particularly in arid developing countries. The global withdrawal of water in the last century is shown in **Figure 1**; a sharp increase is seen in the last 50 years. The total daily fresh water withdrawal in the United States from 1950 to the present follows the general global trend (**Figure 2**). The decrease in the 1980s is primarily due to increased irrigation efficiency and a reduction in water consumed in the thermoelectric power industry as a result of improved power plant technologies and efficiencies. The decrease in industrial water usage, however, is

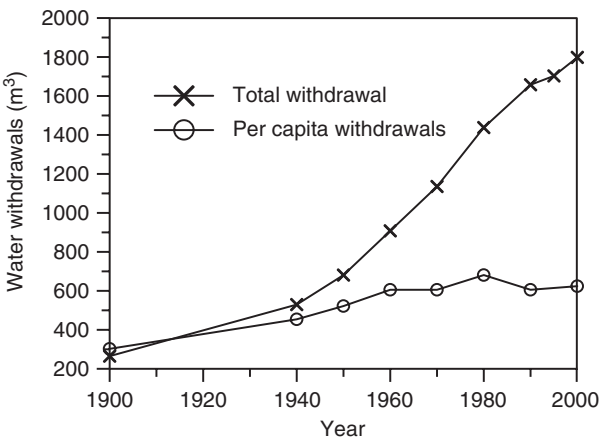

Figure 1 Annual global water withdrawal (data from Gleick 2001).

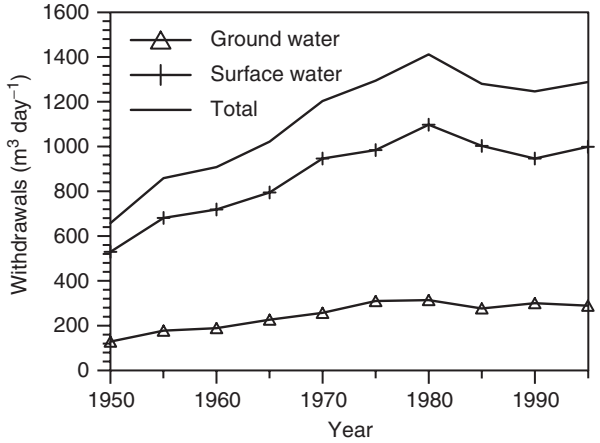

Figure 2 Total daily fresh water withdrawal in the United States (data from Solley *et al.*, 1998).

accompanied by a continual increase in domestic water consumption.

Global Water Distribution

Figure 3 shows the global distribution of water. Of the water on Earth, 97% is stored in the ocean and is too salty to be used directly for human consumption. Ice caps and glaciers, the next largest water reservoir, hold approximately 2% of the global water, which counts for 94% of the total fresh water. The vast domain of the subsurface holds only about 1% of the water on Earth. Furthermore, only the water in the upper few kilometers of the Earth's crust is accessible and fresh enough for human consumption. The salinity of ground water increases with depth and often becomes too high for the water to be useful as a resource below 4 or 5 km. Surface waters, including streams, lakes, and rivers, hold only 0.01% of the global water. Seemingly a small fraction of the water budget, surface water has served as the main resource for civilization,

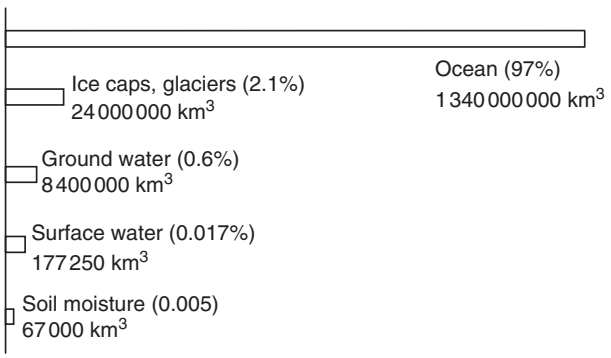

Figure 3 Global water distribution (data from Herschy and Fairbridge 1998). The lengths of the bars are not to scale.

primarily owing to its easy accessibility. The remaining water is distributed in the atmosphere and biosphere.

Hydrological Cycle

Powered by solar energy, the hydrological cycle is the endless movement of water from one reservoir to another in the Earth system (**Figure 4**). Water evaporates into the atmosphere from open waters such as oceans and lakes, from soil moisture in the unsaturated zone, and from the water table. Plants lose water to the atmosphere through the process of transpiration. These two processes, evaporation and transpiration, are collectively known as evapotranspiration. Water falls back to the Earth's surface as precipitation in the form of snow or rain. Upon reaching the surface, water flows overland as runoff to streams or infiltrates to the subsurface to become ground water. In the subsurface, water infiltrates through soils, recharges the ground water table and joins the ground water flow system. Ground water takes its course through geological basins of various scales, and some eventually makes its way to the oceans while some accumulates in inland aquifers. The rates of water flow between reservoirs within the hydrological cycle vary spatially and temporally in the Earth's system. As a result, the residence time of water – the time water remains in a reservoir since recharge – in different reservoirs varies from hours in the near-surface soil to tens of thousands of years in rocks several kilometers deep in the crust.

Water Budget Balance

The primary objective in studying ground and surface water is to understand the spatial and temporal variations of water storage and movement. One basic principle governing these variations is the conservation of mass, or water balance. This principle requires that the amount of water entering minus the amount of water exiting equals the change in storage of a controlled volume during a specified time period. A watershed, or drainage basin, is a geographic region over which the principle of water balance is applied. The watershed is defined as the area over which all the water flows into a given section of a stream. Watershed delineation is based on topography, where topographic divides, or ridges, form the boundaries of a watershed. Equation [1] gives a water balance averaged over a long period.

$$P + G_{\text{in}} - (Q + ET + G_{\text{out}}) = 0 \qquad [1]$$

In eqn [1], P is the precipitation, G_{in} is the ground water inflow, Q is the stream outflow, ET includes evaporation and transpiration, and G_{out} is the ground

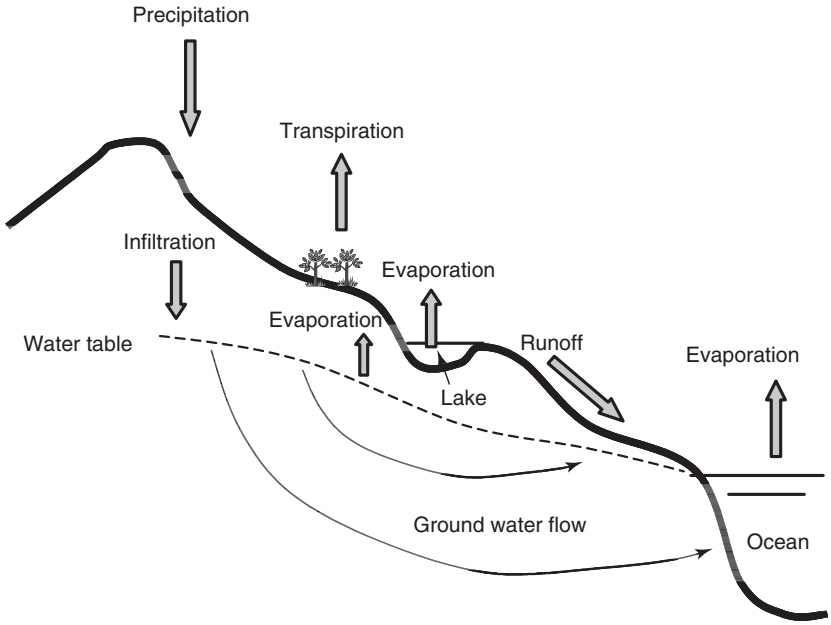

Figure 4 The hydrologic cycle. The water table is the boundary between the unsaturated zone above and the saturated zone below. Upon reaching the land surface, precipitation either infiltrates soil to replenish ground water or flows overland as runoff to open water bodies. Water evaporates from open water bodies at the Earth's surface, soil moistures in the unsaturated zone, and the water table. Transpiration occurs over vegetated lands. Ground water flows through the vast domain of the subsurface and returns to the oceans.

water outflow. All quantities have the dimension $[L\,t^{-1}]$.

Subsurface Hydrological System Characterization

Water Table

In near-surface soils, water is in the form of moisture in the unsaturated zones where void spaces are partially filled with water. At deeper locations, void spaces are completely filled with water, which forms saturated zones. The boundary between the unsaturated and the saturated zone marks the water table. The shape of a water table generally follows the topography, near the surface in low lands but deeper in high elevations. The depth of the water table varies in both space and time. In humid areas, the water table can be at or near the surface, whereas in drier regions the water table can be hundreds of meters below the land surface. The water table can rise in wet seasons as precipitation percolates through unsaturated zones to the water table, and can drop in dry seasons when more water is lost through evaporation and transpiration. The most reliable way to locate water table depth is to drill wells. Because ground water beneath the water table flows in the general down-slope direction of the water table, knowledge of water table shape and position gives useful basic information on the direction of ground water flow.

Porosity, Permeability, and Hydraulic Conductivity

The primary factors controlling ground water occurrence and movement in the subsurface are the hydrological properties of the geological material of interest and the hydrological driving force. The most important material properties are porosity and permeability. Porosity, defined as the fraction of void space in a unit volume, may vary from 0–5% for tight igneous and metamorphic rocks to 25–50% for sands or fractured rocks. Clay can have a porosity as high as 70%. Porosity is a direct measure of the geological material's capacity for holding water.

Permeability, on the other hand, is a measure of the material's ability to transmit water. It is defined as follows: $k = Cd^2$, where k is the intrinsic permeability $[L^2]$, C is a constant of proportionality related to grain size, sediment sorting, and packing arrangement, and d is the average grain diameter [L]. The ease of flow differs depending on the type of fluid flowing through a system. One can imagine that a fluid that is sticky like honey would flow much more slowly than would clean water in the same medium. It is necessary to consider not only the medium but also the fluid properties. Consequently, a hydraulic conductivity is introduced, defined as $K = k\rho g/\mu$, where K is the hydraulic

conductivity [$L\,t^{-1}$], ρ is the fluid density [$M\,L^{-3}$], g is the acceleration [$L\,t^{-2}$], and μ is the dynamic viscosity of the fluid [$M\,L^{-1}\,t^{-1}$]. Values of hydraulic conductivity vary over several orders of magnitude from 10^{-13} $m\,s^{-1}$ for unfractured rocks to $1\,m\,s^{-1}$ for sands and gravels. The most reliable means of obtaining the hydraulic conductivity values is *in situ* well testing. When water is withdrawn from or injected into a well, the rate of water level recovery in the well can be monitored by a pressure transducer and recorded by a data logger. The rate of water level recovery, as an indicator of the material permeability, is used to compute the hydraulic conductivity of the material surrounding the well. Laboratory tests on core samples are conducted to obtain hydraulic conductivity values valid at centimeter scales. Computer modeling is also commonly practiced as an indirect means of inferring hydraulic conductivity at different scales.

Aquifers, Aquitards, and Aquicludes

A geological unit that is highly permeable and that can transmit a significant amount of ground water is called an aquifer. When an aquifer is confined between two less-permeable units, it is called a confined aquifer. When bounded at the top by the water table, the aquifer is called an unconfined aquifer. Water pressure in confined aquifers is usually higher than in unconfined aquifers. Thus, when a well is drilled into a confined aquifer, the water level in the well can rise to above the land surface to form artesian flow. An aquitard is a less permeable geological unit, while the term aquiclude refers to an impermeable unit. As absolutely impermeable geological material is rare, the term aquiclude is infrequently used in hydrological characterizations. Aquifer and aquitard are more commonly used to characterize geological systems. For distinction, aquifers and aquitards are often identified in a relative sense. Gravel, sandy materials, or highly fractured rock make good aquifers, whereas clay-rich, poorly sorted sediments, and unfractured rocks often form aquitards.

Physical Hydrological Processes

Water in Soils

Near the land surface at shallow depths, soils are often partially saturated; such a region is known as the unsaturated zone. The degree of saturation is defined as the fraction of pores that contain water, and varies from 0, representing a dry condition, to 1, representing a fully saturated condition. The water in the partially saturated soils clings to soil particle surfaces and is sustained by suction or tension. Pore pressure in the unsaturated zone is conventionally expressed in negative values, reflecting the use of atmospheric pressure as the zero reference pressure. The pore pressure distribution and the rate of moisture movement vary spatially depending on soil types and weather conditions and temporally in response to rainstorms, seasonal changes, and long-term climate change.

Infiltration is an important process in the unsaturated soil zones and involves downward movement of moisture under wet climatic conditions. The infiltration rate over a small area can be measured using a ring infiltrometer. This is a portable cylindrical ring, with a diameter from a few centimeters to 20 cm, extending several centimeters above and below the surface of the soil. The rate of water dissipating from the ring infiltrometer into the soil can easily be converted to an infiltration rate.

In contrast to infiltration, evaporation and transpiration draw moisture upward under dry climatic conditions. Evaporation causes water loss from surface waters, such as lakes and rivers, and from shallow-depth soils. Water evaporates as a vapor diffusion process that is largely controlled by the energy exchange between radiation or sensible heat from the atmosphere or ground, and the heat energy change in the evaporating body. A direct method for determining the evaporation rate has been developed and is known as the pan-evaporation approach. It involves exposing a cylindrical pan of water to the atmosphere in clearings where precipitation can be monitored. The standard US National Weather Service Class A pan is 1.22 m in diameter and 25.4 cm deep. Transpiration is a process whereby water is lost to the atmosphere through the vascular systems of plants. The transpiration process works by absorption of water by plant roots, translocation of liquid through the plant vascular system, and transpiration into the atmosphere through openings in the leaf surface. Although transpiration is also considered a diffusion process, water is first pulled through the plant by a potential energy gradient before diffusing into the air in response to a vapor pressure difference.

Hydraulic Head

Hydraulic head is one of the key parameters describing the mechanical energy state of a hydraulic system. It represents the mechanical energy per unit weight of the fluid in the system. Hydraulic head is defined as $h = h_p + h_z$, where h is the hydraulic head, h_p is the pressure head, and h_z is the elevation head. All three quantities have the dimension [L]. The pressure head represents the energy due to pore fluid pressure, and the elevation head represents the gravitational energy arising from elevation. Water flows from high to low

hydraulic heads. Knowledge of the hydraulic head distribution in an area allows one to estimate the direction and rate of flow according to Darcy's law, which is described in the following section. *In situ* measurement of hydraulic head is accomplished by measuring the water levels in wells, using electric sounding water tapes or similar devices.

Darcy's Law

The basic theory governing ground water movement is Darcy's law, named after Henry Darcy, a French engineer. In Dijon, France, in 1856, Darcy performed a water purification experiment using a simple setup involving water flow through a cylindrical sand column. The experimental data led to an empirical relationship between water flux and the experimental parameters. That relationship later became the well-known Darcy's law. In a one-dimensional system, Darcy's law is expressed as eqn [2].

$$Q = -AK\frac{\mathrm{d}h}{\mathrm{d}x} \qquad [2]$$

In eqn [2], Q is the volumetric flow rate $[\mathrm{L}^3\,\mathrm{t}^{-1}]$, A is the cross-sectional area of flow $[\mathrm{L}^2]$, K is the hydraulic conductivity $[\mathrm{L}\,\mathrm{t}^{-1}]$, h is the hydraulic head [L], and x is the distance coordinate [L]. The dimensionless quantity $\mathrm{d}h/\mathrm{d}x$ is called the hydraulic gradient. It is clear that the hydraulic gradient is the primary driving force for ground water flow. The negative sign in the equation denotes that ground water flow is in the direction of decreasing hydraulic head.

Ground Water Flow Patterns

The movement of ground water normally occurs as slow-moving seepage through the pore spaces in soils and rocks or as relatively fast flow through rock fractures. The velocities are generally much slower than stream flow, and may be as low as $10^{-9}\,\mathrm{m\,s^{-1}}$. A ground water velocity on the order of $10^{-6}\,\mathrm{m\,s^{-1}}$ would be considered high. The low velocity of ground water has important implications for geological processes such as metamorphisms and ore formation as well as for contaminant movement because it leads to longer residence times. The extent of ground water systems varies from local hill slopes at meter scales to regional geological basins at kilometer scales. Flow paths are shallower in local flow regimes and deeper in regional flow systems. As a result, the residence time of ground water also varies significantly ranging from hours or days in shallow and small systems to tens of thousands of years in deep and large geological basins.

Wells

Wells serve as a window to the subsurface in the study of ground water. The primary need for wells is to withdraw water from the ground. Some wells are used for monitoring water levels and also for sampling water for chemical analyses. When a well is pumped, a depression cone forms around the well as the water level declines. Water level response can be quite different depending on whether the pumping occurs in a confined or in an unconfined aquifer. Under the same pumping rate, a larger water level decline is expected in a confined aquifer and a smaller decline in unconfined aquifers. This is because less water is available in confined aquifer storage, while unconfined water may receive water from drainage of the water table.

Surface Water Dynamics

Although small in volume in the global water distribution, surface water has been the main water source for our civilization. A stream is a body of water flowing down-slope along a natural passage. A river is a stream with a significant amount of flow. A stream with no tributaries is designated as a first-order stream; the confluence of two first-order streams is the beginning of a second-order stream; two second-order streams are the beginning of a third-order stream, and this pattern can continue to form higher-order streams. The branching patterns of stream orders have been studied through a fractal approach that provides a mathematical framework for treatment of similar geometric characteristics over a range of scales. Streams are vital geological agents that incise valleys, transport tonnes of solid and dissolved materials to the sea, and reshape our landscape over time. Stream behavior is affected by a variety of factors such as the steepness of the stream, the cross-sectional area of the stream, the velocity of water in the stream, and sediment load. The Reynolds number, Re, is a convenient parameter for describing the state of flow as laminar or turbulent. The dimensionless Reynolds number is defined as $Re = \rho UY/\mu$, where ρ is the density of water $[\mathrm{M\,L^{-3}}]$, U is the average flow velocity $[\mathrm{L\,t^{-1}}]$, Y is the average flow depth [L], and μ is the dynamic viscosity of the fluid $[\mathrm{M\,L^{-1}\,t^{-1}}]$. Laminar flow occurs when Re is less than 500. Turbulent flow occurs when Re is greater than 2000 and circulating eddies form in turbulent regions. Transitional flow lies between laminar and turbulent flows. Actual stream flow is seldom laminar, but when the degree of turbulence is small, flow is often considered in the laminar range. The following mathematical formulation can then be used. The equation of motion can be expressed as $v = K_{\mathrm{ch}}(\mathrm{d}h/\mathrm{d}l)$, where v

is the flow velocity [$\mathrm{L\,t^{-1}}$], K_{ch} is the flow channel conductivity [$\mathrm{L\,t^{-1}}$], and $\mathrm{d}h/\mathrm{d}l$ is the head gradient, the change in head h with respect to change in distance l. The head represents the energy of the system and the gradient provides the driving force. The energy includes gravitational, kinetic, and pressure components. For turbulent flow, the flow dynamics is more complex and nonlinear and the equation of motion becomes $v = K_{\mathrm{ch}}(\mathrm{d}h/\mathrm{d}l)^{1/2}$.

Surface Water and Ground Water Interaction

In order to effectively manage water resources, we must understand the interaction between surface water and ground water. Conjunctive use of surface and ground water has increasingly become common practice in water management, particularly in arid regions. The basic concept of conjunctive use is to store water in aquifers under wet climate conditions when streams flows are high and to withdraw water from the aquifers under drier conditions when demand is high but stream flows are low. The success of a conjunctive-use project depends heavily on the dynamics of the interaction between the surface water and the ground water. Streams can either gain water from or lose water to a ground water system. In reality, many streams do both at different locations and at different times. The rate and direction of flow into or out of the stream can also vary as the elevations of the water table and stream surface fluctuate. Pumping of ground water can change the quantity and direction of flow between an aquifer and a stream. As a result, stream flow is reduced, which can be of particular concern during droughts. Moreover, the interaction between ground water and surface water affects water quality. When the ground water in shallow aquifers is contaminated (for example, from agricultural practices of applying fertilizer and pesticides) the shallow aquifers can contaminate surface water as the ground water flows toward a stream. The opposite can also occur when a stream is heavily contaminated (for example, from mine waste drainage in mountainous regions); and relatively clean ground water can dilute the contaminant level as the ground water enters the stream from the gaining sections of the contaminated stream. Mixing of ground and surface water also has an effect on other natural environments such as wetlands and aquatic environments when acidity, temperature, and dissolved oxygen are altered by the interaction. Streams may become acidic as they receive atmospheric deposition of chemicals, such as sulfate and nitrate. Acidic precipitation directly affects the well-being of aquatic ecosystems. However, significant ground water flow into a stream may help neutralize and reduce the stream's acidity to tolerable levels for aquatic organisms.

Applied Aspects of Hydrology

While the scientific aspect of hydrology seeks understanding of the role that water plays in natural processes and mechanisms of water movement in the Earth system, the applied aspect relates to using the scientific knowledge to understand the impact and consequences of improper water use, and to better protect water resources. Although only two areas are discussed below, applied hydrology impacts on our society and environment far beyond what is included in this section.

Water Contamination

Water contamination has increasingly become a concern in modern times. Application of pesticides and fertilizer in agricultural practices can result in areal contamination as excess irrigation water percolates through soils and carries the chemicals into the ground water systems. Contamination sources that spread over large areas are known as nonpoint sources; sources confined to small areas are called point sources. Multiple closely spaced point sources can form a nonpoint source. Landfills are major point sources of water contamination. Aging and leaking liners around landfills allow leachate, a mixture of water and dissolved chemicals, to leak into water systems. Wastes from mines and nuclear facilities and petroleum spills have all contributed to the contamination of streams and ground water, thus degrading water quality. While it is necessary to identify sources of contamination, understanding the contaminant transport is equally important so that the future behavior of contaminant plumes can be predicted and effective remediation procedures can be designed. Three major mechanisms control the transport of contaminants in water: dispersion, advection, and retardation. Dispersion involves the dissolved contaminants diffusing into and mixing with clean waters. Advection – the transport of contaminants by moving water – is often the dominant mechanism once the contaminants make their way into permeable rocks. While dispersion and advection are the physical aspects of transport, retardation takes into consideration of the chemical reactions occurring between the contaminant and the media during its migration. Slowdown of the advancing plume can be caused by various chemical reactions such as sorption, precipitation, oxidation, ion exchange, or biological activities.

Land Subsidence

Extraction of ground water plays a direct role in land subsidence. Uneven subsidence of the historic tower of Pisa in Italy has created a tourist attraction. However, in many cities and towns subsidence has been a problem when building foundations and road surfaces become cracked and tilted as the ground subsides. When a large amount of water is withdrawn from the ground, void spaces in rock or soil collapse, the soil is compacted, and the ground subsides. The Santa Clara Valley and nearby San Joaquin Valley in northern California have experienced regional subsidence due to excessive ground water pumpage to sustain productive agriculture. As Las Vegas Valley turns into a fast-growing metropolitan area, ground water is rapidly being depleted and this area has also suffered problems with land subsidence and ground fissures due to horizontal movement. Subsidence may also occur from drainage of soils that are rich in organic carbon as microbial decomposition converts organic carbon to carbon dioxide gas and water. Subsidence at the rate of 20–80 mm per year has been observed as a result of the decomposition of the remains of shallow-water sedges and reeds in California's Sacramento–San Joaquin Delta and in Florida's Everglades. More catastrophic subsidence takes place with the formation of sinkholes associated with localized collapse of subsurface cavities. Often triggered by a decline in the ground water level, sinkholes typically form in areas underlain by carbonates (e.g., limestone and dolomites) and evaporites containing easily dissolved minerals such as salt, gypsum, and anhydrite.

Role of Ground Water in Geological Processes

Water exists in pore spaces in soil and rock from near the surface to a depth of more than 10 km. The amount of pore water decreases with depth in response to a general decrease of rock porosity with depth. Ground water plays an essential role in mineral dissolution and precipitation, and thus impacts metamorphic processes by altering the mineral compositions of rocks. As water flows through deep sections of the crust or passes through thermally active regions, such as in the vicinity of cooling igneous rocks solidifying from magma, the heated waters become hydrothermal waters. Hot springs emerge at the locations of hydrothermal water discharge. Ground water carries dissolved minerals and transports them to ore-forming locations. Petroleum is matured, in natural gas or crude oil states, at a greater depth of burial. Ground water may then transport petroleum to shallow locations, a process known in petroleum system studies as secondary migration. The mechanical interaction between ground water and rock deformation has been thought to contribute to the triggering of earthquakes. As pore pressures in faults and the surrounding area increase, faults become lubricated and the fault is weakened, setting the stage for an earthquake. The best-known example is that of the documented earthquakes between 1962 and 1972 in the Denver area when liquid waste was injected underground into fractured granites a few kilometers deep. The time and frequency of the earthquakes were correlated strongly with the time and volume of waste injection.

See also

Agricultural Meteorology and Climatology. **Climate:** Overview. **Hydrology:** Modeling and Prediction; Overview. **Palmer Drought Index**.

Further Reading

Alley WM, Reilly TE and Franke OL (1999) *Sustainability of Ground-Water Resources*. US Geological Survey Circular 1186. Washington, DC: US Geological Survey.

Dingman SL (1994) *Physical Hydrology*. New York: Macmillan.

Domenico PA and Schwartz FW (1998) *Physical and Chemical Hydrogeology*. New York: Wiley.

Fetter CW (2001) *Applied Hydrogeology*. Englewood Cliffs, NJ: Prentice-Hall.

Fetter CW (1999) *Contaminant Hydrogeology*. Englewood Cliffs, NJ: Prentice-Hall.

Freeze AR and Cherry J (1979) *Groundwater*. Englewood Cliffs, NJ: Prentice-Hall.

Gleick PH (2001) Making every drop count. *Scientific American* February, p. 41.

Herschy RW and Fairbridge RW (1998) *Encyclopedia of Hydrology and Water Resources*. Boston: Kluwer Academic.

Hornberger GM, Raffensperger JP, Wiberg PL and Eshleman KN (1998) *Elements of Physical Hydrogeology*. Baltimore, MD: The Johns Hopkins University Press.

Rodriguez-Iturbe I and Rinaldo A (2001) *Fractal River Basins*. New York: Cambridge University Press.

Solley WB, Pierce RR and Perlman HA (1998) *Estimated Use of Water in the United States in 1995*. US Geological Survey Circular 1200. Washington, DC: US Geological Survey.

Tindall JA and Kunkel JR (1999) *Unsaturated Zone Hydrology*. Englewood Cliffs, NJ: Prentice-Hall.

Vorosmarty CJ, Green P, Salisbury J and Lammers RB (2000) Global water resources: vulnerability from climate change and population growth. *Science* 289: 284–288.

Winter T (1998) *Ground Water and Surface Water, A Single Resource*. US Geological Survey Circular 1139. Washington, DC: US Geological Survey.

Modeling and Prediction

Z Yu, University of Nevada–Las Vegas, Las Vegas, NV, USA

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

Hydrological models have become an indispensable tool for study of hydrological processes and the impact of modern anthropogenic factors on the hydrological system. Mathematical models that are governed by the laws for conservations of mass and momentum are used to describe the temporal and spatial variation of a hydrological system in the field on the basis of information concerning climate, land use and land cover, and hydrology. Modeling the hydrological response to various natural and anthropogenic forcings has the potential to contribute to the understanding of these physical processes, such as flow and transport in the surface and subsurface and the atmosphere–land surface interaction.

Two types of hydrological models have been used in most applications: lumped conceptual models and physically based models. A lumped model is generally applied in a single point or a region for the simulation of various hydrological processes. The parameters used in the lumped model represent spatially averaged characteristics in a hydrological system and are often unable to be directly compared with field measurements. In general, lumped models use simple bookkeeping procedures to quantify physical processes by simulating the temporal variation of various physical processes in a hydrological system. The advantage of these models over physically based models is that the conceptual parameterization in the models is simple and computation is efficient. With the availability of spatially distributed digital and remotely sensed data sets of features such as precipitation, elevation, vegetation, etc., many distributed lumped models have been developed in recent years. These kinds of models have been widely used in most climate and meteorological studies to model hydrological processes.

Many physically based distributed-parameter models have been developed to facilitate various hydrological and climatic applications over recent years. These models represent hydrological processes in a physically rigorous manner because they use process-based partial differential equations (PDEs) to describe the spatial variability of hydrological processes. Within the framework provided by these equations, it is possible to estimate governing parameters through field studies. One disadvantage of such models is that the representation of physical processes in these models is often too crude and the scales of measurement for many hydrological parameters are incompatible with the scales used in the models. Studies have indicated that the uniform effective parameters (e.g., saturated hydraulic conductivity) cannot represent the ensemble heterogeneity, resulting in a significant underestimation of hydrological variables (e.g., streamflow). These physically based distributed models can be referred to as conceptual lumped models in some sense. The physically based models are more complex and require more computing time for solving PDEs numerically and considerable effort to master all their intricacies, such as model calibration. For simulating the hydrological response (e.g., soil moisture) to climate forcing (e.g., storms and human-induced global warming), these models currently offer no advantage over the traditional conceptual lumped water-balance models.

Physically based hydrological models are an important evolutionary step in representing hydrological processes and spatially distributed data. At present, the ability to represent spatially varying processes is outstripping our ability to collect various data sets. The need for the research on the better representation of physical processes in space and time is evident given the availability of digital products (e.g., distributions of elevation, soil, vegetation) and remotely sensed data (e.g., soil moisture, vegetation), along with new technologies for measuring temporal and spatial variability in precipitation.

Research on data assimilation and analysis, sub-grid-scale variability in precipitation and hydraulic parameters, sub-grid-scale model calibration, prediction uncertainty analysis, and inclusion of fine-resolution surface and subsurface hydrology in various hydrological and climatic models is being conducted in various research institutes to facilitate an interactive link between the hydrology and atmosphere. Simulation and prediction of various hydrological systems are among the principal areas of study in the current fields of hydrology and climate.

Hydrological Simulation in Meteorological and Climatic Models

Climate variability has substantial impact on hydrological, biological, and ecological systems such as water availability and quality, floods, and droughts. The effects of climate variability on economic vitality and the quality of life (e.g., the 1988 North American

drought and the 1993 Mississippi flood) indicate that future climate change will be of considerable global, national, and regional importance. The current emphasis on the climate impact analysis and hydrological modeling requires improved specifications of the interactions between the atmosphere and hydrosphere. The difficulty in reconciling the discordant scales in climate change and hydrological research hampers understanding of how regional hydrological systems respond to climate forcing. Part of the difficulty can be attributed to insufficient progress on the theoretical side, related to deficiencies in our abilities to couple atmospheric and hydrological processes that occur over spatial scales ranging from microscopic to global (specifically ranging from 10^{-4} to 10^{6} m) and over time scales ranging from 10^{-4} to 10^{7} s. Various parameterization schemes have been used to reconcile the discordant scales of hydrological and atmospheric models. The approach involves numerical modeling to bridge the gap in scale between general circulation models (GCM) and hydrological models by embedding a regional climate model (RCM) within a GCM so that the natural variability at the regional scale can be represented within the system (**Figure 1**).

The nested climate models used for studying the effect of climate change on the hydrological system indicate that changes in the hydrological system induced by a climatic change could lead to a further change in climate, ecosystems, agriculture, and hydrology. Many comprehensive hydrological models such as the Biosphere–Atmosphere Transfer Scheme (BATS) have been developed to integrate hydrology into climate system by coupling the physical processes from different systems. Such an integrated hydrological model system, including components from soil hydrology, surface water hydrology, and ground water hydrology, is necessary to understand how perturbed climate conditions can modify regional hydrological systems and, in turn, how the modified hydrological conditions can modify the climatic system.

Traditionally, most hydrological models have been developed to simulate individual hydrological processes such as rainfall–runoff partitioning, soil moisture flow, overland flow, and ground water flow, while some comprehensive models are used to simulate all the processes within the hydrological cycle. In general, any use of hydrological models follows five main steps (**Figure 2**). After the purpose of hydrological modeling is established, the conceptual model can then be

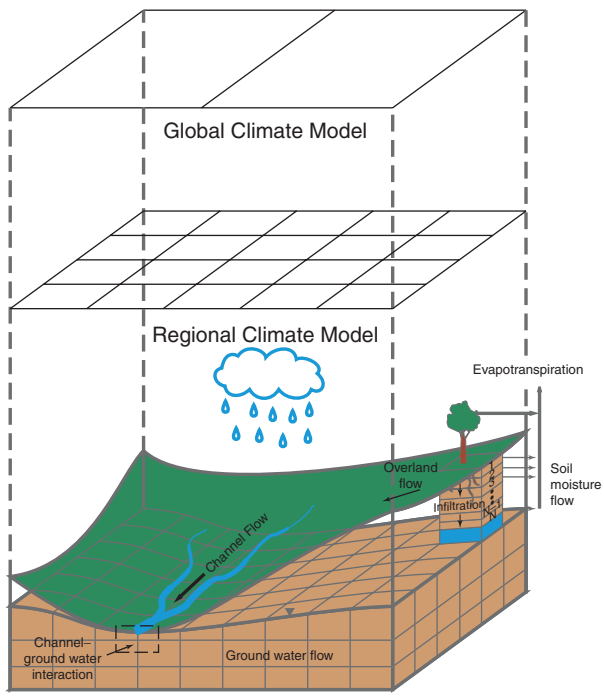

Figure 1 Frame structure of coupled climate and hydrology models. (Reproduced from Yu *et al.* (2000).)

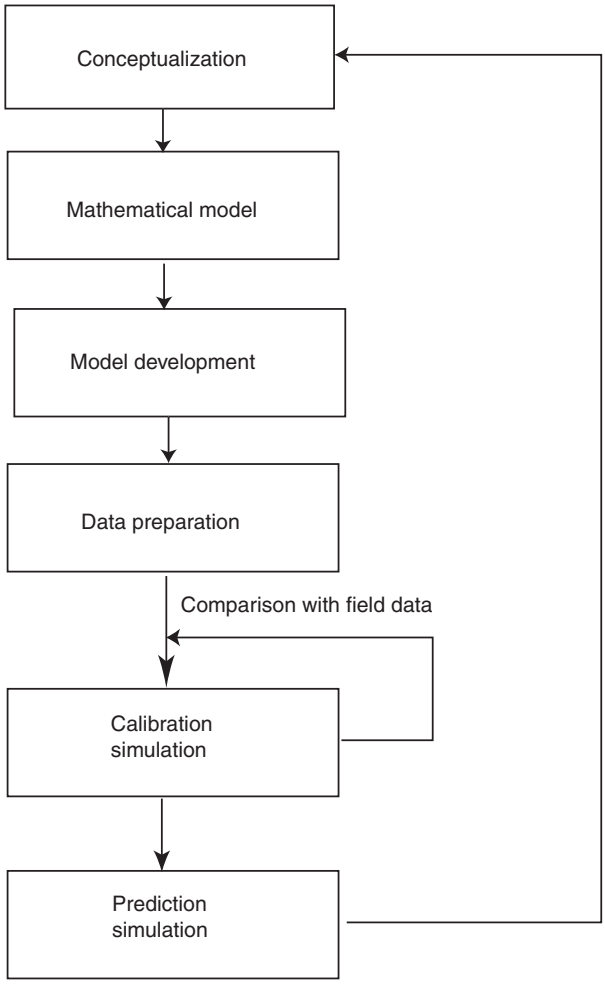

Figure 2 Flow chart of hydrological model application.

constructed to describe the hydrological system that needs to be simulated. In the second step, the various mathematical models can be used to describe individual processes within the hydrological system and computer codes in various computer languages (e.g., C, Fortran) can be developed to solve the PDEs in the mathematical models. In general, the numerical solution of the PDEs must be verified by comparison with known or analytic solutions. The second step can also be completed by selecting existing computer codes that have been verified in previous applications. The third step, data preparation, is to compile and analyze available meteorological and hydrological data for the hydrological simulation. Additional data can be obtained through laboratory studies and field work. After all the necessary data have been integrated into the models, the models should be calibrated to available field-observed data (i.e., streamflow, ground water level). The calibrated models can then be used to predict the hydrological response to various external forcings (e.g., future water usage, land use–land cover change, and climate change).

Conceptualization, Mathematical Models, and Model Development

There are many surface and subsurface hydrological processes in the hydrological system (**Figure 1**). Many conceptual and mathematical models have been developed to describe these physical processes in time and space and various numerical schemes are available to solve various equations for the simulation of hydrological variables. The following sections describe some of major flow processes such as soil water flow, overland flow, and ground water flow, while other processes such as sediment and contaminant transport will not be described here.

When the precipitation reaches the ground, it partitions into components of surface runoff and infiltration. After the water enters the soil column, the soil water flow in the unsaturated zone can be described by the following form of the Richards equation (eqn [1]).

$$-\frac{\partial q(z,t)}{\partial z} = \frac{\partial}{\partial z}\left[K(z,t)\frac{\partial \psi(z,t)}{\partial \theta}\frac{\partial \theta(z,t)}{\partial z}\right] + \frac{\partial K(z,t)}{\partial \theta}\frac{\partial \theta(z,t)}{\partial z} \qquad [1]$$

In eqn [1], where q is the vertical moisture flux; θ is the volumetric water content; z is depth, t is time; K is the unsaturated hydraulic conductivity; and ψ is the hydraulic potential. The Richards equation can be solved numerically using the Crank–Nicholson numerical scheme and a finite difference scheme of forward in time and backward in space, or other numerical methods. Various schemes are available for relating K and ψ to the volumetric water content because the K at a given time is a function of constant saturated hydraulic conductivity and soil moisture status. Infiltration and evaporation can be treated as a source or a sink, respectively, in the Richards equation or the upper boundary condition of the soil profile. The calculation of evaporation on the bare soil and evapotranspiration on the vegetation canopy can be completed with various methods (e.g., the Penman–Monteith method).

The overland flow will form soon after the rainfall–runoff partitioning and can be formulated as a kinematic wave with a flow-direction algorithm to account for the overland-flow delay and storage in each grid cell. Mathematically, one-dimensional overland flow may be described by eqn [2],

$$\frac{\partial d}{\partial t} + \frac{\partial q}{\partial x} = i_e, \quad q = \alpha d^m \qquad [2]$$

where $d = d_{(x,t)}$ is the depth of overland flow, $q = q_{(x,t)}$ is the rate of overland flow, i_e is the excess rainfall rate, α is conveyance, and m is surface roughness. The overland flow is discretized into a large number of smaller segments to solve the kinematic-wave flow and the equation can be solved numerically for the one-dimensional flow. In general, a conservative numerical scheme is implemented in the model to maintain the numerical stability in some extreme cases.

The overland flow quickly reaches nearby streams to form channel flow. The channel flow can be simulated hydraulically and hydrologically. The Muskingum–Cunge method or other channel routing schemes can be used for channel-flow routing through the channel networks, which are derived from digital elevation models (DEMs). For instance, the extended finite difference form of the governing equation can be written in the Muskingum channel routing method as eqn [3], where Q_x is the inflow to a given stream reach, Q_{x+1} is the outflow from a given stream reach, k is a travel time parameter, and ε is a weighting factor.

$$k\frac{\mathrm{d}}{\mathrm{d}t}[\varepsilon Q_x + (1-\varepsilon Q)_{x+1}] = Q_x - Q_{x+1} \qquad [3]$$

The routing equations are produced using a finite difference form of the above equation. The outflow of a stream reach is calculated at each time step and routed to downstream reach along DEM-derived channel network, eventually to the basin outlet.

The second-order PDE of eqn [4] is used to describe ground water flow in an aquifer, where T is transmissivity, h is hydraulic head, S is storativity, t is

time, and Q_{net} is net ground water withdrawal rate.

$$\frac{\partial}{\partial x}\left(T\frac{\partial h}{\partial x}\right)+\frac{\partial}{\partial y}\left(T\frac{\partial h}{\partial y}\right)=S\frac{\partial h}{\partial t}+Q_{net} \qquad [4]$$

Q_{net} includes ground water recharge from infiltration, evaporation of shallow ground water, withdrawal of ground water from wells, and other sources or sinks, such as induced infiltration of ground water from the stream network.

In general, the simulation domain for a basin is discretized into a set of rectangular cells with dimensions Δz, Δx, and Δy. Δz is the vertical thickness of the aquifer, and Δx and Δy are the cell dimensions in the x- and y-coordinate directions (**Figure 1**). A finite difference or finite element scheme can be used to discretize the hydrological system. Many iterative methods such as the method in MODFLOW (developed by US Geological Survey) are available for obtaining solutions for the above PDE.

The interaction between a stream system and ground water can be simulated with Darcy's law. The calculated inflow/outflow is a function of the hydraulic head difference between the ground water and the river stage at a given stream reach. When the ground water level is lower than the stream bed, water flows from the stream to the ground water. When the ground water level is lower than the stream bed, the stream will dry out and eventually there will be no channel–ground water interaction.

After the implementation of numerical schemes for solving various PDEs in the hydrological system, the developed model codes need to be tested against known and analytic solutions before they can be used for modeling applications. For specific applications, the simulation domain must be set up with defined boundary conditions, so that the given forcings (i.e., meteorological data and water usage) can be used to drive the simulation of various hydrological processes.

Data Preparation

Data preparation is a major fundamental part of hydrological modeling. The work involves compiling various available field and digital data sets and processing and integrating these data sets into models for hydrological simulations. Sources of data can be from various government agencies and institutes. With the availability of new technology, ground-based and remotely sensed data sets as well as data collected through traditional field methods have been used for hydrological modeling.

For topography-based hydrological models, digital elevation models (DEMs) are routinely used to describe the land-surface topography, which contains crucial information for the surface water flow and the interaction of surface water and ground water. After DEMs have been conditioned to remove spurious sinks and peaks, they can be used to determine the flow direction of surface water at a grid cell by comparing the elevation at a grid cell to the elevations of neighboring grid cells within the simulation domain, assuming that the water always flows from high to low. With DEMs, a flow accumulation data set can be obtained in which the contributing area that drains to a grid cell is determined for each grid cell. The DEMs and derived data sets of flow direction and flow accumulation can then be processed to delineate drainage and stream networks, overland flow paths, watershed and subwatershed boundaries, and other features. Derived stream networks facilitate the channel–shallow ground water interaction, while the derived overland flow paths provide accurate prediction of the rainfall–runoff partitioning and soil moisture status over overland flow paths.

Simulations of various hydrological processes require spatially distributed information on land use and land cover, soil type, and hydraulic properties in addition to the topography. Many digital soil data sets are available from various sources. For instance, the Natural Resources Conservation Service (NRCS) of the US Department of Agriculture compiled the State Soil Geographic (STATSGO) soils data. In general, these digital data sets are stored in a Geographical Information System (GIS) format. The hydraulic parameters (i.e., the average saturated hydraulic conductivity and the average capillary suction) data sets can be derived by assigning a value for each grid cell based on various parameterization schemes of hydraulic variables and the soil texture from STATSGO soil data. These derived data sets of various hydraulic parameters can then be integrated into models for the hydrological simulation.

With the development of remote sensing and satellite technology, these data sets hold great potential for practical application to regional ecology, hydrology, and planning. The satellite data can be used to derive a variety of surface parameters, such as radiant surface temperature and vegetation fraction; these variables can be used for both spatial and temporal comparisons, and for regional hydrological modeling of important hydrological processes. The remotely sensed data such as vegetation classification can be obtained from the AVHRR (Advanced Very High Resolution Radiometer) and Landsat TM images. Spatially distributed data of meteorological variables (e.g., precipitation, temperature, wind speed) are available from NEXRAD (Next Generation Weather Radar), radar and satellite images at a time resolution from minutes to days. With the improved resolution of

these data sets, more accurate simulation of various hydrological processes is expected in both temporal and spatial scales.

Different types of field-observed data are available for hydrological simulation and model calibration and will be discussed in the next section. There are networks of rain gauges across the world. At outlets of many river systems, gauge stations are available to measure daily streamflow rates and other hydrological variables. Ground water levels are also available from many wells and many other data such as evapotranspiration and moisture content are obtainable from local specific regions. For instance, the US Geological Survey (USGS) provides infrastructure for monitoring stations and collecting measured data of streamflow and ground water levels while the National Oceanic and Atmospheric Administration (NOAA) maintains the collection of meteorological data.

Model Calibration

Complexity in hydrological modeling over large space and long times has prompted a significant need for model calibration. Model calibration is a demonstration that the model is capable of reproducing field-observed values of various hydrological variables (e.g., streamflow, soil moisture, and well observed ground water level) (**Figure 3**). Prediction of various hydrological variables on the basis of an uncalibrated flow model are sterile and indefensible. Generally, the goodness-of-fit between simulated and measured variables on the basis of the initial values of hydrological and hydraulic parameters used in the model is not satisfactory. The goodness-of-fit can be improved by adjustment of the values of these parameters until the difference between simulated and measured variables is satisfactory during the model calibration. The adjustment process is most commonly based on trial-and-error changes in a parameter while other parameter values are held constant. Some numerical models are now equipped with a semiautomated or automated procedure. The range of adjustment to values of hydrological and hydraulic parameters must be constrained by plausible site-specific field data such as streamflow, water levels, hydraulic conductivity, etc. The difficulty in achieving a good calibration is that boundary conditions and values of hydrological and hydraulic parameters are always subject to uncertainty.

Goodness-of-fit calibration can be evaluated through visual comparison and statistical measures. Visual comparison includes scatterplots of simulated versus measured variables, simulated and field-based temporal and spatial distribution, and spatial distribution of residuals. Statistical measures consist of mean error, absolute mean error, and root mean squared error between simulated and observed variables.

The trial-and-error calibration involves manually adjusting parameters to match the simulated results (e.g., streamflow and ground water level hydrographs) with field-observed historical records. The fit between simulation and observation can be evaluated visually and statistically after each simulation with changed parameters. Knowledgeable modelers can conduct acceptable model calibration using this approach. On the other hand, the calibration process can be a very frustrating and time-consuming practice for many untrained modelers and sometimes even for

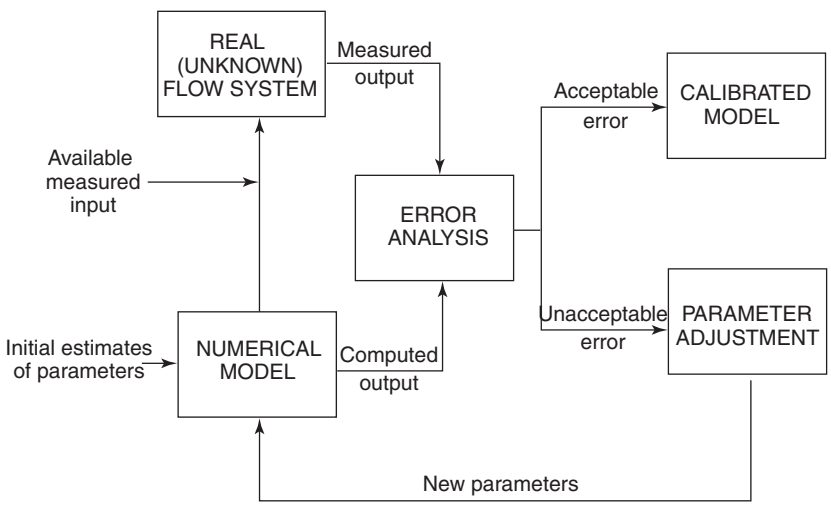

Figure 3 The procedure of model calibration. (Reproduced from Freeze and Cherry (1979).)

experienced modelers when many parameters are changed simultaneously. The disadvantage of such approach is the low efficiency and the lack of standard measures in evaluating the performance of model calibration.

While the trial-and-error calibration is still the most common method for calibrating models of all kinds, many automatic or semiautomatic direct or indirect techniques are available for solving an inverse problem in various hydrological systems. Automatic direct or indirect model calibrations require the setting up of objective functions, development of optimization algorithms, setting up of termination criteria, and of course, using observed data as calibration targets. The direct approaches use an inverse operator to solve the problems directly, while the indirect approaches use an iterative scheme. For the indirect approaches, various sampling procedure schemes (e.g., Monte Carlo and Latin Hypercube samplings) are available for sampling that cover the range of adjustable variables systematically. In effect, the simulation problem is solved many times over with coverage of the potential solution space. Ultimately, the process finds the optimum solution, providing a set of parameters that yield the best fit between the observed data and simulated results.

For any model calibration, objective functions need to be set up for variables (e.g., streamflow, ground water level) that will be optimized during the calibration process. The least-squares method, a common method in the model calibration, is equivalent to the minimization of the sum of squares of the residuals (also called objective function or fitness measure). The general form for a steady state is given by eqn [5],

$$\mathrm{Obj} = \sum_{i=1}^{n} w_i [s_i^{\mathrm{o}} - s_i^{\mathrm{c}}]^2 \qquad [5]$$

where w_i is a weighting factor, s_i^{o} is the observed data, s_i^{c} is a simulated result, i is the index for location points, and n is the number of location points.

The weighting factor represents the proportional importance allocated to a variable at particular locations. If the sum of all weighting factors is 1.0, eqn [5] becomes the simple least-squares function. It is feasible to zone spatially for both steady state and transient state, although execution times will increase significantly, especially for the transient state, as the degrees of freedom in parameter values increase.

The objective function can be constructed for hydrological variables that can be used as calibration targets (e.g., streamflow and ground water level). Objective functions, based on a weighted least-square error criterion, can then be used to drive the calibration process. Different combinations of various changes in different optimizing parameters are selected for each simulation run. The objective is to minimize the difference between measured data and simulated results. The objective function for a transient simulation can also be formed as eqn [6], where m is the number of time steps, j is the index for time steps, and other variables are as defined previously.

$$\mathrm{Obj} = \sum_{j=1}^{m} \sum_{i=1}^{n} w_{i,j} [s_{i,j}^{\mathrm{o}} - s_{i,j}^{\mathrm{c}}]^2 \qquad [6]$$

The transient calibration requires much more computational time than the steady-state calibration as a result of the additional time dimension. There are other alternatives to the least-squares method for objective functions, such as the maximum likelihood approach.

The direct approaches use an inverse operator to solve the problems directly. Let h^{o} be the observed hydrological variables and h^{c} be the simulated hydrological variables. Then, according to a Taylor's series expansion, the equation can be written as eqn [7].

$$\sum_{j=1}^{m} \frac{\partial h_i^{\mathrm{c}}}{\partial p_j} \delta p_j = h_i^{\mathrm{o}}(p + \delta p) - h_i^{\mathrm{c}}(p), \quad i = 1, \ldots, n \qquad [7]$$

In matrix form, eqn [7] is written as eqn [8],

$$\mathbf{J}\,\delta\mathbf{p} = \mathbf{r} \qquad [8]$$

where $\mathbf{J}$ is a rectangular Jacobian matrix, $\mathbf{p}$ is the parameter vector to be inverted, $\delta\mathbf{p}$ is the parameter correction vector, and $\mathbf{r}$ is the residual vector between the measured and simulated hydraulic heads. In order to solve the equation, both sides of the equation are multiplied by the transpose of matrix $\mathbf{J}$. The result is a system of equations with m unknowns. There is a problem because the solution of the matrix system is unstable. Marquardt's approach, which involves adding terms to the diagonals of the matrix $\mathbf{J}^{\mathrm{T}}\mathbf{J}$, can overcome this problem. The Levenberg–Marquardt method or Marquardt method (iterative procedures) can be used to solve this system.

After the model calibration, a sensitivity analysis is needed to investigate the uncertainty in the calibrated model caused by the uncertainty in the estimation of various parameters. The sensitivity analysis can be conducted by evaluating the hydrological flow system in response to various parameter disturbances based on the calibrated hydraulic parameters. The analysis involves perturbing the values of the hydraulic

parameters with respect to the best estimates and examining how the simulated hydrological variables change. The common approach is to evaluate one parameter at a time; all other parameters are kept constant while a given parameter is being evaluated. Then various parameters are varied simultaneously to examine the behavior of the hydrological flow system under different situations. In this way, the nonuniqueness of solutions of the parameter estimations for the hydrological flow system can be evaluated and other issues along with the model uncertainty can be assessed.

Prediction

With the set of optimized hydraulic parameters from the model calibration, the calibrated model can be used to conduct predictive simulations. The predictive simulations can be driven by external forcings (e.g., climate change scenarios) to derive the system response to future events. Some climate and hydrology problems require the hydrological modeling to predict the system response for as many as thousands of years. For instance, the risk assessment of contaminant transport in the storage of low-level nuclear wastes in arid regions requires prediction of the water flow in the unsaturated and saturated zones for 10 000 years.

How good predictions are depends on how well the model is calibrated and how credible are future external forcings that are used to drive predictive simulations. Because of the nonlinearity of natural hydrological systems and simplified model structures as compared to the actual complex natural physical processes, the calibration process cannot guarantee that the global minimum can be found, and so the calibrated model could produce unrealistic results beyond the confident period. In general, the calibrated model should not be used to predict the system in the future longer than twice the period of model calibration with available field-observed data. In some cases, owing to the lack of available long-period historical data for model calibration and the nature of some applications that require an extended period of prediction, there will always be uncertainty in predictive simulations. For this reason, users must conduct a detailed assessment of the uncertainty of any future prediction of system response as well as the uncertainty associated with model calibration and future external forcings.

The hydrological cycle plays an important role in the climate system. The two-way interaction between hydrology and climate is a crucial element in our current and future modeling of hydrology and climate for an understanding of how perturbed climate conditions can modify regional hydrological systems and, in turn, how the modified hydrological conditions can influence the climatic system. Data assimilation and analysis of remotely sensed and satellite-derived information should be integrated into hydrological models to better represent sub-grid-scale spatial and temporal variability in meteorological and hydrological parameters. Model calibration and sub-grid-scale model calibration remain important steps in the better simulation of various hydrological components. In addition, prediction uncertainty analysis and inclusion of fine-resolution surface and subsurface hydrology are important concerns in such an interactive link.

Glossary

Calibration The process for selecting proper parameters for model simulation.

Conceptualization A pictural representation of a hydrological flow system.

Darcy's law Darcy's law states that the discharge flowing through a porous medium is proportional to the hydraulic gradient.

DEM Digital elevation model. DEMs contain elevation data for describing the topography of a region.

Evapotranspiration The processes that return water vapor to the atmosphere directly from the soil and by the action of plants.

Ground water Water found in the saturated zone of the soil and bedrock.

Hydraulic conductivity The ability of a porous medium to transmit fluid.

Hydrological cycle The global-scale process that circulates water between the atmosphere, continents, and oceans.

Hydrology The study of the occurrence and movement of surface and subsurface water.

Infiltration The movement of water (e.g., precipitation, melting snow) into the soil at the land surface.

Model Any device that represents an approximation of a field situation.

Muskingum A method for routing streamflow from upstream to downstream.

Numerical method A method for solving partial differential equations by transforming them into a set of algebraic equations that can be solved with a computer.

Overland flow The water flowing on the surfaces of hillslopes after rainfall.

Prediction The simulation for speculating the response of a system to potential changes in the system.

Richards equation An expression of mass conservation incorporating Darcy's law in the unsaturated zone.

Runoff Excess precipitation water on the surface after rainfall.

Saturated zone A subsurface region where pores are completely filled with water.

Sensitivity analysis A process for identifying the sensitivity of the model response to changes in input parameters.

Simulation Use of computer codes to model field-observed physical processes.

Soil moisture Water held in soils and rocks in the weathered unsaturated zone.

See also

Agricultural Meteorology and Climatology. **Climate:** Overview. **Flooding**. **General Circulation:** Models. **Global Change:** Human Impact of Climate Change. **Hydrology:** Ground and Surface Water; Overview; Soil Moisture. **Land–Atmosphere Interactions:** Overview. **Numerical Models:** Methods. **Weather Prediction:** Regional Prediction Models.

Further Reading

Anderson MP and Woessner WW (1992) *Applied Groundwater Modeling*. New York: Academic Press.

Beven KJ and Moore ID (1993) *Terrain Analysis and Distributed Modeling in Hydrology*. New York: Wiley.

Freeze RA and Cherry JA (1979) *Groundwater*. Englewood Cliffs, NJ: Prentice-Hall.

Maidment DR (1993) *Handbook of Hydrology*. New York: McGraw-Hill.

Neuman SP (1973) Calibration of distributed parameter ground water flow models viewed as a multiple-objective decision process uncertainty. *Water Resources Research* 9: 1006–1021.

Singh VP (1995) *Computer Models of Watershed Hydrology*. Water Resource Publications.

Yu Z and Schwartz FW (1999) Automated calibration applied to constrained ground water flow modeling. *Hydrological Processes* 13: 191–209.

Yu Z, Barron EJ and Schwartz FW (2000) Retrospective simulation of a storm event: a first step in coupled climate/hydrologic modeling. *Geophysical Research Letters* 27: 2561–2565.

Yu Z, Lakhtakia MN, Yarnal B, *et al*. (1999) Simulation of the hydrologic response to atmospheric forcing in large river basins: linking a mesoscale meteorological model and a hydrologic model system. *Journal of Hydrology* 218: 72–91.

Soil Moisture

A Robock, Rutgers University, New Brunswick, NJ, USA

Copyright 2003 Elsevier Science Ltd. All Rights Reserved.

Introduction

Soil moisture is the amount of water in the active layer of the soil, typically the top 1–2 m. It is extremely important because it is the main source of water for agriculture and natural vegetation. Near-surface soil moisture also controls the partitioning of available energy at the surface into sensible and latent heat exchange with the atmosphere, thus linking the water and energy balances through the moisture and temperature states of the soil. Soil moisture is the source of water that evaporates and transpires from the soil and vegetation into the atmosphere, thus affecting the distribution of clouds and precipitation. Surface temperature is controlled by soil moisture, as a wetter surface will be cooler, with more of the available energy going into evapotranspiration (evaporation and transpiration) rather than heating the surface. Soil moisture also affects runoff, determining how much precipitation or snowmelt goes immediately into rivers and streams, or in extreme cases into flooding. A deficit of soil moisture is often connected to drought, and soil moisture interactions with the atmosphere may be important in maintaining droughts. Soil moisture, along with snow cover, are also the most important components of meteorological memory for the climate system over the land. Thus, the soil moisture state is an important predictor of monthly to seasonal climate variations.

Soil moisture is measured in a variety of ways, from the simple gravimetric technique to more complex electronic instruments. For the top 1 m soil layers, the temporal scale of soil moisture variation in the Northern Hemisphere midlatitudes is 1.5 to 2 months and the spatial scale about 500 km. Despite the importance of soil moisture, parameterization schemes used in weather forecast and climate simulation models, in general, do not capture the observed soil moisture variations when forced with either model-generated or observed meteorology. During the twenty-first century, a significant vulnerability for

human society may be summer drying in the midlatitudes. Observations over the past several decades, however, show that for the stations with the longest records, summer soil moisture in the top 1 m has increased while surface temperatures have risen. The increasing trend in precipitation so far has more than compensated for the enhanced evaporation.

Definition of Soil Moisture

Soil moisture is the amount of water in the upper layer of soil that interacts with the atmosphere. The depth of this active layer varies with soil type and vegetation, but is typically about the top 1 m. Soil moisture can be expressed in different units. The most common are as plant-available volumetric (W) or total volumetric soil moisture (W_T), expressed as the depth of a column of water contained in a given depth of soil, or as the volumetric percentage of water in a given soil depth. A fraction, typically less than half, of soil consists of pores that can be filled with air or water. This fraction is called the porosity (P). If this fraction were completely filled with water, the soil would contain its total water-holding capacity (W_0), and the water table would be at the surface. For any layer of depth (thickness) D, $W_0 = PD$. Were the soil saturated, so that $W_T = W_0$, and then gravitational drainage allowed to occur until it was negligible, the amount of water left in the soil would be the field capacity (W_f). If vegetation then extracted as much water as possible until it wilted, the remaining soil moisture would be the wilting level (W_*), and this amount of water is unavailable to plants. The plant-available soil moisture is $W = W_T - W_*$.

Measurement of Soil Moisture

There are many different techniques for measuring soil moisture. The choice of a particular method depends on the application and the resources available. Here, the principal ones are described briefly.

Gravimetric

The gravimetric method, also called the thermostat-weight technique, has been in use for a long time. Soil samples are taken using coring devices or augers at required depths and locations. Typically (in the Russian method) segments 10 cm long are extracted down to a depth of 1 or 1.5 m and a smaller sample removed from each segment. The sample is weighed, oven-dried, and weighed again. The difference in mass gives the total soil moisture in the sample, which is converted to volumetric units using the density of the soil. The wilting level is then subtracted, giving plant-available soil moisture, expressed as depth of liquid water.

Under the leadership of the State Hydrological Institute, the Russians established a system of soil moisture observations throughout the former Soviet Union, with a network of several thousand stations using the gravimetric technique. Typically, samples are taken at four different locations at a station each 10 days. After 10 days, the cores are replaced in the holes and samples taken from other holes. In this way, after some time the effects of sampling disappear and the same location can be reused. In very cold regions, observations were taken either once a month or not at all. Influenced by the Soviet Union, other countries also set up soil moisture observation programs, and extensive data sets now also exist for China and Mongolia.

The gravimetric method is low-tech and simple, making it excellent for long, homogeneous climatological records. As it is labor-intensive, and somewhat destructive to its site, new electronic methods are being introduced which are indirect and require calibration and theoretical assumptions. With suitable parallel measurements, the new methods can produce useful long records, and are briefly described below. However, all the electronic methods suffer from some problems and require extensive calibration with gravimetric measurements that span the time scales of the seasonal cycle and interannual variations. Furthermore, hysteresis effects may give ambiguous readings and some techniques may be less accurate in certain soils and for extremely dry or wet soil.

Neutron Probe

The neutron probe is relatively easy to use, accurate, and capable of measurements in real time. A probe with a fast neutron source is placed on the surface or lowered in an access tube (transparent to the neutrons), and the backscattered slow neutrons measured. The backscattered flux of slow neutrons is proportional to the density of hydrogen atoms. Water is the major source of hydrogen atoms that changes with time, so the neutron probe provides a good measure of soil water content. Calibration of slow neutron counts with gravimetric samples of soil moisture content and bulk densities yields a relationship to estimate the volumetric soil moisture content. Since radioactive scattering occurs over a spherical domain, a neutron probe samples a volume of soil rather than a point. As this volume of influence depends on soil moisture content, there are differences in the soil volumes sampled in dry and wet soils. These differences are generally small compared with the total volume

sampled by the probe, but they influence the depth resolution of the probe. The probe's relatively large volume of influence makes observations at shallow depths prone to errors, as adjoining air is also sampled. Disadvantages of neutron probes include that they are also labor-intensive, the need for precautions associated with handling radioactive material, and the relatively high costs. Starting in 1981, an extensive data set of soil moisture observations from Illinois was made with neutron probes, with observations made two to three times per month year round.

Heat Dissipation Sensors

These sensors make point measurement of soil moisture tension by measuring temperature changes in response to a heat pulse. A small ceramic block with an embedded sensor is briefly heated. The rate of heating is affected by the ability of the ceramic block to dissipate the heat. This is related to its soil moisture content, which has equilibrated with the surrounding soil. The measured heating rate must be calibrated, and soil moisture tension related to volumetric water content, by gravimetric observations for each location or with theoretical relationships. These sensors are relatively inexpensive, and can produce measurements every 30 minutes. An extensive network in Oklahoma now uses these sensors. Once installed and calibrated, a very labor-intensive effort, an advantage of these sensors is that they are automatic and can take observations of both soil temperature and soil moisture with high temporal resolution.

Other Sensors

Other soil moisture measurement techniques include: the tensiometer (a bulb of porous ceramic material is placed inside the soil and connected to a water-filled tube, which is used to measure soil moisture tension after allowing the system to equilibrate); the gypsum block (small cylindrical gypsum blocks embedded with electrodes are buried at required depths in the soil, and measure the electrical resistance, which is related to the water content); time domain reflectometry (TDR; based on monitoring changes in the dielectric properties of the soil at microwave frequencies); frequency domain reflectometry (FDR; similar to TDR, except that it derives soil moisture content based on changes in the frequency of signals due to the dielectric properties of the soil); and gamma densitometry (based on the relatively greater gamma radiation attenuation factor of water compared with other soil components). Each technique has limitations and advantages.

Soil Moisture Observations

Scales of Temporal and Spatial Variations of Soil Moisture

It is well known that the complex topography of natural landscapes, with spatially variable vegetation and soil types, and gravitational drainage and infiltration of water after heavy rains, is responsible for very small-scale spatial (tens of meters) and temporal (up to few days) variability in the soil moisture field. This scale of soil moisture variation is of intense interest to hydrologists, who connect it to variations of the water budget within catchments. In the study of climate change, however, a larger scale is of interest, driven by fields of precipitation and evapotranspiration. In addition to this small-scale component of soil moisture variability, analysis of spatial fields and time series of soil moisture observations in the Northern Hemisphere midlatitudes also finds a long-term (about 1–4 months) and large-scale (about 400–800 km) signal related to atmospheric forcing. The meteorological component of soil moisture field variability has been found in observations and later received theoretical explanation. For example, the temporal scale is equal to the field capacity divided by the potential evaporation. The spatial structure of actual soil moisture obviously depends on the distribution of topography, soils, and vegetation on all scales, but the scale of soil moisture variations can be related to the scale of atmospheric forcing.

Climatology

Figure 1 shows the mean seasonal cycle from eight different regions where actual soil moisture observations exist *in-situ* (**Figure 2**). For each region, optimal averaging was used to produce one value representative of the area. Optimal averaging takes into account the spatial scale of soil moisture and gives less weight to repeated information and more weight to independent information. This is particularly important when there is missing information at particular times, producing a changing distribution of stations. It also produces an estimate of the error of the estimated average, which is also shown on the figure.

Western Russia, Illinois, and Iowa show a typical midlatitude seasonal cycle in a climate with precipitation distributed uniformly throughout the year. Soil moisture is high in the winter, and when the snow melts in the spring some of the water infiltrates into the soil, producing the peak soil moisture for the year. During the summer, when evapotranspiration exceeds precipitation, soil moisture falls with the minimum value at the end of the summer. In the autumn, evapotranspiration falls and soil moisture increases until the winter. When the ground is covered by snow,

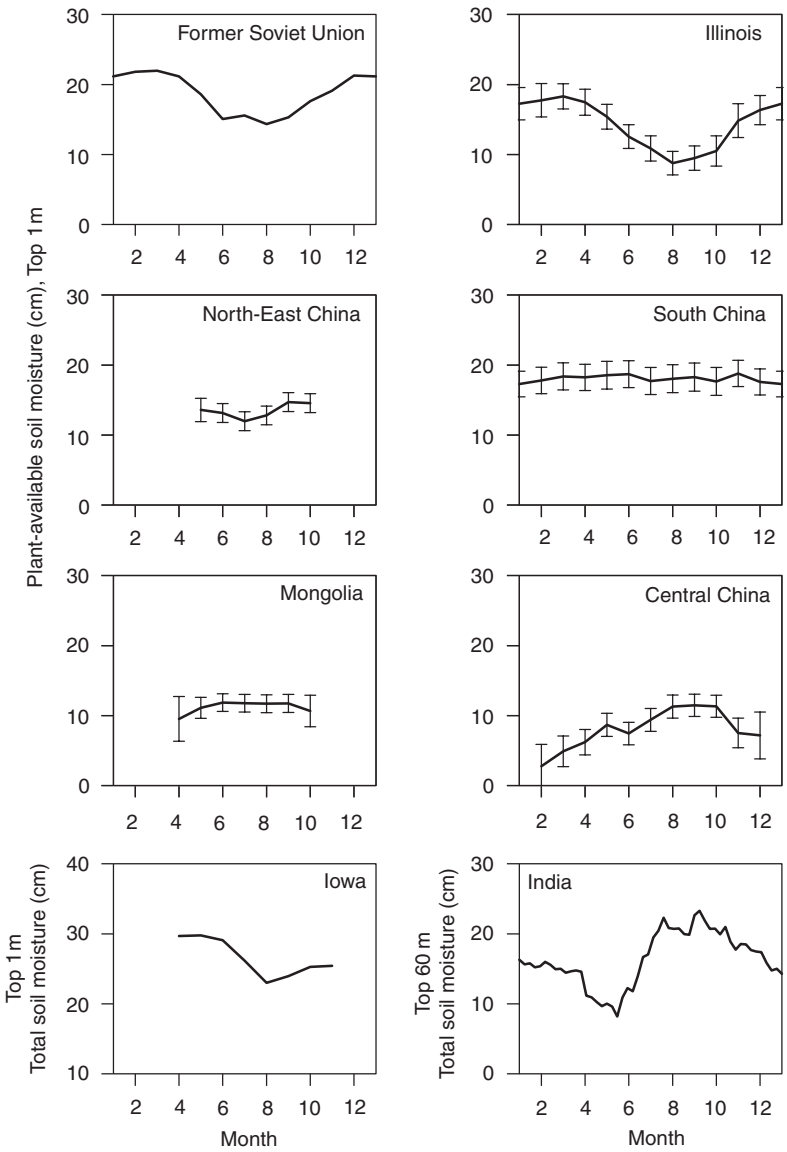

Figure 1 Average seasonal cycles of soil moisture, averaged for the entire length of each data set (**Figure 2**). Units are cm of water in the top 1 m, except the top 60 cm for India. The shape depends on the seasonal cycles of precipitation and evapotranspiration. For those regions where optimal averaging was used, whiskers indicate the estimated one standard deviation error range associated with the spatial distribution of the stations. (Reproduced with permission from Robock A, Vinnikov KY and Srinivasan G, *et al.* (2000) The global soil moisture data bank. *Bulletin of the American Meteorological Society* 81: 1281–1299 © 2000 American Meteorological Society (AMS).)

there is less change in soil moisture. Vapor exchange and infiltration through cracks and worm holes are still important in frozen soils.

The seasonal cycle in eastern Asia, on the other hand, shows a different character, as there is a large summer precipitation maximum associated with the summer monsoon. In Mongolia, during the half-year for which we have data, soil moisture stays almost constant. The increased evapotranspiration is almost exactly matched by the increased precipitation. In India, where evapotranspiration is much larger, soil moisture falls from the end of the rainy season and reaches a minimum just before the onset of the summer monsoon precipitation.

Trends

Figure 3 shows the trends of summer (June, July, August) soil moisture from the longest soil moisture time series available. Although all the stations or

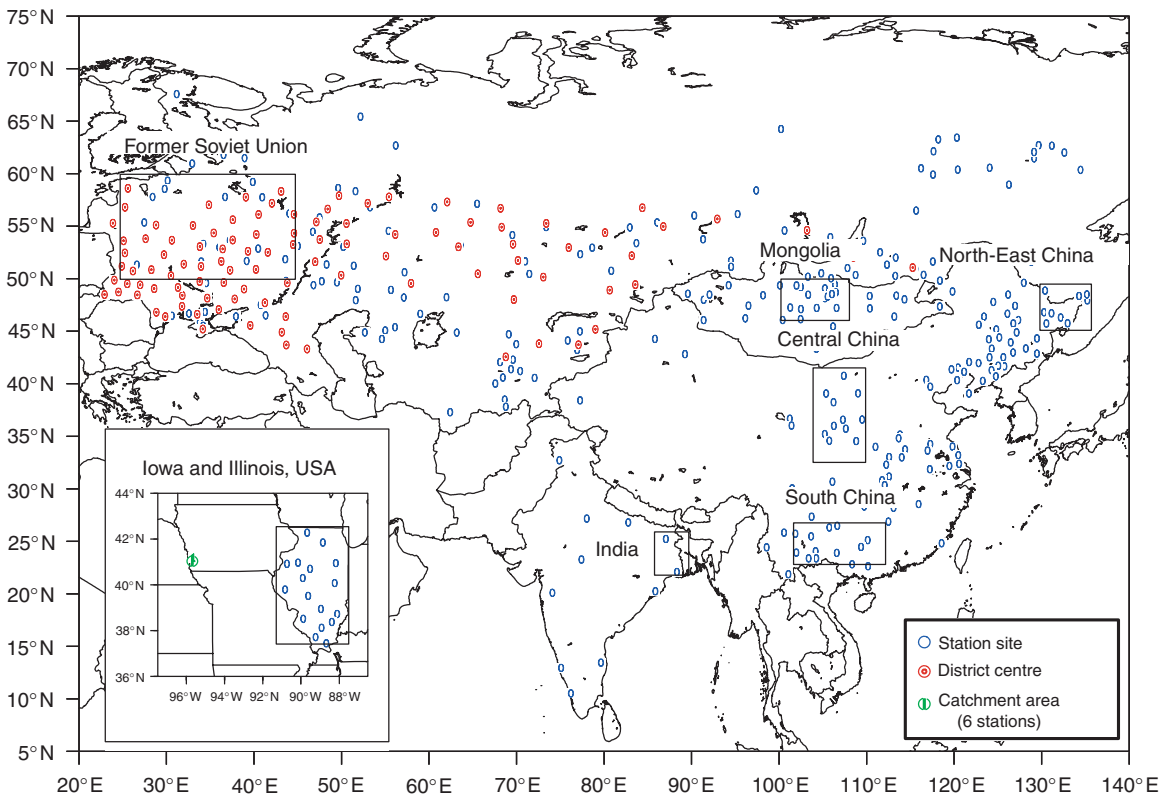

Figure 2 Map of the distribution of the stations in the Global Soil Moisture Data Bank and location of the regions used for sample seasonal cycle plots (**Figure 1**). Part of the Russian data represent averages for administrative districts, rather than individual stations and are indicated as circled dots. (Reproduced with permission from Robock A, Vinnikov KY and Srinivasan G, *et al.* (2000) The global soil moisture data bank. *Bulletin of the American Meteorological Society* 81: 1281–1299 © 2000 American Meteorological Society (AMS).)

regions have an upward trend in temperature during the period (not shown), they all show an upward trend in soil moisture.

Virtually all general circulation model (GCM) simulations of the future climate show summer drying of the midlatitudes. This summer desiccation is caused by evapotranspiration going up faster than precipitation in a greenhouse-gas-warmed world. Observations, however, do not show this effect yet. Better models (discussed below) and longer observational time series will be needed to decide whether this prediction is valid.

Modeling of Soil Moisture

For many purposes, including accurate climate modeling, seasonal prediction, and water resource management, it is crucial to have accurate land surface models. The soil moisture component driven by atmospheric forcing may be modeled using routine meteorological observations at regular meteorological stations. Small-scale variability of the soil moisture field is unpredictable and appears as a stochastic process in this context. However, land surface schemes, whether driven by GCMs or by observations, currently do an imperfect job of simulating the mean seasonal cycle and interannual variation of soil moisture, when compared with observations.

Soil moisture (W) variations in the active soil layer can be calculated with the following equation:

$$\frac{\mathrm{d}W}{\mathrm{d}t} = P + M + WT - E - R \qquad [1]$$

where P is liquid precipitation, M is snowmelt, WT is water table contributions, E is evapotranspiration, and R is runoff. Sources of water for the soil are rain, melting snow, and possible subsurface contributions from a rising water table. Soil moisture is lost by evaporation from bare soil, transpiration through plants, and runoff either from the surface or through subsurface drainage. The dependence of evapotranspiration on soil moisture is a function of soil moisture, incoming energy, and rooting depth. When the soil is close to saturation or very dry, when the incoming

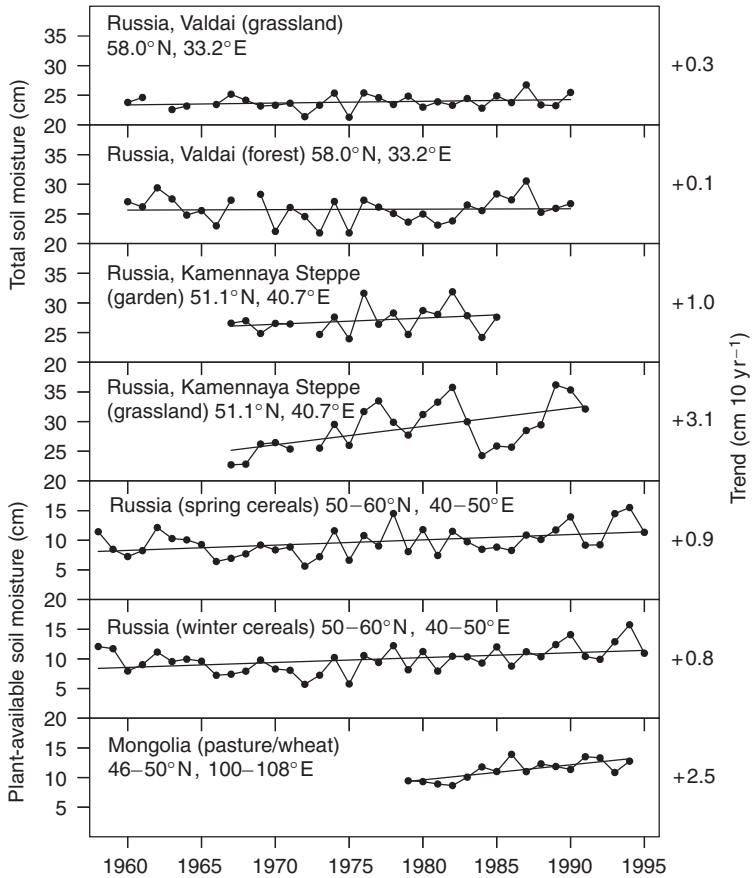

Figure 3 Trends of summer soil moisture from stations or regions with the longest records. (Reproduced with permission from Robock A, Vinnikov KY and Srinivasan G, *et al.* (2000) The global soil moisture data bank. *Bulletin of the American Meteorological Society* 81: 1281–1299 © 2000 American Meteorological Society (AMS).)

energy is low (such as in high latitudes), or when plants have a shallow rooting depth, soil moisture has only a small influence on evapotranspiration. Models of soil moisture changes differ in their treatment of evapotranspiration, runoff, snow, frozen soil, and water table. Some models also explicitly contain a parameter that gives a temperature threshold to separate incoming liquid precipitation from solid precipitation.

The first soil moisture schemes in GCMs were based on the Manabe bucket model with 15 cm field capacity, implicit vegetation, and runoff only with bucket overflow. More recently, models have been developed with explicit consideration of vegetation that account for canopy interception of a portion of the precipitation, and explicit stomatal resistance to transpiration. The latest models also incorporate fluxes of CO_2 and other gases as part of the plant and soil biology and physics. In the past, it was convenient to refer to bucket-type models with implicit vegetation as simplistic, and models with explicit consideration of vegetation as sophisticated, complex, and presumably more accurate. But these more complex models have so far not shown themselves to be more accurate than the simple ones in climate model applications. Runoff can also be modeled in a more complex manner, including partitioning a fraction of heavy precipitation into immediate runoff and subsoil drainage. Different models have different field capacities and rooting depths, and divide the soil into different numbers of vertical layers for temperature and soil moisture computations.

Weather forecast models now incorporate land surface schemes to predict latent and sensible heat fluxes from the Earth's surface, as well as the radiative balance. In the process, soil moisture and other hydrological quantities, such as runoff, are calculated. Archives of climate-model-produced soil moisture, output from GCM runs, are also now becoming available. The soil moisture results from different types of simulations will be different, depending on the way the calculation was done. Some of these come from reanalyses of the global climate system, but the

soil moisture calculations in these reanalyses so far do not conserve water, adding or subtracting moisture to make up for atmospheric model deficiencies. Therefore it is important to not confuse actual observations with model calculations. If a soil moisture data set is to be used to initialize a land surface scheme, then that data set must be calculated with the same land surface scheme, even if the land surface scheme is known to produce a bias in soil moisture. Otherwise, there will be spin-up problems, with the model trying to adjust from the initial values to its own climatology.

There are several international programs now under way to improve land surface models in controlled experiments evaluated by actual soil moisture observations. These include the Project on Intercomparison of Land-surface Parameterization Schemes (PILPS), the Atmospheric Model Intercomparison Project (AMIP), the Global Soil Wetness Project (GSWP), and the Land Surface Data Assimilation Schemes (LDAS) project.

Remote Sensing of Soil Moisture

While satellite microwave data have been used to produce global observations of surface skin wetness (water resting on the surface), there are no existing global soil moisture data sets measured from remote sensing. There are various ideas about how satellite measurements can be used to infer surface (top few centimetres) soil moisture, but none have been shown to be robust products globally yet. Passive and active microwave studies have demonstrated the potential for microwave remote sensing of soil moisture in regions without snow or tall vegetation. Visible and infrared radiation can also be used for indirect satellite soil moisture monitoring, but cannot make measurements when cloudy. Microwave remote sensing offers the most promise for future global data sets but will have to combined with validated land surface models using data assimilation to produce soil moisture fields for the entire active soil layer.

See also

Boundary Layers: Neutrally Stratified Boundary Layer. **Climate Prediction (Empirical and Numerical)**. **Climate Variability:** Seasonal to Interannual Variability. **Global Change:** Biospheric Impacts and Feedbacks. **Hydrology:** Ground and Surface Water. **Observations for Chemistry (Remote Sensing):** Microwave. **Palmer Drought Index**. **Snow (Surface)**. **Weather Prediction:** Seasonal and Interannual Weather Prediction.

Further Reading

Budyko MI (1956) *Heat Balance of the Earth's Surface*. Leningrad: Gidrometeoizdat. (In Russian).

Delworth T and Manabe S (1993) Climate variability and land surface processes. *Advances in Water Resources* 16: 3–20.

Manabe S (1969) Climate and the ocean circulation, 1. The atmospheric circulation and the hydrology of the earth's surface. *Monthly Weather Review* 97: 739–774.

Manabe S, Wetherald RT and Stouffer RJ (1981) Summer dryness due to increase of atmospheric CO_2 concentration. *Climatic Change* 3: 347–384.

Robock A, Vinnikov KY, Srinivasan G, *et al*. (2000) The Global Soil Moisture Data Bank. *Bulletin of the American Meteorological Society* 81: 1281–1299.

Sellers PJ (1992) Biophysical models of land surface processes. In: Trenberth KE (ed.) *Climate System Modeling*, ch. 14, pp. 451–490. London: Cambridge University Press.

Vinnikov KYa and Yeserkepova IB (1991) Soil moisture: empirical data and model results. *Journal of Climate* 4: 66–79.

Vinnikov KY, Robock A, Speranskaya NA and Schlosser CA (1996) Scales of temporal and spatial variability of midlatitude soil moisture. *Journal of Geophysical Research* 101: 7163–7174.